



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

COMMON SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

I/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards)

Under Choice Based Credit System

GROUP – B (ECE, EEE & Mechanical) Branches

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ENG 1101	English *	4	3	1	--	4	30	70	100
B16 ENG 1102	Mathematics-I *	4	3	1	--	4	30	70	100
B16 ENG 1103	Mathematics-II *	4	3	1	--	4	30	70	100
B16 ENG 1105	Physics	4	3	1	--	4	30	70	100
B16 ENG 1107	Engineering Graphics	4	2	--	3	5	30	70	100
B16 ENG 1109	Professional Ethics & Moral Values	2	2	--	--	2	30	70	100
B16 ENG 1111	Physics Lab	2	--	--	3	3	50	50	100
B16 ENG 1113	Workshop	2	--	--	3	3	50	50	100
B16 ENG 1114	NCC/NSS (Audit)	--	--	--	--	3	--	---	--
Total		26	16	4	9	32	280	520	800

Note: Add-On Course: Technical English *

* Common to both Group - A and Group - B

ENGLISH
(Common to All Branches)

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

COURSE OBJECTIVES:

Reading Skills

1. Addressing explicit and implicit meanings of a text on current topics.
2. Understanding the context.
3. Learning new words and phrases.
4. Using words and phrases in different contexts.

Writing Skills

1. Using the basic structure of a sentence.
2. Applying relevant writing formats to create paragraphs, essays, letters, emails, reports and presentations.
3. Retaining a logical flow while writing.
4. Planning and executing an assignment creatively.

Interactive Skills

1. Analyzing a topic of discussion and relating to it.
2. Participating in discussions and influencing them.
3. Communicating ideas effectively.
4. Presenting ideas coherently within a stipulated time.

Life Skills and Core Skills

1. Examining self-attributes and identifying areas that require improvement: self-diagnosis and self-motivation.
2. Adapting to a given situation and developing a functional approach to finding solutions: adaptability and problem solving.
3. Understanding the importance of helping others: community services and enthusiasm.

COURSE OUTCOMES:

1. The overall performance of the students will be enhanced after the course; they will be in a position to make presentations on topics of current interests – politics, famous personalities, science and technology, tourism, work and business environment, with increased public speaking skills.
2. Students will be able to read, listen, speak and write effectively in both academic and non-academic environment.
3. The students will be updated with certain real life situations, which they can handle when come face to face.

SYLLABUS

Listening Skills

Conversations: Life in a Hostel – Eating Away those Blues – Meeting Carl Jung – A Documentary on the Big Cat – A Consultant Interviewing Employees – A Conversation about a Business Idea.

Speaking Skills

Your Favorite Holiday Destination – Describe Yourself – Why we need to save our Tiger – A Dialogue – Your First Interview – Pair Work: Setting up a New Business.

Reading Skills

Reading Comprehension: Famous People – What is Personality, Personality based on Blood Groups – News Report, Magazine Article, Mobile Towers and Health – An Excerpt from a Short Story, An Excerpt from a Biography – Open Letter to Prime Minister, Business Dilemmas: An Email Exchange – A Review of IPL: The Inside Story, Mark Zuckerberg: World's Youngest Billionaire.

Writing Skills

Letter Writing, Essay Writing, Email Writing, Report Writing, Paragraph Writing, Drafting a Pamphlet, Argument Writing, Dialogue Writing.

Grammar

Types of Sentences, Articles, Prepositions, Gerunds and Infinitives, Conjunctions, Tense, Quantifiers, Punctuations, Correction of Sentences, Fill-in the Blanks.

Vocabulary

Synonyms, Antonyms, Idioms, One Word Substitution.

Life Skills and Core Skills

Self Awareness and Self Motivation – Communication, Adaptability – Motivation, Problem Solving – Personal Presentation Skills, Stress Management – Professionalism – Ethics.

TEXT BOOK:

1. Life Through Language: A Holistic Approach to Language Learning. Board of Editors, Pearson Publishers, India 2013.

REFERENCE BOOKS:

1. Basic Vocabulary. Edgar Thorpe, Showick Thorpe. Pearson P. 2008
2. Quick Solutions to Common Errors in English, Angela Bunt. MacMillan P. 2008
3. Know Your English (Volume 1 & 2), by Dr. S. Upendra, Universities Press, India 2012
4. Business Communication Strategies. Maathukutty Monipally. Tata Mc Grahill P. 2009.

MATHEMATICS – I
(Common to All Branches)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

Students learn

1. How to find partial derivatives , Jacobians, how to change from one set of variables to another set of variables
2. Know Geometrical interpretation of partial derivatives
3. Learn applications to errors and approximations, maxima and minima
4. How to form ordinary differential equations (ODE) and solve equations of first order and first degree.
5. Applications of first order ODE to orthogonal trajectories, simple electrical circuits, Newton's law of cooling, Law of natural growth and decay
6. How to solve higher order linear differential equations with constant coefficients, Cauchy's linear equation, Legendre's linear equation and simultaneous first order ODEs
7. How to obtain Fourier Series for a periodic function of period 2π and Period $2C$ (C is a constant)
8. How to obtain Half range Cosine & Sine series and also Parseval's formula

COURSE OUTCOMES:

Students will be able to

1. Find partial derivatives, expand a function of more than one variable in a Taylor series and utilize them for errors and approximations, maxima and minima.
2. Solve a first order ODE and also find orthogonal trajectories and solve problems related to simple applications.
3. Solve a given higher order ODE, an equation with constant coefficients, a Cauchy's equation or a Legendre's equation.
4. Utilize knowledge of Fourier series for solving partial differential equations and also in understanding courses like Signals & Systems

SYLLABUS

Partial Differentiation

Functions of two or more variables – Partial derivatives – Homogeneous Functions – Euler’s Theorem – Total Derivative – Change of Variables – Jacobians – Geometrical Interpretation: Tangent Plane and Normal to a Surface.

Applications of Partial Differentiation

Taylor’s Theorem for functions of two variables – Errors and Approximations – Total Differential – Maxima and Minima of functions of two variables – Lagrange’s Method of Undetermined Multipliers – Differentiation Under the Integral Sign – Liebnitz’s Rules.

Ordinary Differential Equations of First Order and First Degree

Formation of ordinary differential equations (ODEs) – Solution of an ordinary differential equation – Equations of the First Order and First Degree – Linear Differential Equation – Bernoulli’s Equation – Exact Differential Equations – Equations Reducible to exact equations.

Applications of Differential Equations of First Order

Orthogonal Trajectories – Simple electric (LR & CR) Circuits – Newton’s Law of Cooling – Law of Natural growth and decay.

Linear Differential Equations of Higher Order

Solutions of Linear Ordinary Differential Equations With Constant Coefficients – Rules for finding Complimentary Function – Rules for finding the particular integral – Method of variation of parameters – Cauchy’s linear equation – Legendre’s Linear Equation – Simultaneous linear equations.

Fourier series

Introduction - Euler’s Formulae - Conditions for a Fourier Expansion - Functions having points of discontinuity - Change of Interval - Odd and Even Functions - Expansions of Odd or Even Periodic Functions, Half-Range Series - Parseval’s Formula.

TEXT BOOK:

1. Scope and Treatment as in “Higher Engineering Mathematics”, by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company
5. Higher Engineering Mathematics by Dr. M. K. Venkataraman, The National Publishing Company.

MATHEMATICS – II
(Common to All Branches)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

Students learn

1. The concept of rank, Normal form of a matrix, consistency of a system of linear algebraic equations
2. Eigen values and Eigen vectors of a matrix, the Cayley Hamilton theorem
3. Orthogonal reduction of a quadratic form.
4. Complex matrices and their properties
5. Laplace transform, existence, properties.
6. Inverse Laplace transform, convolution theorem, how to solve Ordinary Differential Equations (ODE) and simultaneous ODE
7. Difference equations and their methods of solution.
8. Z-transform, its properties, important results and its use to solve difference equations.

COURSE OUTCOMES:

Students will be capable of

1. Utilizing the knowledge of matrices for solving linear simultaneous equations, find Eigen values and Eigen vectors and handle quadratic forms
2. Utilizing the knowledge of Laplace Transforms to find transforms of important functions that arise in applications and also solve ODE
3. Utilizing the knowledge of Laplace Transforms in courses like Net Works, Signals & Systems and Control Systems
4. Utilizing the knowledge of difference equations and Z-transforms in understanding courses like Discrete Mathematical Structures and also Signals & Systems.

SYLLABUS

Matrices – I

Rank of a matrix - Normal Form – Solutions of Linear System of Equations- Consistency of Linear System of Equations – Rouche's Theorem(statement) - Direct Methods: Gauss Elimination Method, LU Factorization Method – Eigen Values and Eigen Vectors of a Matrix – Properties - Cayley – Hamilton Theorem – Inverse and Powers of a Matrix using Cayley – Hamilton Theorem.

Matrices – II

Diagonalization of a Matrix – Quadratic Forms – Reduction of Quadratic Form to Canonical Form – Nature of a Quadratic Form – Complex Matrices: Hermitian, Skew-Hermitian and Unitary Matrices and their Properties.

Laplace Transforms-I

Introduction – Existence Conditions – Transforms of Elementary Functions – Properties of Laplace Transforms – Transforms of Derivatives – Transforms of Integrals – Multiplication by 't' – Division by t – Evaluation of Integrals by Laplace Transforms – Laplace Transforms of Unit Step Function, Unit Impulse Function and Periodic Functions.

Laplace Transforms-II

Inverse Laplace Transform – different methods - Convolution Theorem – Applications of Laplace Transforms to Ordinary Differential Equations, Simultaneous Linear Differential Equations with Constant Coefficients.

Difference Equations

Definition - order and solution of a difference equation - Formation of difference equations - Linear difference equations - Rules for finding C.F. - Rules for finding P.I.- Simultaneous difference equations with constant coefficients. Application to deflection of a loaded string.

Z-transforms

Z-transforms – definition - some standard Z-transforms - Linear property, Damping rule - some standard results - shifting rules - initial and final value theorems - convolution theorem - Evaluation of inverse transforms - Applications of Z-transform to difference equation.

TEXT BOOK:

1. Scope and Treatment as in “Higher Engineering Mathematics”, by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company

PHYSICS
(Common to ECE, EEE & Mechanical)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To understand the basic and advanced concepts of classical and modern physics – Identify, analyze, formulate and solve a wide variety of real world problems.
2. To develop scientific reasoning and problem solving methods.
3. To bring awareness of latest technologies (physics based) – lasers and fiber optics.

COURSE OUTCOMES:

1. Students learn in depth about the topics of Lasers, fiber optics, quantum mechanical theory and classical theories of thermodynamics and electromagnetism.
2. Students understand the classical and modern concepts.

SYLLABUS**Thermodynamics**

Introduction, Heat and Work, First Law of Thermodynamics and applications, Reversible and Irreversible Process, Carnot Cycle and Efficiency, Second Law of Thermodynamics, Carnot's Theorem, Entropy, Second Law in terms of entropy, Entropy and disorder, Third Law of Thermodynamics (Statement Only).

Electromagnetism

Effect of Magnetic Field on – moving charges, current in long straight wire and rectangular Current Loop, Hall Effect, Biot-Savart's Law, B near a Long Wire, B for a Circular Current Loop, Ampere's Law, B for a Solenoid, Faraday's Law of electromagnetic induction, Lenz's law, Inductance of a solenoid, L-R Circuit, Displacement Current, Maxwell's Equations (integral form) and their significance (without derivation).

Interference

Principle of Super Position – Young's Experiment – Coherence – Inference in thin transparent films, Newton's Rings, Michelson Interferometer and its applications.

Lasers

Introduction, spontaneous and stimulated emissions, requirements of laser device, Ruby Laser, Gas Laser (He-Ne Laser), Semiconductor diode Laser, Characteristics and applications of Lasers.

Optical Fibers

Introduction, Principles of light propagation in optical fiber, Acceptance angle, Numerical aperture, types of fiber, Applications of Optical Fibers, Optical Fiber in Communications, advantages.

Ultrasonics

Definition, Production of Ultrasonics by Magnetostriction and Piezoelectric methods, detection methods, acoustic grating, characteristics and applications of Ultrasonics.

Modern Physics

Introduction, de Broglie concept of matter waves, Properties of matter waves, experimental verification (Davisson - Germer experiment), Heisenberg uncertainty principle, Wave function and its physical significance, Schrodinger time independent wave equation, application to a particle in a box, Band theory of Solids, Kronig - Penney model (qualitative treatment), Origin of energy band formation in solids, Classification of materials into conductors, semi conductors and insulators .

Nanophase Materials

Definition, Synthesis – Synthesis methods, Condensation and Ball milling, Chemical vapour deposition method – sol-gel methods, Characterisation, analysis and applications of nano materials.

TEXT BOOKS

1. Physics by David Halliday and Robert Resnick – Part I and Part II, Wiley Eastern India(pvt.) Ltd.
2. Engineering Physics by M.N. Avadhanulu & P.G. Kshirasagar; S. Chand & Company Ltd

REFERENCE BOOKS

1. Modern Engineering Physics by A.S. Vasudeva, S. Chand & Company Ltd
2. University Physics by Young and Freedman, Addison-Wesley
3. Engineering Physics by R.K. Gaur and S.L. Gupta, Dhanpat Rai & CO

ENGINEERING GRAPHICS
(Common to ECE, EEE & Mechanical)

Theory	: 2 Periods	Sessionals	: 30
Lab	: 3 Periods	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To highlight the significance of universal language of engineers.
2. To impart basic knowledge and skills required to prepare engineering drawings.
3. To impart knowledge and skills required to draw projections of solids in different contexts.
4. To visualize and represent the pictorial views with proper dimensioning and scaling.

COURSE OUTCOMES:

Students will be able to

1. Apply principles of drawing to represent dimensions of an object.
2. Construct polygons and engineering curves.
3. Draw projections of points, lines, planes and solids.
4. Represent sectional views of solids.
5. Develop the surfaces of regular solids.
6. Draw the isometric views of solids and combination of solids.

SYLLABUS**Introduction**

Lines, Lettering and Dimensioning. Geometrical Constructions.

Curves

Conic sections: General construction of ellipse, parabola and hyperbola. Construction of involutes. Normal and Tangent.

Projections of Points

Principal or Reference Planes, Projections of a point situated in any one of the four quadrants

Projections of Straight Lines

Projections of straight lines parallel to both reference planes, perpendicular to one reference plane and parallel to other reference plane, inclined to one reference plane and parallel to the other reference plane. Projections of straight line inclined to both the reference planes:

Projections of Planes

Projection of Perpendicular planes: Perpendicular to both reference planes, perpendicular to one reference plane and parallel to other reference plane, perpendicular to one reference plane and inclined to other reference plane. Projection of Oblique planes. Introduction to Auxiliary Planes.

Projections of Solids

Types of solids: Polyhedra and Solids of revolution. Projection of solids in simple positions: Axis perpendicular to horizontal plane, Axis perpendicular to vertical plane and Axis parallel to both the reference planes, Projection of Solids with axis inclined to one reference plane and parallel to other and axes inclined to both the reference planes.

Projections of Section of Solids

Section Planes: Parallel and inclined section planes, Sections and True shape of section, Sections of Solids: Prism, Pyramid, Cylinder and Cone .

Development of Surfaces

Methods of Development: Parallel line development and radial line development. Development of a cube, prism, cylinder, pyramid and cone.

Isometric Views

Introduction to Isometric projection, Isometric scale and Isometric view. Isometric views of simple planes. Isometric view of Prisms, Pyramids, cylinder and cone. Isometric view of a combination of solids.

TEXT BOOK:

1. Elementary Engineering Drawing by N.D.Bhatt, Charotar Publishing House.

REFERENCE BOOK:

1. Engineering Graphics by K.L. Narayana and P. Kannaiah, Tata Mc-Graw Hill.

PROFESSIONAL ETHICS AND MORAL VALUES

(Common to ECE, EEE & Mechanical)

Theory : 2 Periods
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 2

COURSE OBJECTIVES:

1. To inculcate Ethics and Human Values into the young minds.
2. To develop moral responsibility and mould them as best professionals.
3. To create ethical vision and achieve harmony in life.

COURSE OUTCOME:

1. By the end of the course student should be able to understand the importance of ethics and values in life and society.

SYLLABUS**Ethics and Human Values**

Ethics and Values, Ethical Vision, Ethical Decisions, **Human Values** – Classification of Values, Universality of Values.

Engineering Ethics

Nature of Engineering Ethics, Profession and Professionalism, Professional Ethics, Code of Ethics, Sample Codes – IEEE, ASCE, ASME and CSI.

Engineering as Social Experimentation

Engineering as social experimentation, Engineering Professionals – life skills, Engineers as Managers, Consultants and Leaders, Role of engineers in promoting ethical climate, balanced outlook on law.

Safety Social Responsibility and Rights

Safety and Risk, moral responsibility of engineers for safety, case studies – Bhopal gas tragedy, Chernobyl disaster, Fukushima Nuclear disaster, Professional rights, Gender discrimination, Sexual harassment at work place.

Global Issues

Globalization and MNCs, Environmental Ethics, Computer Ethics, Cyber Crimes, Ethical living, concept of Harmony in life.

TEXT BOOKS:

1. Govindharajan, M., Natarajan, S. and Senthil Kumar, V.S., Engineering Ethics, Prentice Hall of India, (PHI) Delhi, 2004.
2. Subramainam, R., Professional Ethics, Oxford University Press, New Delhi, 2013.

REFERENCE BOOK:

1. Charles D, Fleddermann, “Engineering Ethics”, Pearson / PHI, New Jersey 2004 (Indian Reprint)

PHYSICS LAB
(Common to ECE, EEE & Mechanical)

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

LIST OF EXPERIMENTS

1. Sonometer – verification of laws of transverse vibrations in stretched strings.
2. Melde’s Experiment – Determination of the frequency of an electrically maintained tuning fork.
3. Newton’s Rings – Determination of radius of curvature of a convex lens.
4. Diffraction Grating – Determination of Wavelengths of lines of mercury spectrum using spectrometer by normal incidence method.
5. Determination of Cauchy’s constants of the material of the given prism using Spectrometer and mercury light.
6. Wedge Method – Determination of thickness of a paper by forming parallel interference fringes.
7. Variation of magnetic field along the axis of current carrying circular coil – Stewart and Gee’s apparatus.
8. Carey Foster’s bridge – Verification of laws of resistances.
9. Lee’s Method – Determination of coefficient of thermal conductivity of a bad conductor.
10. Calibration of voltmeter using potentiometer.
11. Calibration of low range Ammeter using potentiometer.
12. Laser – Diffraction.

REFERENCE BOOK:

1. Advanced Practical Physics Vol I & II by SP Singh and MS Chauhan, Pragathi Prakasam Publications

WORKSHOP
(Common to ECE, EEE & Mechanical)

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

COURSE OBJECTIVE:

1. To impart hands-on training on basic engineering trades.

COURSE OUTCOMES:

Student will be able to

1. Use various tools to prepare basic carpentry and fitting joints.
2. Fabricate simple components using tin smithy.

SYLLABUS

Carpentry

Bench Work, tools used in carpentry.

Jobs for Class work – half lap joint, mortise and tenon joint, half – lap dovetail joint, corner dovetail joint, central bridle joint.

Sheet Metal

Tools used in sheet metal work, Laying development of the sheet metal jobs, soldering.

Jobs for class works – Square tray, taper tray (sides), funnel, elbow pipe joint, 60° pipe joint.

Fitting

Tools used in fitting work, Different files, chisels, hammers and bench vice.

Jobs for class work – Square, hexagon, rectangular fit, circular fit and triangular fit.

REFERENCE BOOK:

1. Elements of workshop technology, Vol.1 by S. K. and A. K. Choudary, Media Promoters & Publishers Pvt. Ltd.

Code: B16 ENG 1114

NCC/NSS
(ECE, EEE & Mechanical)

Contact Hrs. : 3 Hrs.

This is an Audit Course. Every student should choose either NCC or NSS at the starting of the semester and pursue the same in that semester. It does not carry any Grade or Credit. It will not be included in the Grade Memo / Certificate.

ADD -ON COURSE
TECHNICAL ENGLISH
(Common to All Branches)

Theory: 2 Periods

COURSE OBJECTIVES:

1. To improve the language proficiency of the students regarding the technical aspects.
2. To enhance the awareness about the formal speaking and writing skills in different contexts.
3. Motivating the students to improve their personal skills.

COURSE OUTCOMES:

1. Students improve their language skills in formal/ technical contexts.
2. They enhance their understanding of technical terms.
3. They improve their personal skills.

SYLLABUS

1. Defining Technical Writing.
2. Writing Skills
Business letters, E-mails
Report Writing
Writing Proposals & Business Quotations
Brochures, Newsletters
How to write an abstract & research paper
3. Speaking Skills
Technical Descriptions
PPT Presentations
Paper Presentations
4. Vocabulary
Technical Terms
5. Behavioral / Attitudinal Skills

TEXT BOOK:

1. Effective Technical Communication by Ashraf Rizvi: Tata McGraw Hill.

REFERENCE BOOKS:

1. Handbook for Technical Writing by David A. McMurrey and Joanne Buckley: Cengage Learning.
2. Technical Writing: Process and Product by Sharon J. Gerson and Steven M. Gerson: Pearson Publications

COMMON SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

I/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards)
Under Choice Based Credit System

GROUP – B (ECE, EEE & Mechanical) Branches

II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ENG 1201	Mathematics-III *	4	3	1	--	4	30	70	100
B16 ENG 1203	Chemistry	4	3	1	--	4	30	70	100
B16 ENG 1205	Computer Programming Using C & Numerical Methods	4	3	1	--	4	30	70	100
B16 ENG 1207	History of Science and Technology	2	2	--	--	2	30	70	100
DS	Department Subject #	4	3	1	--	4	30	70	100
B16 ENG 1210	Chemistry Lab	2	--	--	3	3	50	50	100
B16 ENG 1212	Computer programming Using C & Numerical Methods Lab	2	--	--	3	3	50	50	100
B16 ENG 1213	English Language Lab *	2	--	--	3	3	50	50	100
B16 ENG 1214	Sports (Audit)	--	--	--	--	3	--	--	--
Total		24	14	4	9	30	300	500	800

Note: Add-On Course: Technology Course-I *

*** Common to both Group-A and Group-B**

ECE: B16 EC1208: Electronic Devices and Circuits

EEE: B16 EE1208: Circuit Theory

ME: B16 ME1208: Metallurgy and Materials Engineering

Code: B16 ENG 1201

MATHEMATICS – III
(Common to All Branches)

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

COURSE OBJECTIVES:

Students learn

1. Equations of a line, shortest distance between skew lines
2. About sphere, cone and cylinder
3. Double integrals, change of order of integration
4. Beta function, Gamma function, error function
5. Applications of double integrals to Areas, Volumes, mass, centre of gravity, Moment of inertia
6. The concept of Fourier Integral, Fourier Transform, properties.
7. Parseval's formulae

COURSE OUTCOMES:

Students will be able to

1. Utilize knowledge of line, sphere etc. in his engineering subjects
2. Utilize the knowledge of Beta and Gamma functions and multiple integrals to evaluate the integrals they come across in their applications
3. Utilize the knowledge of Fourier Transform in courses like Signals and Systems and in the solution of partial differential equations at a later stage

SYLLABUS

Solid Geometry

Equations of a plane, Normal form, Intercept form, Equations of Straight Line – Conditions for a line to lie in a Plane – Coplanar lines – Shortest distance between two skew lines - Intersection of three Planes – Equations of Sphere – Tangent Plane to a Sphere –Cone – Cylinder.

Multiple Integrals-1

Double Integrals - Change of Order of Integration - Double Integrals in Polar Coordinates - Triple Integrals - Change of Variables.

Multiple Integrals-2

Beta Function - Gamma Function - Relation between Beta and Gamma Functions-Error Function - Area enclosed by plane curves - Volumes of solids - Area of a curved surface - Calculation of mass - Center of gravity of a plane lamina- Moment of inertia.

Fourier Transforms

Introduction – definition - Fourier integral - Sine and Cosine integrals - Complex form of Fourier integral - Fourier transform - Fourier Sine and Cosine transforms -Finite Fourier Sine and Cosine transforms - properties of Fourier transforms, Convolution theorem for Fourier transforms - Parseval's identity for Fourier transforms - Fourier transforms of derivatives of a function.

TEXT BOOK:

1. Scope and Treatment as in “Higher Engineering Mathematics”, by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company
5. Higher Engineering Mathematics by Dr. M. K. Venkataraman, The National Publishing Company.

CHEMISTRY
(Common to ECE, EEE & Mechanical)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To develop full pledged laboratories for the analysis of engineering materials.
2. To develop full pledged lab for pollution, corrosion, polymers and electrochemical cells.
3. Extend the infrastructure facilities of the department for the project works of engineering students and also guide the project work under faculty supervision.
4. To develop full pledged lab for analysis of Water, Cement, Hydrogen Peroxide, pollutants and raw materials.

COURSE OUTCOMES:

1. Students learn in-depth about the topics of desalination of sea water, CNG, LPG Biogas, Semiconductors, Liquid crystals, Conducting polymers, fiber reinforced plastics, building materials.
2. Students understand the basic and advanced applied concepts.
3. Students learn to interrelate the theory and with the relevant experiment.
4. Students learn experimental techniques and understand the theory about experiments.

SYLLABUS**Water Chemistry**

Source of water- impurities- Hardness and its determination by EDTA method- Boiler troubles and their removal. Water softening methods- lime soda, zeolite and ion exchange. Municipal water treatment- Break point chlorination. Desalination of sea water – electro dialysis and reverse osmosis methods.

Building Materials

Portland cement: Manufacture-Chemistry involved in setting and hardening of cement –Cement concrete -RCC –Decay of concrete.

Refractories: Classification-Properties and Engineering applications.

Ceramics: Classification-Properties and uses.

Solid State Chemistry

Classification of solids-Types of crystals-properties-Imperfections in crystals. Band theory of solids. Chemistry of semi conductors –Intrinsic, Extrinsic, compound and defect. Organic semi conductors Purification of solids by Zone refining-Single crystal growth – epitaxial growth. Elementary ideas on Liquid crystals.

Corrosion Chemistry

Definition of corrosion- Types of corrosion-chemical & electrochemical corrosion –Pitting, stress corrosion, Galvanic corrosion, Water line corrosion Factors affecting corrosion – Prevention of corrosion- Cathodic protection. Corrosion inhibitors Protective coatings-Metallic coatings, electro plating, electroless plating, chemical conversion coatings- phosphate coatings, chromate coatings, anodizing. Organic coatings-Paints.

Polymers And Plastics

Definition-Types of polymerization – mechanism of free radical polymerization. plastics-Thermosetting and thermoplastic resins cellulose derivatives, vinyl resins, nylon 6,6, bakelite-Compounding of plastics-Fabrication of plastics. Fiber reinforced plastics – conducting polymers -Engineering applications of polymers.

Fuels And Lubricants

Solid fuels: Coal –Analysis of coal – Metallurgical coke- manufacture-Engineering applications.

Petroleum-refining-Knocking and Octane number of gasoline-Cetane number of diesel oil. Synthetic petrol – LPG-CNG–Applications. Rocket fuels – Propellants-classification.

LUBRICANTS: Principles of lubrications, classification of lubricants and properties of lubricants (any five)

TEXT BOOKS:

1. Engineering chemistry by Jain & Jain 15th edition Dhanpatrai publishing company.
2. Engineering chemistry by Dr. K.AnjiReddy and Dr. M.Sita rama reddy, silicon publications, 2015.

REFERENCE BOOKS:

1. A Text Book of Engineering Chemistry by S.S.Dara; S.Chand & Company Ltd.
2. Engineering Chemistry by B.Siva Sankar Mc.Graw Hill Education(India) Pvt.Ltd.
3. A Text Book of Engineering Chemistry by Sashi Chawla, Dhanpatrai & Co.

COMPUTER PROGRAMMING USING C & NUMERICAL METHODS
(Common to ECE, EEE & Mechanical)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To make the student familiar with programming in C and enable the student to implement the numerical methods described in this course using C as Programming Language.
2. To introduce the programming principles
3. To familiarize the variables construction, declarations, initializations etc.,
4. To introduce the Control statement
5. To introduce the arrays, structures, Unions, Files etc.,
6. To give the overall idea on Pointers
7. To give the idea about dynamic memory allocation
8. The nature of C language is emphasized in the wide variety of examples and applications.
9. To know about some popular programming languages.
10. To solve various numerical Methods
11. Student can write various programs by applying elementary concepts like arrays, structures, Unions, Files etc.,

COURSE OUTCOMES:

1. Student can understand basic terminology used in C programming.
2. Student can write programs by applying elementary algorithms to solve problems in C language.
3. Student can write, compile and debug programs in C language.
4. Student can Write programs to solve numerical methods
5. Student can be familiar with finite precision computation.

SYLLABUS

Introduction To C

Basic structure of C program, Constants, Variables and data types, Operators and Expressions, Arithmetic Precedence and associativity, Type Conversions. Managing Input and Output Operations, Formatted Input, Formatted Output.

Decision Making, Branching, Looping, Arrays & Strings: Decision making with if statement, Simple if statement, The if...else statement, Nesting of if...else statement, the else..if ladder, switch statement, the (?:) operator, the GOTO statement., The while statement, the do statement, The for statement, Jumps in Loops, One, Two-dimensional Arrays, Character Arrays. Declaration and initialization of Strings, reading and writing of strings, String handling functions, Table of strings.

Functions

Definition of Functions, Return Values and their Types, Function Calls, Function Declaration, Category of Functions: No Arguments and no Return Values, Arguments but no Return Values, Arguments with Return Values, No Argument but Returns a Value, Functions that Return Multiple Values. Nesting of functions, recursion, passing arrays to functions, passing strings to functions, The scope, visibility and lifetime of variables. .

Pointers

Accessing the address of a variable, declaring pointer variables, initializing of pointer variables, accessing variables using pointers, chain of pointers, pointer expressions, pointers and arrays, pointers and character strings, array of pointers, pointers as function arguments, functions returning pointers, pointers to functions, pointers to structures-Program Applications

Structure and Unions

Defining a structure, declaring structure variables, accessing structure members, structure initialization, copying and comparing structure variables, arrays of structures, arrays within structures, structures within structures, structures and functions and unions, size of structures and bit-fields- Program applications.

File handling

Defining and opening a file, closing a file, Input/ Output operations on files, Error handling during I/O operations, random access to files and Command Line Arguments- Program Applications.

Numerical Methods

Solutions of Algebraic and Transcendental Equations: Bisection Method, Newton Raphson Method.

Interpolation - Newton's forward and backward Interpolation, Lagrange's Interpolation in unequal intervals.

Solutions of Linear Equations: Gauss Elimination Method, Jacobi and Gauss Seidel Methods. Numerical Integration: Trapezoidal rule, Simpson's 1/3 rule.

Solutions of Ordinary First Order Differential Equations: Euler's Method, Modified Euler's Method and Runge-Kutta Method.

TEXT BOOKS:

1. Programming in ANSI C, E Balagurusamy, 6th Edition. McGraw Hill Education (India) Private Limited.
2. Introduction to Numerical Methods, SS Sastry, Prentice Hall India Pvt. Ltd.

REFERENCE BOOKS:

1. Let Us C, Yashwant Kanetkar, BPB Publications, 5th Edition.
2. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
3. The C –Programming Language' B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt. Ltd.
4. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific

HISTORY OF SCIENCE AND TECHNOLOGY

(Common to ECE, EEE & Mechanical)

Theory : 2 periods
Exam : 3 Hrs.

Sessionals :30
Ext. Marks :70
Credits : 2

COURSE OBJECTIVES:

1. To know the contributions of scientists for the development of society over a period of time.
2. To understand the Science and Technological developments that lead to human welfare.
3. To appreciate the Science and Technological contributions for the development of various sectors of the economy.
4. To identify the technological transfer versus economic progress of the countries.

COURSE OUTCOMES:

1. By the end of this course the students should be able to understand the contribution of Scientific and Technological developments for the benefit of society at large.

SYLLABUS**Historical Perspective of Science and Technology**

Nature and Definitions; Roots of Science – In Ancient Period and Modern Period (During the British Period); Science and Society; Role of Scientist in the Society.

Policies and Plans after Independence

Science and Technology Policy Resolutions; New Technology Fund; Technology Development (TIFAC); Programs aimed at Technological Self Reliance; Activities of Council of Scientific and Industrial Research.

Science and Technological Developments in Critical Areas

Space – The Indian Space Program: India's Geostationary Satellite Services – INSAT System And INSAT Services; **Defense Research and Technology** – Research Coordination, Research efforts and Development of technologies and Spin-off technologies for civilian use; **Nuclear Energy** –Effects of a nuclear explosion and India's safety measures.

Impact of Science and Technology in Major Areas

Ocean Development: Objectives of Ocean Development, Biological and Mineral resources, Marine Research and Capacity Building; **Biotechnology:** Meaning, Biotechnology techniques- Bioreactors, Cell fusion, Cell or Tissue Culture, DNA Fingerprinting, Cloning, Artificial Insemination and Embryo Transfer Technology and Stem Cell Technology; Application of Biotechnology – Medicine, Biocatalysts, Food Biotechnology, Fuel and Fodder and Development of Biosensors.

Technology Transfer and Development

Transfer of Technology – Types, Methods, Mechanisms, Process, Channels and Techniques;
Appropriate Technology - Criteria and Selection of an Appropriate Technology; Barriers of Technological Change. .

TEXT BOOKS:

1. Kalpana Rajaram, Science and Technology in India, Published and Distributed by Spectrum Books (P) Ltd., New Delhi-58.
2. Srinivasan, M., Management of Science and Technology (Problems & Prospects), East –West Press (P) Ltd., New Delhi.

REFERENCE BOOKS:

1. Dr. G.R. Kohli, History of Science and Technology and Environmental Movements in India, Surjeet Publications, New Delhi

ELECTRONIC DEVICES AND CIRCUITS (For ECE)

Theory	: 3 periods	Sessionals	: 30
Tutorials	: 1 period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To give exposure to students on semiconductor physics of the intrinsic and extrinsic semiconductors.
2. To give exposure to students on the basics of semiconductor diodes, special purpose diodes like Zener diode, Photo diode, LED, Schottky barrier diode, PIN diode, varactor diode and tunnel diode.
3. To give exposure to students on rectifier circuits using diodes.
4. To give exposure to students on basics of BJT, JFET and MOSFET and biasing of BJT and FET.
5. To give exposure to students on the analysis of transistor at low and high frequencies.

COURSE OUTCOMES:

After completion of the course the students will be able to

1. Understand the physical structure, principles of operation, electrical characteristics and circuit models of diodes, BJTs and FETs.
2. Use this knowledge to analyze and design basic electronic application circuits.
3. Extend the understanding of how electronic circuits and their functions fit into larger electronic systems.

SYLLABUS

Transport Phenomena in SemiConductors

Mobility and conductivity, intrinsic and extrinsic semiconductors, mass action law, charge densities in a semiconductors, Hall Effect, generation and recombination of charges, drift and diffusion currents, continuity equation, injected minority carrier charge, potential variation in graded semiconductors.

PN junction diode

Open circuited PN junction , PN junction as a rectifier, current components in a PN diode, V-I characteristics and its temperature dependence, transition capacitance, charge control description of a diode, diffusion capacitance, junction diode switching times, Zener diode, Tunnel Diode, Photo diode, Point Contact diode, Schottky barrier diode, varactor diode, PIN diode, LED.

Diode Rectifiers

Half wave, Full wave and Bridge Rectifiers with and without filters, Ripple factor and regulation characteristics

Bipolar junction transistors

Introduction to BJT, operation of a transistor and transistor biasing for different operating conditions, transistor current components, transistor amplification factors: α, β, γ relation between α and β, γ early effect or basewidth modulation, common base configuration and its input and output characteristics, common emitter configuration and its input and output characteristics, common collector configuration and its input and output characteristics, Comparison of CE, CB and CC Configurations, Break- down in transistors, Photo Transistor.

Field Effect transistors

JFET and its characteristics, pinch off voltage, FET small signal model, MOSFET and its characteristics

Transistor Biasing Circuits

The operating point, Bias stability, different types of biasing techniques, stabilization against variation in I_{co} , V_{BE} , & β . Bias compensation, thermal runaway, thermal stability, Biasing of FETs.

Transistors at low and High frequencies

Transistor hybrid model, h-parameters, Analysis of transistor amplifier circuits using h-parameters, comparison of transistor amplifier configurations, analysis of single stage amplifier, effects of bypass and coupling capacitors, frequency response of CE amplifier, Emitter follower, High frequency model of transistor.

TEXT BOOKS:

1. Integrated electronics analog and digital circuits and systems by Jacob Millman, C Halkias, Chetan D Parikh, Tata McGrawhill publications.
2. Electronic Devices and Circuits Theory by Boylestad, Prentice Hall Publications.

REFERENCE BOOKS:

1. Electronic Principles by Albert Malvino, David J Bates, McGrawhill publications.
2. Electronic Devices and Circuits an Introduction by Allen Mottershead, Prenticehall India.
3. Electronic Devices and Circuits by Salivahanan, Tata McGrawhill publications.

CIRCUIT THEORY
(For EEE)

Theory	: 3 periods	Sessionals	: 30
Tutorials	: 1 period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To enlighten and enrich the students to acquire knowledge about the basics of circuit analysis, concepts of AC circuits, DC/AC network theorems, coupled & three phase circuits.
2. To inculcate problem solving skills and understanding of circuit theory through the application of techniques and principles of Electrical Circuits to common circuit problems.

COURSE OUTCOMES:

1. Able to develop an understanding of the basic fundamental electrical laws, elements of electric Networks and learn the techniques to measure voltage and current.
2. Develops the ability to apply circuit theorems to DC and AC circuits.
3. Able to analyse the coupled & three phase circuits.

SYLLABUS**Fundamentals of Electric Circuits**

Concepts of Electric circuit: EMF, Current, Potential difference, Power and Energy; Concepts of Network: Active and Passive elements, classification of Linear, Non-linear, Unilateral, Bilateral Lumped and Distributed elements; Reference directions for current and voltage; Voltage and Current sources; Voltage-Current Relations of R, L, C elements; Voltage and Current division; Series and Parallel combinations of Resistance, Inductance and Capacitance

D.C Circuits

Kirchhoff's Laws; Nodal Analysis, Mesh Analysis, Source Transformation, Linearity and Superposition Theorem, Thevenin's And Norton's Theorems, Reciprocity theorem, Maximum Power Transfer Theorem, Star-Delta Transformation.

AC Circuits

The Sinusoidal Forcing Function, Phasor Concept, Average and Effective Values of Voltage and Current, Instantaneous and Average Power, Complex Power, Steady State Analysis using Mesh and Nodal Analysis, Resonance.

Three Phase Circuits

Advantages of Three Phase Circuits, Balanced and Unbalanced systems, Relation between Line and Phase Quantities in Star and delta connected circuits, Analysis of Balanced & Unbalanced Three Phase Circuits, Measurement of Power in Three Phase Power Circuits.

Magnetic Circuits

Magneto motive force(MMF), Reluctance, Magnetic flux; Analysis of magnetic circuit, Analogy between Electric & Magnetic circuits, Series Magnetic circuits, Magnetic leakage, B-H curve, Faraday's Laws of Electromagnetic Induction, Induced EMF, Dynamically Induced EMF, Statically Induced EMF, Self-Inductance, Mutual Inductance.(simple numerical problems)

TEXT BOOKS:

1. Engineering Circuit Analysis By W.H. Hayt Jr & J.E. Kemmerly, 5th Ed., Mc.Graw Hill.

REFERENCE BOOKS:

1. Basic Electrical Engineering By V.K Mehta & Rohit Mehta- S.Chand Pub.
2. Sudhakar & Syam Mohan "Network Analysis", Mcgraw Hill

METALLURGY AND MATERIALS ENGINEERING
(For Mechanical Engineering)

Theory	: 3 periods	Sessionals	: 30
Tutorials	: 1 period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

PREREQUISITES:

Basic knowledge in solid state chemistry, alloy systems.

COURSE OBJECTIVES:

The overall objective of the course is to impart knowledge about the engineering materials and their properties and predict their behavior under different working conditions and methods to enhance the mechanical properties of the material Specific objectives include:

1. This course will help in learning crystal structures, crystal defects, space lattices.
2. This course will be helps students to understand systems of solid solutions and their nature.
3. To acquaint the knowledge about the cooling curves and phase diagrams of different alloy systems, calculation of weight percent of phases in alloy system by applying Lever rule.
4. To acquaint knowledge about ferrous alloys, particularly Steel and Cast Irons.
5. To impart knowledge about Equilibrium and Non-Equilibrium diagrams.
6. To impart knowledge about different heat treatment and surface hardening methods in improving the mechanical properties of steels.
7. To impart knowledge about the plastic deformation of materials.
8. To impart knowledge about composite materials.

COURSE OUTCOMES:

Upon completion of this course, students should be able to:

1. Understand crystalline solids and their atomic structures.
2. Suggest and recommend necessary engineering materials for specific applications keeping in view of the cost, design, reliability, life, working conditions and properties of the products.
3. Understand different phase transformations in Iron-Iron Carbide diagram and distinguish between steels and cast irons.
4. Select different materials for tools and components based on functional requirements.
5. Use composite materials for different engineering applications like aerospace, automobile, ship building industry, sports item etc.

SYLLABUS

Structure of crystalline solids

Atomic structure & bonding in solids- Crystal structures-calculations of radius, Coordination Number and Atomic Packing Factor for different cubic structures - Imperfection in solids, point defects, Linear defects, Planar defects and Volume defects- Concept of Slip & twinning.

Phase diagrams

Basic terms- phase rule- Lever rule & free energy of phase mixtures cooling curves- Phase diagram & phase transformation - construction of phase diagrams- binary phase diagrams - Brass, Bronze, Al-Cu and AlSi phase diagrams- Invariant reactions, eutectic, peritectic, eutectoid, peritectoid, metatectic & monotectic reactions, Iron carbon phase diagram & microstructures of plain carbon steel & cast iron

Heat treatment

Heat treatment of steel- Annealing, and its types, normalizing, hardening, tempering, martempering, austempering - TTT diagrams, drawing of TTT diagram, TTT diagram for hypo- & hypereutectoid steels, effect of alloying elements, CCT diagram- Martensitic transformation, nature of martensitic transformation- Surface hardening processes like case hardening, carburizing, cyaniding, nitriding Induction hardening, hardenability, Jominy end-quench test, Age hardening of Al & Cu alloys Precipitation Hardening

Engineering Alloys

Properties, composition, microstructure and uses of low carbon, mild medium & high carbon steels. stainless steels, high speed steels, Hadfield steels, tool steels - Cast irons, gray CI, white CI, malleable CI, SC iron- The light alloys- Al & Mg & Titanium alloys- Copper & its alloys: brasses & bronzes- super alloys, Smart materials- Nano materials.

Composite Materials

Classification of composite materials, dispersion strengthened, particle reinforced and fiber reinforced composite laminates properties of matrix and reinforcement materials and structural applications of different types of composite materials.

TEXT BOOKS:

1. "Materials Science & Engineering- An Introduction", William D. Callister Jr. Wiley India Pvt. Ltd. 6th Edition, 2006, New Delhi.
2. Physical Metallurgy, Principles & Practices", V Raghavan. PHI 2nd Edition 2006, New Delhi.

REFERENCE BOOKS:

1. Introduction to Physical Metallurgy by Sidney H Avner, Tata McGraw-Hill Education 1997
2. Materials Science And Engineering: A First Course By V. Raghavan Phi 5th Edition 2011, New Delhi

CHEMISTRY LAB
(Common to ECE, EEE & Mechanical)

Lab : 3 periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

LIST OF PRACTICAL EXPERIMENTS:

1. Estimation of Sodium hydroxide with HCl (Na_2CO_3 primary standard).
2. Estimation of Iron as Ferrous iron in an Ore Sample.
3. Estimation of Oxalic acid by a redox method
4. Estimation of Calcium in a sample of Portland cement.
5. Estimation of Volume strength of Hydrogen Peroxide.
6. Estimation of Mohr's salt by Potassium dichromate
7. Determination of Hardness of an underground water sample.
8. Estimation of Zinc by EDTA method.
9. Determination of Alkalinity of water sample.

DEMONSTRATION EXPERIMENTS:

10. Determination of Viscosity and Viscosity index of a lubricant.
11. Printed Circuit Board
12. Determination of dissolved oxygen in given water sample.
13. Potentiometric titrations.
14. P^{H} Determination by using P^{H} meter.

REFERENCE BOOKS:

1. Essentials Experimental Engineering Chemistry by Sashi Chawla, Dhanpatrai & Co Pvt. Ltd.
2. Laboratory Manual on Engineering Chemistry by Dr. Sudha Rani, Dhanpatrai Publishing Company.
3. Engineering Chemistry Laboratory Manual by Dr. K.Anji Reddy, Tulip Publications.

COMPUTER PROGRAMMING USING C & NUMERICAL METHODS LAB
(Common to ECE, EEE & Mechanical)

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

LIST OF PROGRAMMES:

1. Write a program to read x, y coordinates of 3 points and then calculate the area of a triangle formed by them and print the coordinates of the three points and the area of the triangle. What will be the output from your program if the three given points are in a straight line.
2. Write a program which generates 100 random numbers in the range of 1 to 100. Store them in an array and then print the array. Write 3 versions of the program using different loop constructs (eg. for, while and do-while).
3. Write a set of string manipulation functions eg. for getting a sub-string from a given position, copying one string to another, reversing a string and adding one string to another.
4. Write a program which determines the largest and the smallest number that can be stored in different data types like short, int, long, float and double. What happens when you add 1 to the largest possible integer number that can be stored?
5. Write a program which generates 100 random real numbers in the range of 10.0 to 20.0 and sort them in descending order.
6. Write a function for transporting a square matrix in place (in place means that you are not allowed to have full temporary matrix).
7. First use an editor to create a file with some integer numbers. Now write a program, which reads these numbers and determines their mean and standard deviation.
8. Implement bisection method to find the square root of a given number to a given accuracy.
9. Implement Newton Raphson Method to determine a root of polynomial equation.
10. Given a table of x and corresponding f(x) values, write a program which will determine f(x) value at an intermediate x value using Lagrange's Interpolation.
11. Write a function which will invert a matrix.
12. Implement Simpson's $1/3^{rd}$ rule for numerical integration.
13. Implement Trapezoidal rule for numerical integration.
14. Write a program to solve a set of linear algebraic equations.
15. Write a program to solve a differential equation using Runge-Kutta Method.

REFERENCE BOOKS:

1. Programming in ANSI C, E Balagurusamy, 6th Edition. McGraw Hill Education (India) Private Limited.
2. Introduction to Numerical Methods, SS Sastry, Prentice Hall India Pvt.Ltd

ENGLISH LANGUAGE LAB
(Common to All Branches)

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

The Language Lab focuses on the production and practices of sounds of language and familiarizes the students with use of English in everyday situations and contexts.

COURSE OBJECTIVES:

1. To make students recognize the sounds of English through Audio-Visual aids.
2. To help students build their confidence and help overcome their inhibitions and self consciousness while speaking in English. *The focus shall be on fluency.*
3. To familiarize the students with stress and intonation and enable them to speak English effectively.

COURSE OUTCOMES:

1. Students will be sensitized towards recognition of English sound pattern.
2. The fluency in speech will be enhanced.

SYLLABUS

1. English Sound Pattern – Letters
2. Sounds of English
3. Pronunciation
4. Stress and Intonation

Laboratory Practice Sessions:

1. Letters and Sounds, Worksheet-1
2. Interactions-1, Worksheet-2
3. The Sounds of English, Worksheet-3
4. Interactions-2, Worksheet-4
5. Pronouncing Words-Some Important Patterns, Worksheet-5
6. Interactions-3, Worksheet-2
7. Stress and Intonation, Worksheet-2

REFERENCE BOOKS:

1. Speak Well, Board of Editors, Orient Black Swan Publishers, Hyderabad, 2012
2. Cambridge English Pronouncing Dictionary, Cambridge University Press, India, 2012.
3. A Textbook of English phonetics for Indian students by T. Balasubramanian, Macmillan publisher, 1981.

Code: B16 ENG 1214

SPORTS
(Common to ECE, EEE & Mechanical)

Contact Hrs. 3 Hrs.

This is an Audit Course. It does not carry any Grade or Credit. It will not be included in the Grade Memo / Certificate.

ADD ON COURSE
TECHNOLOGY COURSE – I
(Professional C Programming)
(Common to All Branches)

Total Hrs: 30

COURSE OBJECTIVES:

1. On successful completion of course students will be able to reinforce their learning of C, read code, and write good procedural code.

COURSE OUTCOMES:

1. Students will be able to solve a series of graduated problems
2. Students will be able to do projects in 'C' Language.

SYLLABUS

Introduction to Linux

Introduction to C with problem solving

Functions usage and application with solved problems

Introduction to pointers and applications with examples

Structures and Unions and file handling with program applications

Project using C

RESOURCES:

Talent Sprint Problem Bank, Project Euler



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

II/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards)

Under Choice Based Credit System

MECHANICAL ENGINEERING

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ENG 2101	Mathematics-IV	4	3	1	--	4	30	70	100
B16 ME 2101	Mechanics of Solids	4	3	1	--	4	30	70	100
B16 ME 2102	Thermodynamics	4	3	1	--	4	30	70	100
B16 ME 2103	Manufacturing Process	4	4	--	--	4	30	70	100
B16 ME 2104	Engineering Mechanics	4	3	1	--	4	30	70	100
B16 ME 2105	Mechanical Engineering Drawing	4	--	--	4	4	30	70	100
B16 ME 2107	Mechanical Engineering Lab	2	--	--	3	3	50	50	100
B16 CE 2108	Mechanics of Solids Lab	2	--	--	3	3	50	50	100
B16 ENG 2104	English Proficiency	2	1	1	--	2	50	50	100
B16 ME 2108	Auto CAD	1	--	--	2	2	50	--	50
Total		31	17	5	12	34	380	570	950

Code: B16 ENG 2101

MATHEMATICS – IV
(Common to CIV,ECE,EEE & ME)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. The concepts of Gradient, Divergence, Curl, Directional derivative, solenoidal and Irrotational fields
2. Green's, Stokes' and Divergence theorems
3. Classification of 2nd order Partial Differential Equations as well as solution of 1-Dimensional Wave equation and 1-Dimensional Heat equation
4. the concept of Analytic function, CR equations
5. Cauchy's Integral Theorem and Integral Formula
6. Taylor and Laurent series, Residues and Residue theorem

Course Outcomes:

Students will be able to

1. Apply the concepts of Gradient, Divergence, Curl, Directional derivative, solenoidal and Irrotational fields
2. Determine scalar potential, circulation and work done
3. Evaluate integrals using Green's, Stokes' and Divergence theorems
4. Obtain the solution of 1-D wave equation and 1-D heat equation
5. Determine the zeroes and poles of functions and residues at poles
6. Evaluate certain real definite integrals that arise in applications by the use of Residue theorem

SYLLABUS

Vector Calculus-1

Definitions of Scalar and Vector point functions, Differentiation of vectors, Vector differential operator del, Del applied to scalar point function – gradient, Del applied to vector point function- divergence and curl, physical interpretation of gradient, divergence and curl(without proof), Del applied twice to a point function, Del applied to product of two functions, Irrotational and Solenoidal Fields, scalar potential

Vector Calculus-2

Integration of vectors, line integral, circulation, work done, surface integral, Flux, Green's, Stokes' and Gauss Divergence Theorems (Without proofs). Introduction to orthogonal curvilinear coordinates, cylindrical polar coordinates and spherical polar coordinates.

Applications Of Partial Differential Equations

Classification of second order partial differential equations, Method of separation of variables, One –dimensional wave equation- vibrations of a stretched string (no derivation)-, one-dimensional heat equation – Heat flow along a long horizontal bar (no derivation) (problems on heat equation involving homogeneous end conditions only), two dimensional Laplace equation in Cartesian coordinates.

Complex Variables-1

Review- Cartesian form and polar form of a complex variable, Real and imaginary parts of z^n , e^z , $\sin z$, $\sinh z$ and $\log z$.

Limit and continuity of a function of the complex variable, derivative, analytic function, properties of Analytic functions, Cauchy- Riemann equations, Harmonic functions and Orthogonal system, application of analytic function to flow problems, geometric representation of $w=f(z)$, conformal mapping – Bilinear transformation only.

Complex Variables-2

Integration of complex functions, Cauchy's theorem, Cauchy's integral formula (statements only) . Taylor and Laurent series expansions of functions (statement of theorems only), zeros and singularities, Residue, calculation of residues, Cauchy's Residue theorem (without proof), Evaluation of real and definite integrals- integration around a unit circle

Text Book:

1. "Higher Engineering Mathematics", by Dr.B.S.Grewal, 43rd Edition, Khanna Publishers.

Reference Books:

1. Advanced Engineering Mathematics, by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N.P.Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics, by H.K.Dass, S.Chand Company.
4. Higher Engineering Mathematics, by B.V.Ramana, Tata Mc Graw Hill Company.
5. Higher Engineering Mathematics, by Dr. M.K.Venkatraman, The National Publishing Company.

MECHANICS OF SOLIDS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To impart the knowledge on internal behavior of mechanical elements under the action of applied loads.

Course Outcomes:

1. Fundamental understanding of the concepts of stress and strain in mechanics of solids and structures and material properties.
2. Apply the fundamental concepts of principle of superposition, equilibrium, compatibility, force-deformation, and stress-strain relationships to the solid and structural mechanics problems.
3. Analyze determinate bars, beams, to determine axial forces, torques, shear forces, and bending moments.
4. Physical insight into distribution of stresses and strains in structural members by determining stress, strain, and deformation of bars, and beams, and performing stress and strain transformations.
5. Basic understanding of the method of superposition, flexibility method, and stiffness method as applied to statically determinate axial and torsional members, and beams.
6. Ability to design structural members given the dimensions, material properties such as force-displacement relationships, boundary conditions, loading, allowable stresses, and factor of safety.

SYLLABUS**Simple Stresses:**

Stress, Strain, Stress- Strain curve, Lateral strain, Relationship between elastic constants, Bars of varying cross-section, Compound bars, Temperature stresses in bars.

Complex Stresses:

Stresses on an inclined plane under different uniaxial, biaxial and general case of plane stress systems, Principal planes and principal stresses, Mohr's circle, Strain energy, Impact loading.

Bending Moments and Shear Forces:

Beam - Types of loads, Types of supports, S.F. and B.M. diagrams for Cantilever, Simply supported and over hanging beams.

Deflections of Beams:

Relation between curvature, slope and deflection - simple cases in Cantilever, Simply supported and Over hanging beams.

Stresses in Beams:

Theory of bending, Flexural formula, Determination bending stresses – section modulus of rectangular and circular sections (Solid and Hollow), I,T, Angle and Channel sections – Design of simple beam sections, Shear stresses in beams, Shear stress distribution across various beams sections like rectangular, circular, triangular, I, T, angle sections.

Torsional Stresses in Shafts:

Analysis of torsional stresses, Power transmitted, combined bending and torsion.

Cylinders and Spherical Shells:

Stresses and strains in thin cylinders, Thin seamless cylindrical shells – Derivation of formula for longitudinal and circumferential stresses –hoop, longitudinal and Volumetric strains – changes in dia, and volume of thin cylinders and Thin spherical shell.

Text Books:

1. Analysis of Structures, by Vazirani and Ratwani, Vol. 1, 1993 edition.
2. Mechanics of Materials by James M. Gere , Stephen P. Timoshenko , CBS Publishers
3. Solid Mechanics, by Popov

Reference Books:

1. Strength of Materials, by Timoshenko
2. Strength of Materials -By Jindal, Umesh Publications.
3. Analysis of structures by Vazirani and Ratwani.
4. Mechanics of Structures Vol-III, by S.B.Junnarkar.
5. Strength of Materials by Andrew Pytel and Ferdinand L. Singer Longman

THERMODYNAMICS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

1. To educate students about the behavior of real gases and the significance of ideal gas theory
2. To educate the students about the properties of ideal gas and their relationship
3. To familiarize the students about the behavior of ideal gases under heating, cooling, compression and expansion processes
4. To educate the students about the working principle of combustion engines (internal and external) and their cycles such as Otto, Diesel, Atkinson, Ericson, Brayton, etc., and their comparison

Course Outcomes

1. Students realize the practical importance of ideal gas theory and the use of real gases in combustion engines such as IC Engines and Gas turbines
2. Students are able to calculate the properties of the gases such as internal energy, enthalpy and entropy.
3. Students are able to estimate the losses which occur during operation of the heat engines, and their maximum possible operating efficiencies under STP conditions.
4. Students can estimate the maximum work-output delivered by the heat engines and maximum work consumed by the reversed heat engines

SYLLABUS**Introduction:**

Basic concepts; Thermodynamic systems; Micro & Macro systems; Homogeneous and heterogeneous systems; Concept of continuum; Pure substance; Thermodynamic equilibrium; State; Property; Path; Process; Reversible and irreversible cycles; Work; Heat; Point function; Path function; Heat transfer;

Perfect gas laws-

Equation of state- Universal gas constant, various non-flow processes-Properties of end states- Heat transfer and work transfer- Change in internal energy-throttling and free expansion- Flow processes- Deviations from perfect gas model-Vanderwall's equation of state- Compressibility charts- Variable specific heats.

Zeroth and First law of thermodynamics:

Concept of equality of temperatures- Joule's experiments-First law of thermodynamics- Isolated systems and steady flow systems- Specific heats at constant volume and pressure - Enthalpy- First law applied to flow systems- Systems undergoing a cycle and change of state- First law applied to steady flow processes-Limitations of first law of thermodynamics.

Second law of thermodynamics-

Kelvin Plank statement and Clasius statement and their equivalence, Corollaries- Perpetual motion machines of first kind and second kind-Reversibility and irreversibility- Cause of irreversibility- Carnot cycle- Heat engines and heat pumps- Carnot efficiency- Clasius theorem- Clasius inequality- Concept of entropy-Principles of increase of entropy- Entropy and disorder.

Air Standard Cycles

Only P-V and T-S diagrams along with air standard efficiencies of Otto cycle, Diesel cycle, Dual combustion cycle, Comparison of Otto-Diesel and Dual cycles based on same compression ratio- same maximum pressure and same maximum temperature.

General Relations, Availability and Unavailability

Helmholtz function and Gibbs function, Maxwell's equations- Tds relations, relation between specific heats, Available energy, unavailable energy, Available and unavailable forms of energy for a flow and non-flow process.

Gases and Vapour Mixtures –

Daltons law and Gibbs-Dalton law, Volumetric Analysis of gas mixtures, Apparent molecular weight and gas constant, specific heats of gas mixture, Adiabatic mixing of perfect gases, Gas and vapour mixtures.

Text Books:

1. Engineering Thermodynamics, by P.K. Nag, Tata McGraw-Hill Publications Company.
2. Thermal Engineering by R.K Rajput, Laxmi publications.
3. Applied Thermodynamics-I by R. Yadav, Central Book House.

References Books:

1. Engineering Thermodynamics by Rathakrishnan, Prentice - Hall India.
2. Engineering Thermodynamics by Y.V.C. Rao.
3. Engineering Thermodynamics by K. Ramakrishna, Anuradha agencies.
4. Engineering Thermodynamics Work and Heat Transfer, by G.F.C Rogers and Y.R. Mayhew, ELBS publication.

MANUFACTURING PROCESS

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

1. To provide an understanding and appreciation of the different manufacturing and fabrication methods to the students.
2. To expose the students to various applications of the manufacturing process in the real life articles/products.

Course Outcomes:

1. Student will be able to recognize various manufacturing materials, manufacturing process and types of productions.
2. Student will be able to identify various casting processes, metal forming process and welding process.
3. Student will be able to design of gating system, patterns and cores for various casting processes.
4. Student will be able to apply knowledge of casting process for manufacturing of products.
5. Student will be able to apply knowledge of rolling, forging, extrusion for manufacturing of products.
6. Student will be able to apply knowledge of welding, brazing and soldering for joining of metals.

SYLLABUS**Manufacturing concepts:**

Product cycle, Job, batch and mass production, Primary and secondary manufacturing processes.

Metal Casting Process:

Principle of metal casting, Pattern: Materials, Allowances and Types, Core boxes, Moulding sands: ingredients, properties, preparation, types, Moulding tools, Sand testing, Sand moulding, Machine moulding, Core making, Melting and pouring-Classification of furnaces, Cupola furnace, pouring laddels; Element of gating system.

Special Casting Techniques:

Permanent mould casting, Pressure die casting, Centrifugal casting, Shell mold casting, Investment casting and CO₂ process, casting defects.

Metal Forming:

Hot & Cold working, Rolling, Extrusion, metal spinning, Drawing, Piercing.

Sheet Metal Forming:

Concept of spring back, Materials, tools, operations, embossing, coining, stretch forming, Progressive and Compound Dies.

Forging Processes:

Forgability, Forging Materials, Classification: smith, drop, press and machine forging, Forging tools, Forging Operations, High energy rate forming, Swaging.

Welding Processes:

Welding metallurgy, Weldability, Classification: Plastic welding (Forge, Resistance & Thermit welding), Fusion welding (Gas, Arc & Thermit welding), Solid state welding (Friction, Ultrasonic, Diffusion and Explosive welding), Soldering and Brazing, Weld defects, Weld inspection and testing.

Text Books:

1. Elements of Workshop Technology Vol-1: Manufacturing Processes by S.K. Hajra Choudhury, A.K. Hajra Choudhury, Nirjhar Roy, MPP, Pvt. Ltd.
2. Manufacturing Technology- Foundry, Forming and Welding by P.N. Rao, Tata McGraw- Hill Publishing Company.

Reference Books:

1. Process and Materials of Manufacture (4th Edition) by Roy A. Lindberg, Prentice-Hall of India Private Limited.
2. Manufacturing Engineering & Technology by Kalpak Jain, Addison Wesley Edition.
3. Materials and Processes in Manufacturing by De Margo, Black and Kohsen, Prentice Hall of India.
4. Principles of Metal Casting by Hein and Rosenthol, Tata Mc-Graw Hill India.

ENGINEERING MECHANICS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. The course uses the Laws of Mechanics to predict forces in and motions of machines and structures.
2. The course is the key prerequisite course to sequences of courses dealing with mechanics of machines, stress analysis and design of mechanical systems.
3. Comprehensive, theory based understanding of the underpinning natural and physical sciences and the engineering fundamentals applicable to the engineering discipline.
4. In-depth understanding of specialist bodies of knowledge within the engineering discipline.
5. Application of established engineering methods to complex engineering problem solving.
6. Application of systematic engineering synthesis and design processes.

Course Outcomes:

Upon successful completion of this course student should be able to:

1. Use scalar and vector analytical techniques for analyzing forces in statically determinate structures.
2. Apply fundamental concepts of kinematics and kinetics of particles to the analysis of simple, practical problems.
3. Apply basic knowledge of maths and physics to solve real-world problems

SYLLABUS**Basic Concepts:**

Scalar and vector quantities- Representation vectors- Free vector force, Specification of force- Effect of force on rigid body- Free body diagram. Concurrent Forces and Parallel Forces in a Plane: Principles of statics-Resolution and Composition of forces in a plane-Equilibrium of concurrent forces in a plane- Method of projections- Equilibrium of three forces in a plane Method of moments.

Centroid & Moment of Inertia - Centroid & M.I – Area & Mass M.I – Radius of Gyration, Parallel axis– Perpendicular axis theorem – Simple Problems.

General Case of Forces in a Plane: Resultant and equilibrium of general case of parallel forces in a plane, Statically determinate plane trusses-Method of joints and Method of sections.

Friction – Coulombs laws of dry friction – Limiting friction, Problems on Wedge friction, Belt Friction-problems.

Dynamics of Particles - Rectilinear Motion – Kinematics Problems, D'Alembert's principle, Kinetics – Problems, Work & Energy – Impulse Moment, Direct Central Impact – coefficient of restitution.

Curvilinear Motion – Projectile Motion, Moment of momentum, Work & Energy in Curvilinear motion.

Dynamics of Rigid Bodies - Rigid body rotation – Kinematics - Kinetics, Problems – Work & Energy in Rigid body rotation, Plane Motion – Kinematics, Problem – Instantaneous center of rotation, work-energy principle in plane motion.

Text Book:

1. Engineering Mechanics by S.Timoshenko and D.HYoung McGraw-Hill.

Reference Books:

1. Engineering Mechanics, Vol.1 & 2 by J.L. Meriems and L.G. Kraige.
2. Engineering Mechanics by Singer.
3. Engineering Mechanics by K.L. Kumar, Tata Mc-Graw Hill.
4. Engineering mechanics by Bhavikatti. New age international.

MECHANICAL ENGINEERING DRAWING

Theory/Practice	: 4 Periods	Sessionals	: 30
Exam	: 3 Hrs.	Ext. Marks	: 70
		Credits	: 4

Course Objectives

1. To provide an understanding and draw the assembly and production drawing of various engine components and machine tool components to the students.
2. To expose the students to draw various screw fastenings (Screw, Riveted, welded etc.), Bearings, couplings, key, dimensional and geometrical tolerances etc.

Course Outcomes

Students will be able to

1. Know drawing of Screw threads and Screw Fastenings using standard Empirical formulae.
2. Draw Riveted joints, Keys, Cotter-joint, Draw Couplings (Shaft couplings: Box and split muff couplings, Flanged, Flexible, Universal and Oldham couplings).
3. Draw the dimensional and geometrical tolerances and surface roughness symbols.
4. Draw Assembly and production drawings of various engine components and machine tool components.

SYLLABUS

Screw threads, Screw Fastenings, keys, and Riveted joints using standard Empirical formulae.

Cotter-joints, Shaft couplings: Box and split muff couplings, Flanged, Flexible, Universal and Oldham couplings,

Assembly drawing of various engine components and machine tool components (Simple eccentric, swivel bearing, plumber block, Screw Jack, Stuffing Box).

Conventional representations, Limits, Fits and Tolerances, Geometrical Tolerances, Indication of surface roughness, Production Drawings.

Text Books:

1. Machine Drawing, by N.D.Bhatt, Charotal Publishing House.g
2. Production Drawing by K.L Narayan, P. Kannaiah and K. Venkata Reddy, New Age.
3. Engineering Drawing, by A.C.Parkinson, Wheeler Publishing.

Reference Book:

1. Machine Drawing by K.L Narayan, P. Kannaiah and K. Venkata Reddy, New Age.

MECHANICAL ENGINEERING LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To understand the principle and functioning of various mechanical devices such as boilers, engines etc.
2. Ability to understand the working of two stroke and four stroke engines.
3. Acquiring the knowledge of operation of a reciprocating compressor and to assess the pressure gauge performance.
4. The way of determination of flash and fire points of oil samples and carbon residue and their importance is acquired.
5. The procedure for determination of calorific values of the fuels and viscosities of oil samples can be understood.
6. Practically the procedure for moment of inertia of fly wheel, connecting rod and modulus of rigidity is acquired.

Course Outcomes:

1. Students are now aware of the use of drawing valve timing diagrams of an engine and method to evaluate the volumetric efficiency of air compressor.
2. They are also aware of method of calibrating pressure gauge, the importance of flash and fire points and calorific values of fuels.
3. The importance and application by calculating viscosities of oil samples are understood.
4. The use of moment of inertia and modulus of rigidity is understood.
5. They are also now able to identify the parts of boiler and engines etc.

LIST OF EXPERIMENTS

1. Study and valve timing diagrams for four-stroke and study & PTD of two-stroke engines.
2. Determination of volumetric efficiency of the given air compressor by (i) plate orifice method and (ii) tank capacity method.
3. Calibration of the given pressure gauge.
4. Determination of flash and fire points and b) Canradsons carbon residue test.
5. Determination of calorific value of flues (solid, liquid and gaseous) by Bomb calorimeter/Gas calorimeter.
6. Determination of the kinematic and absolute viscosity of the given sample oils.
7. Determination of inertia of the given flywheel and connecting rod.
8. Determination of modulus of rigidity of the given wire with torsion pendulum.
9. Study of boilers, various mountings and accessories.
10. Assembling of the given two-stroke petrol engine. (Instead of engine, any mechanical unit can be given for this experiment.)

Reference Books:

1. Engineering Mechanics by S.Timoshenko and D.HYoung McGraw-Hill.
2. Engineering Mechanics by Singer.
3. Internal Combustion Engines by V. Ganesan, McGraw-Hill.

MECHANICS OF SOLIDS LAB

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

Course Outcomes:

1. Ability to identify different types of loads and measure them.
2. Ability to measure material properties of different materials using different methods.
3. Ability to measure bulking property and fineness of sand grains.

Course Objectives:

1. To understand the different types of loading and measure the loads.
2. To understand the material properties of different materials and the ways of finding them.
3. To understand the bulking property and fineness of sand grains and the methods of finding them.

LIST OF EXPERIMENTS

1. To study the stress strain characteristics (tension and compression) of metals by using UTM.
2. To study the stress strain characteristics of metals by using Hounsefield Tensometer.
3. Determination of compression strength of wood.
4. Determination of hardness using different hardness testing machines-Brinnels, Vickers and Rockwell's.
5. Impact test by using Izod and Charpy methods.
6. Deflection test on beams using UTM.
7. Tension shear test on M.S. Rods.
8. To find stiffness and modulus of rigidity by conducting compression tests on springs.
9. Torsion tests on circular shafts.
10. Bulking of sand.
11. Punch shear test, hardness test and compression test by using Hounsefield tensometer.
12. Sieve Analysis and determination of fineness number.

Reference Books:

1. Strength of Materials, by Timoshenko
2. Strength of Materials -By Jindal, Umesh Publications.
3. Strength of Materials by Andrew Pytel and Ferdinand L. Singer Longman.

ENGLISH PROFICIENCY
(Common to All Branches)

Theory	: 1 Period	Sessionals	: 50
Tutorial	: 1 Period	Ext. Marks	: 50
Exam	: 3 Hrs.	Credits	: 2

AIM:

Enriching the communicative competency of the students by adopting the activity-based as well as the class-oriented instruction with a view to facilitate and enable them to enhance their language proficiency skills.

Course Objectives:

Students be able to

1. Understand the importance of professional communication.
2. Learn language skills and vocabulary in order to improve their language competency.
3. Know and perform well in real life contexts.
4. Identify and examine their self-attributes which require improvement and motivation.
5. Build their confidence and overcome their inhibitions.
6. Improve their strategies in reading skills.

Course Outcomes:

1. Students enhance their vocabulary and use it in the relevant contexts .
2. They improve speaking skills.
3. They learn and practice the skills of composition writing.
4. They enhance their reading and understanding of different texts.
5. They enrich their communication both in formal and informal contexts.
6. They strengthen their confidence in presentation skills.

SYLLABUS

Speaking Skills

PPT

Describing event/place/thing

Picture Description

Extempore

Debate

Telephonic Skills

Analyzing Proverbs

Vocabulary

Affixes

Pairs of Words

Reading Skills

Reading Comprehension

Reading/Summarizing News Paper Article

Writing Skills

Designing Posters

Essay writing

Resume Writing

Reference Books:

1. Interchange (4th edition) Student's books 1&2 by Jack C. Richards, CUP.
2. Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP
3. English and Communication Skills for Students of Science and Engineering, by S.P.
4. Dhanavel, Orient Blackswan Ltd. 2009
5. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
6. The Oxford Guide to Writing and Speaking by John Seely OUP

(***Note: Sessional Marks will be evaluated based on Continuous Comprehensive Evaluation of the students' Performance - 40M, Attendance – 10M and External Marks will be evaluated based on Presentation Skills – 30M, Project 20M)

Auto CAD

Lab	: 2 Periods	Sessionals	:50
Exam	: 3 Hrs.	Credits	: 1

Course Objectives:

1. Increase ability to communicate with people
2. Learn to sketch and take field dimensions.
3. Learn to take data and transform it into graphic drawings.
4. Learn basic Auto Cad skills.
5. Learn basic engineering drawing formats
6. Prepare the student for future Engineering positions

Course Outcomes

Up on completion of the course the student shall be able to gain knowledge on:

1. Auto CAD screen and various Tool bars and menus and Explain about Dimensioning and Hatching
2. Draw the 2D – drawings like knuckle joint, screw jack, flange coupling, lathe tool post, eccentric etc.,
3. Explain about 3D solids and solids tool bar options and Drawing of 3D – components like bolt & nut, screw jack
4. Rendering of 3D images

LIST OF EXERCISES

1. Study the Auto CAD screen, various toolbars and menus
2. Exercises on usage of Draw and modify tool bar.
3. Exercises on mirror, rotate, array and move commands
4. Exercises on dimension and hatching
5. Draw the 2D knuckle joint with full details & dimensioning
6. Draw the screw jack 2D drawing
7. Study the 3D solids (primitives) and solids tool bar options
8. Draw bolt and nut in 3D
9. Draw various parts of screw jack in assemble them as 3D component
10. Render the 3D images already generated and apply materials and light.

(Note: Total Marks will be evaluated based on Continuous Evaluation - 25 Marks, Design Contest -25 Marks)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

II/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards)
Under Choice Based Credit System

MECHANICAL ENGINEERING

II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ME 2201	Advanced Strength of Materials	4	3	1	--	4	30	70	100
B16 ME 2202	Thermal Engineering	4	3	1	--	4	30	70	100
B16 ME 2203	Metal Cutting & Machine Tools	4	4	--	--	4	30	70	100
B16 ENG 2202	Engineering Economics	4	4	--	--	4	30	70	100
B16 EE 2204	Basic Electrical & Electronics Engineering	4	4	--	--	4	30	70	100
B16 ENG 2201	Environmental Studies	4	3	1	--	4	30	70	100
B16 ME 2205	Manufacturing Process Lab	2	--	--	3	3	50	50	100
B16 EE 2206	Basic Electrical & Electronics Engineering Lab	2	--	--	3	3	50	50	100
B16 ME 2206	Industry Oriented Technology Lab	1	--	--	2	2	50	--	50
Total		29	21	3	8	32	330	520	850

ADVANCED STRENGTH OF MATERIALS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To enrich the student on the concept of fixed beams with uniform Moment of inertia as well as Non uniform Moment of inertia both under stability of beam supports and under sinking & rotation of the supports.
2. To make the student capable of evaluating the deflection, slope and stress of the fixed beam with uniform Moment of inertia as well as Non uniform Moment of inertia both under the stability of beam supports and under the sinking & rotation of the supports.
3. To make the student understand the concept of continuous beams with uniform Moment of inertia as well as Non uniform Moment of inertia both under stability of supports as well as sinking of supports

Course Outcomes:

1. Students are able to evaluate the stresses across the cross-sections of the curved beam.
2. Calculate the radial stress and circumferential stress for rotating circular disc (both hollow and solid) of uniform thickness
3. Modeling the thickness of circular rotating disc having uniform strength
4. Calculate the radial and circumferential stress for both thick and compound cylinders under different pressurized conditions
5. Evaluate the deflection and slope of simply supported beams and cantilever beams using different energy methods

SYLLABUS

Fixed Beams:

Fixing moments for a fixed beam of uniform and variable sections, Effect of sinking support, slope and deflection.

Continuous beams:

Analysis of continuous beam, Reactions at the supports, Effect of sinking of supports.

Columns and Struts:

Columns with one end free and the other fixed, Both ends fixed, One end fixed and other hinged, Limitation of Euler's formula, Column carrying eccentric load, Empirical formulae.

Bending of Curved Bars:

Stresses in bars of circular, rectangular and trapezoidal sections.

Stresses due to rotation:

Wheel rim, disc of uniform thickness, disc of uniform strength.

Thick cylinders subjected to internal and external pressure and compound cylinders.

Text Books:

1. Analysis of Structures, Vol. 1, 1993 edition, by Vazirani and Ratwani.
2. Chapter VI from Advanced Topics in Strength of Materials, by Prof. L.B.Shah and Dr.R.T.Shah.

Reference Book:

1. Strength of Materials, by Timoshenko.

THERMAL ENGINEERING

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To gear the student with basic principles of steam properties.
2. To prepare the student for industrial application of steam.
3. The student is taught to design the steam equipment so that R&D in industry is improved.

Course Outcomes:

1. The student gets complete knowledge of steam and its properties.
2. The student learns the complete calculation procedures for designing steam turbines, steam condensers, nozzles etc. used in thermal power plants, steam engines, water turbines and many other industrial applications.
3. The student is prepared to work in industry immediately after his course

SYLLABUS**Properties of Pure Substance:**

Definition of pure substance, phase change of a pure substance, p-T (Pressure-Temperature) diagram for a pure substance, p-V-T(Pressure-Volume-Temperature) surface, phase change terminology and definitions, property Diagrams in common use, Formation of steam, Important terms relating to steam formation, Thermodynamic properties of steam and steam tables, External work done during evaporation, Internal latent heat, Internal energy of steam, Entropy of water, Entropy of evaporation, Entropy of wet steam, Entropy of superheated steam, Enthalpy-Entropy (h-s) charts for Mollier diagram, Determination of dryness fraction- Tank or bucket calorimeter, throttling calorimeter, separating and throttling calorimeter.

Vapor Power Cycles

Vapor power cycle- Rankine cycle- Reheat cycle- Regenerative cycle- Thermodynamic variables effecting efficiency and output of Rankine and Regenerative cycles- Improvements of efficiency, Binary vapor power cycle.

Steam Nozzles:

Type of nozzles- Flow through nozzles- Condition for maximum discharge- Nozzle efficiency- Super saturated flow in nozzles- Relationship between area velocity and pressure in nozzle flow- Steam injectors.

Steam Turbines:

Classification of steam turbines- Impulse turbine and reaction turbine- Compounding in turbines- Velocity diagrams in impulse and reaction turbines- Degree of reaction- Condition for maximum efficiency of reaction turbines- Effect of friction on turbines constructional features governing of turbines.

Condensers:

Classification of condenser- Jet, Evaporative and surface condensers- Vacuum and its measurement- Vacuum efficiency- Sources of air leakage in condensers- Condenser efficiency- Daltons law of partial pressures- Determination of mass of cooling water- Air pumps.

Steam Boilers –

Working principle of various boilers their accessories and mountings (Simple vertical, Cochran, Babcock & Wilcox and Lancashire Boiler), Performance of boilers (simple problems)

Text Books:

1. A Treatise on Heat Engineering by Vasandhani and Kumar.
2. Applied Thermodynamics-II by R. Yadav.
3. Thermal Engineering, by R. K. Rajput.

References Books:

1. Fundamentals of Engineering Thermodynamics by E. Radhakrishna, PHI.
2. Fluid Flow Machines, by M.S. Govinda Rao, Tata McGraw Hill publishing company Ltd.
3. Refrigeration and Air-conditioning, by C.P.Arora and Domokundwar.
4. Thermal Science and Engineering by D.S. Kumar, S.K. Kataria and Sons.
5. Refrigeration and Air-conditioning, by Ahamadul Ameen, PHI.

METAL CUTTING & MACHINE TOOLS

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

1. To give a clear understanding of the mechanism of machining to the students.
2. To describe the mechanisms of the various machine tools, types of machine tools, various operations that can be performed on them, machining time and force calculations etc to the students.

Course Outcomes:

1. Students will be able to describe the mechanisms of metal cutting.
2. Students will be able to calculate cutting forces, tool life and machining parameters.
3. Students will be able to design the single point and multi point cutting tools.
4. Students will be able to demonstrate the working of various machine tools like lathe, milling machine and grinding machine etc.
5. Students will be able to identify different micro finishing operations.
6. Students will be able to assess the advantages, limitations and applications of unconventional methods of machining.

SYLLABUS**Mechanics of Metal Cutting:**

Orthogonal and oblique cutting, mechanics of chip formation, types of chips; classification, nomenclature, signature (ASA & ISO systems) of single point cutting tools, tool materials; tool wear and tool life; Cutting forces-Merchant's circle, Machinability, Cutting fluids.

Machine tools using Single point cutting tools:

Engine lathe; Capstan and turret lathe, shaper, planner, Slotter and boring-Types, Parts, Specifications, Mechanisms, Operations and machining parameters.

Machine tools using Multi point cutting tools:

Drilling machine-Types, Parts, Specifications, Mechanisms, Types of drills, Nomenclatures of twist drill, Operations and machining parameters

Milling machine-Types, Parts, Specifications, Mechanisms, Attachments, Types of Milling cutters, Nomenclature of plain milling cutter, Operations, machining parameters and Indexing methods.

Broaching machine-Types, Parts, Specifications, Types of Broaches, Nomenclature of pull broach, Operations and machining parameters

Machine tools using Abrasive wheels:

Grinding Machine- Types, Parts, Specifications, Manufacturing of grinding wheel-bonding processes, grit, grade and structure, selection of grinding wheels, mounting of grinding wheels, glazing, loading, dressing and truing of grinding wheel, Operations and machining parameters

Micro finishing Operations-Lapping, honing, super finishing, polishing and buffing

Unconventional Methods of Machining:

Process, Characteristics, Advantages, Limitations, Applications of Abrasive Jet Machining (AJM), Ultrasonic Machining (USM), Water Jet Machining (WJM), Electro Discharge Machining (EDM), Wire-cut EDM, Electron Beam Machining (EBM), Plasma Arc Machining (PAM), Laser Beam Machining (LBM), Chemical milling; Photochemical milling, Electro Chemical Machining (ECM), Electro Chemical Grinding (ECG)

Text Books:

1. Elements of Workshop Technology Vol-2: Machine Tools by S.K. Hajra Choudhury, A.K. Hajra Choudhury, Nirjhar Roy, MPP, Pvt. Ltd.
2. Metal cutting and Machine tools by P.N. Rao, Tata McGraw- Hill Publishing Company.
3. Process and Materials of Manufacture (4th Edition) by Roy A. Lindberg, Prentice-Hall of India Private Limited.

Reference Books:

1. Fundamentals of Metal Machining and Machine Tools by Geoffrey Boothroyd, International Student Edition, McGraw-Hill Book Company.
2. Metal Cutting Principles by M.C. Shaw, MIT Press, Cambridge.
3. Advanced Methods of Machining by J. A. McGeough, Chapman & Hall Publishers.
4. Metal Cutting-Theory and Practice by Amitabha Bhattacharya, Central Book Publishers.
5. Production Engineering by P.C. Sharma, S. Chand and Company.

ENGINEERING ECONOMICS

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

1. To understand the concept of economics and to discuss about demand analysis.
2. To know different types of economic systems and to learn factors required for production.
3. To examine different market structures and their price determination.
4. To determine and examine business cycles and to understand inflation.
5. To describe the elements of cost and to know about break-even analysis and break-even chart.
6. To explain forms of business organizations along with types of public enterprises.
7. To know how to prepare financial statement and examine the methods used for depreciation.

Course Outcomes:

1. Awareness about how resources should be allocated and utilized efficiently and types of demand.
2. Determine types of economic systems with their respective pros & cons and how factors of production will help engineers to achieve their goals.
3. Develop the capability to understand different market structures and act accordingly.
4. Understand the stages of business cycles and causes and effects of inflation.
5. Examine the nature of cost and learn how to construct a break-even chart to know no profit – no loss point.
6. Evaluate forms of business organization along with their pros and cons.
7. Construct a financial statement to know the financial position and calculation of depreciation by using different methods.

SYLLABUS**Introduction to Economics and Demand Analysis:**

Definition of economics; Micro and Macro Economics. Demand-Law of Demand; Elasticity of Demand – Measurement of elasticity of demand and types of elasticity of demand.

Economic Systems and Factors of Production:

Economic Systems – Capitalism, Socialism and mixed Economy. Factors of Production – Classification of Factors of Production – Meaning and characteristics of land, labor, capital and organization.

Markets:

Perfect Competition – features and price determination under perfect competition; Imperfect Competition-Monopoly, Monopolistic Competition. Duopoly and Oligopoly (in brief).

Business Cycles (trade Cycles) and Inflation:

Business Cycles – Meaning, phases of Business Cycle, causes and consequences of Business Cycle: Inflation – types, causes and effects of Inflation.

Cost classification and Break-even Analysis:

Costs – Classification of Costs, Elements of Cost. Components of Total Cost: Methods of costing – Job Costing, process costing and unit costing. Break-even Analysis-determination of Break-even point and application of Break-even Analysis.

Forms of Business Organizations:

Sole Proprietorship, Partnership, Co-operative Society, Joint Stock Company (Private and Public Ltd) – Features, Merits and Demerits. Public enterprises and their types.

Depreciation and Financial Accounting:

Depreciation – Causes and Methods of Depreciation. Financial Accounting – Preparation of Trading Account, Profit & Loss Account and Balance Sheet of a Sole Proprietor.

Text Books:

1. Managerial Economics and Financial Analysis – By A.R. Aryasri, Tata McGraw Hill Education Private Ltd., New Delhi.
2. Engineering Economics – By Tara Chand. Nem Chand & Bros. Roorke.

Reference Books:

1. Modern Economic – By K.K. Dewett, S. Chand & Co, New Delhi.
2. Principles of Economics – Vrinda Publications (P) Ltd. New Delhi.

BASIC ELECTRICAL & ELECTRONICS ENGINEERING

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives

1. To learn the basic principles of electrical laws and analysis of networks.
2. To understand the principle of operation and construction details of DC machines.
3. To understand the principle of operation and construction details of Transformers.
4. To understand the principle of operation and construction details of three phase Induction motor and Alternators.
5. To study the operation of PN junction diode, Half wave, Full wave rectifiers.
6. To study the operation OP-AMP.
7. To learn the operation of PNP and NPN Transistors and various Amplifiers.

Course Outcomes

1. Able to analyze the various Electrical networks.
2. Able to understand the basics of Magnetic Circuits.
3. Able to understand the operation of DC generators, 3-Point starter and conduct the Swinburne's test.
4. Able to analyze the Performance of Transformers.
5. Able to explain the operation of three phase induction motors and alternator.
6. Able to analyze the operation of Half-wave and Full-wave rectifiers.
7. Able to explain the operation of single stage CE amplifier.

SYLLABUS**Electrical And Magnetic Circuits:**

Basic definitions, Types of network elements, Ohm's Law, Kirchhoff's Laws, series and parallel circuits and star-delta and delta-star transformations-simple problems.

Magnetic flux, MMF, Reluctance, Faradays laws, Lenz's Law, statically induced emf, dynamically induced emf, Hysteresis.

DC Machines:

Principal of operation of DC generation- EMF equation-Types-DC motor Types-Torque equation-Applications-Swinburne's Test, Speed control methods.

Transformers:

Principal of operation of Single phase Transformers- EMF equation-losses-Efficiency and Regulation.

AC Machines:

Principal of operation of three phase Induction motor-Slip-Torque characteristics-Efficiency-applications- Principal of operation of Alternators-EMF equation, regulation of alternator by synchronous impedance method.

Diodes & Rectifiers: PN junction diode-Forward bias and reverse bias operation, V-I characteristics-Diode applications (Half wave, Full wave and bridge rectifier), Zener diode.

Transistors: PNP and NPN junction Transistors, Transistor as an amplifier, single stage CE amplifier, Frequency response of CE amplifier.

Text Books:

1. ELECTRICAL TECHNOLOGY by Surinder Pal Bali, Pearson Publications.
2. ELECTRONIC DEVICES AND CIRCUITS, R.L Boylestad and Louis Nashelsky, 9th edition, PEI/PHI 2006.
3. Principles of Electrical Engineering, V.K mehta, Rohit Mehta, S. Chand Publications.
4. Basic Electrical Engineering by D.P.Kothari, I.J. Nagrath, McGraw Hill.
5. Basic Electrical Engineering, S.N. Singh, PHI.
6. A text book of ELECTRICAL TECHNOLOGY Volume-II, by B.L Theraja & A.K Theraja S.Chand publications.

Reference Books:

1. Basic Electrical Engineering, V.K mehta, Rohit Mehta, S. Chand Publications.
2. Fundamentals of Electrical Engineering and Electronics, Rajendra Prasad, PHI.
3. Principles of electronics Engineering, V.K mehta, Rohit Mehta, S. Chand Publications.

**ENVIRONMENTAL STUDIES
(Common to ECE,EEE & ME)**

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. To develop an awareness and sensitivity to the total environment and its related problems.
2. To participate actively participation in environmental protection and improvement.
3. To develop skills for active identification and development of solutions to environmental problems
4. To evaluate environment programmes in terms of social, economic, ecological and aesthetic factors.
5. To Create a “CONCERN AND RESPECT FOR THE ENVIRONMENT”

Course Outcomes:

Students will be able to

1. Get awareness among the students about the nature and natural ecosystems.
2. Learn sustainable utilization of natural resources like water, land, minerals, air.
3. Learn resource pollution and over exploitation of land, water, air and catastrophic (events) impacts of climate change, global warming, ozone layer depletion, marine, radioactive pollution etc to inculcate the students about environmental awareness and safe transfer of our mother earth and its natural resources to the next generation.
4. Safe guard against industrial accidents particularly nuclear accidents.
5. Learn Constitutional provisions for the protection of natural resources.

SYLLABUS**Global Environmental Crisis:**

Environmental Studies - Definition, Scope and importance, Need for public awareness.
Global Environmental Crisis

Ecosystems:

Basic concepts, Forest Ecosystems, Grassland Ecosystems and Desert Ecosystems, Aquatic Ecosystems

Biodiversity:

Introduction to Biodiversity, Value of Bio-diversity, Bio-geographical classification of India, India as a Mega-diversity habitat, Threats to biodiversity, Conservation of Biodiversity: In-situ and Ex-situ conservation of bio-diversity.

Environmental and Natural Resources Management:

Land Resources: Land degradation, soil erosion and desertification, Effects of modern agriculture **Forest Resources:** Use and over exploitation-Mining and Dams-their effects on forest and tribal people, **Water resources:** Use and over utilization of surface and ground water, Floods, droughts, conflict over water, water logging and salinity, dams – benefits and problems

Energy Resources: Renewable and non-renewable energy sources, use of alternate energy sources-impact of energy use on environment.

Environmental Pollution:

Causes, Effects and Control measures of - Air pollution, Water pollution, Soil pollution, Marine Pollution, Thermal pollution, Noise pollution, Nuclear Hazards; Climate change and global warming, acid rain and Ozone layer depletion.

Environmental Problems in India:

Drinking water, Sanitation and Public health, population growth and environment; Water Scarcity and Ground Water Depletion; Rain water harvesting, Cloud seeding and Watershed management.

Text Books:

1. Environmental Studies (From Crisis to Cure) by R. Rajagopalan, Oxford university Press, 2008
2. Environmental Studies by Anubha Kaushik & C.P. Kauskik, New Age International (P) Ltd, New Delhi, 2006

Reference Books:

1. Environmental Sciences by G.Tyler Miller, JR,10th ed, Thomson publishers, 2004.

MANUFACTURING PROCESS LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To recognize the tools, materials, machines used for making products in foundry, welding and machine shop.
2. To differentiate the different welding techniques used for different materials.
3. To realize the various moulding sands used for making moulds

Course Outcomes:

1. Student will be able to prepare moulds for a given component.
2. Student will be able to apply the knowledge of arc welding to join two metal pieces.
3. Student will be able to practice plain turning, facing, step turning, taper turning, and thread cutting operations on the lathe machine.
4. Student will be able to generate horizontal, vertical and angular surfaces on a given work piece using shaper.
5. Student will be able to generate spur gear on milling machine.
6. Student will be able to demonstrate Capstan and Turret lathe, cylindrical grinder and surface grinding machine.

LIST OF EXPERIMENTS

1. Use of basic tools and operations of the following trades

S.No.	Trade/Machine	No. of exercises
1	Moulding	3
2	Welding	3
3	Lathe Machine	3
4	Milling Machine	1
5	Shaping Machine	1

2. Moulding sand testing (Not for examination only for demonstration purpose)
3. Forging (Not for examination only for demonstration purpose)

Reference Books:

1. Elements of Workshop Technology Vol-1: Manufacturing Processes by S.K. Hajra Choudhury, A.K. HajraChoudhury, Nirjhar Roy, MPP, Pvt. Ltd.
2. Elements of Workshop Technology Vol-2: Machine Tools by S.K. Hajra Choudhury, A.K. HajraChoudhury, Nirjhar Roy, MPP, Pvt. Ltd.

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. Information to supplement to the Electrical & Electronics Engineering courses.
2. The ability to conduct testing and experimental procedures on Circuits.

Course Outcomes:

Students will be able to

1. Apply the concepts of Theorems for a given electrical circuit.
2. Evaluate the efficiency and regulation of a single phase transformer.
3. Relate physical observations and measurements involving electrical circuit to theoretical principles.
4. Design amplifier circuit using NPN transistor

LIST OF EXPERIMENTS**Part-A: Electrical Engineering**

1. Verification of KCL and KVL
2. Verification of Ohms law (draw the V-I characteristics for a particular resistor)
3. Swinburne's test on D.C shunt Machine (predetermination of efficiency when working as motor and generator)
4. Brake test on D.C shunt motor. (determination of performance characteristics)
5. Brake test on D.C series motor. (determination of performance characteristics)
6. Brake test on three phase induction motor. (determination of performance characteristics)
7. Speed control of D.C shunt by (a) Armature voltage control (b) Field flux control Method.
8. OC and SC test on single phase Transformer (predetermination of efficiency and regulation at given power factor).

Part- B: Electronics Engineering

1. PN junction Diode Characteristics (a) Forward bias (b) Reverse bias. (cut in voltage and resistance calculations)
2. Half wave rectifier with and without filters.
3. Full wave rectifier with and without filters.
4. Transistor CE characteristics (Input and Output)
5. Characteristics CE amplifier
6. Zener diode characteristics
7. Regulation characteristics of Zener diode.

Reference Books:

1. Electrical Technology by Surinder Pal Bali, Pearson Publications.
2. Electronic Devices And Circuits, R.L Boylestad and Louis Nashelsky, 9th edition, PHI/PHI 2006.

Code: B16 ME 2206

INDUSTRY ORIENTED TECHNOLOGY LAB

Lab	: 2 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Credits	: 1

CATIA

Course Objectives:

1. Increase ability to communicate with people
2. Learn to sketch and take field dimensions.
3. Learn to take data and transform it into graphic drawings.
4. Learn basic CATIA skills.
5. Learn basic engineering drawing formats
6. Prepare the student for future Engineering positions

Course Outcomes

Up on completion of the course the student shall be able to gain knowledge on:

1. CATIA screen and various Tool bars and menus and Explain about Dimensioning and Hatching
2. Draw the 2D – drawings like knuckle joint, screw jack, flange coupling, lathe tool post, eccentric etc.,
3. Explain about 3D solids and solids tool bar options and Drawing of 3D – components like bolt & nut, screw jack.
4. Rendering of 3D images.

LIST OF EXERCISES

1. Study the CATIA CAD screen, various toolbars and menus
2. Exercises on usage of Draw and modify tool bar.
3. Exercises on mirror, rotate, array and move commands
4. Exercises on dimension and hatching
5. Draw the 2D knuckle joint with full details & dimensioning
6. Draw the screw jack 2D drawing
7. Study the 3D solids (primitives) and solids tool bar options
8. Draw bolt and nut in 3D
9. Draw various parts of screw jack in assemble them as 3D component
10. Render the 3D images already generated and apply materials and light.

(Note: Total Marks will be evaluated based on Continuous Evaluation - 25 Marks, Record/Report-10 Marks, Exam-10 Marks and Attendance-5 Marks)



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

III/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards)

Under Choice Based Credit System

MECHANICAL ENGINEERING

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ME 3101	Operations Research	4	3	1	--	4	30	70	100
B16 ME 3102	Fluid Mechanics	4	3	1	--	4	30	70	100
B16 ME 3103	IC Engines & Gas Turbines	4	3	1	--	4	30	70	100
B16 ME 3104	Industrial Measurements & Metrology	4	3	1	--	4	30	70	100
B16 ME 3105	Kinematics of Machines	4	3	1	--	4	30	70	100
#ELE-I	Elective-I	4	3	1	--	4	30	70	100
B16 ME 3112	Industrial Metrology & Mechatronics Lab	2	--	--	3	3	50	50	100
B16 ME 3113	IC Engines Lab	2	--	--	3	3	50	50	100
B16 ENG 3102	Verbal & Quantitative Aptitude-I	2	5	--	--	5	100	--	100
# M-I	MOOCS -I	2	4	--	--	4	100	--	100
# M-II	MOOCS -II	2	4	--	--	4	100	--	100
Total		34	31	6	6	43	580	520	1100

#ELE-I	B16 ME 3106	Finite Element Analysis	B16 ME 3109	Non Conventional Energy Resources
	B16 ME 3107	Automation in Manufacturing	B16 ME 3110	Production Planning and Control
	B16 ME 3108	Tool Design	B16 ME 3111	Rapid Prototyping

# M-I	B16 ME 3114A	Introduction To Abrasive Machining And Finishing Processes
	B16 ME 3114B	Principles of Casting Technology
	B16 ME 3114C	Principles of Metal Forming Technology
	B16 ME 3114D	Mechanics of Machining
# M-II	B16 ME 3115A	Manufacturing Of Composites
	B16 ME 3115B	Processing of Polymers and Polymer Composites
	B16 ME 3115C	Robotics
	B16 ME 3115D	Analysis and Modelling of Welding

OPERATIONS RESEARCH

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

This Course aims to:

1. Introduce students to use quantitative methods and techniques for effective Decisions–making; model formulation and applications that are used in solving business and industrial decision making problems.
2. Familiarize the students with the use of practice oriented mathematical applications for optimization functions in an organization.

Course Outcomes:

Upon completion of this course,

- Students on an overall perspective will be able to:
 1. Formulate a real time situation into a mathematical model.
 2. Identify and develop operational research models from verbal description of real system.
 3. Formulate simple reasoning, learning and optimization problems, in terms of the representations and methods presented.
 4. Demonstrate the hand execution of basic reasoning and optimization algorithms on simple problems.
 5. Formulate more complex, but still relatively simple problems, and apply implementations of selected algorithms to solve these problems.
 6. Apply and analyze mathematical optimization functions to various applications.
- Students will specifically able to:
 1. Find the best use of an organization's resources.
 2. Design an optimum distribution schedule of products from different sources to different destinations.
 3. Allocate various resources to various activities on a one to one basis.
 4. Assign a right job to a right person using job sequencing.
 5. Design optimum schedules for projects.
 6. Make right decisions and strategies in operations management using game theory and queuing theory.
 7. Define optimum inventory policies suitable to a given situation.

SYLLABUS

Introduction to OR: Definition of OR, Characteristics and phases of OR, Scope of OR, OR Models, General Methods for Solving OR Models, Role of Computers in OR.

Linear Programming: Formulation, Graphical Solution, Simplex Method, Artificial Variable Technique – Big M method, Duality.

Transportation Model: Mathematical Formulation, Tabular Representation, Balanced and unbalanced transportation Problems - Initial Solution by VAM and Optimality test by MODI Method, Degeneracy in TP.

Assignment Model: Mathematical Formulation, Hungarian Algorithm, Balanced and unbalanced Assignment Problems, Travelling Salesman Problem.

Job Sequencing: Introduction, Assumptions, Johnson's Algorithm for Sequencing n - jobs through 2 machines, Problem with n - jobs and 3 machines, Problems with n - jobs and m - machines, Graphical Solution for 2 - Jobs and m - machines problem.

Inventory Models: Definition of inventory, costs associated with inventory problems, classification of inventory models, Deterministic inventory models - EOQ model without and with shortages, Production inventory model without and with shortages, Inventory models with price - breaks.

Games Theory: Introduction, Basic definitions, Two - Person zero - sum games - Minimax (maximin) criterion, saddle point, value of a game, Solution of games with Saddle point, Mixed strategy games - Arithmetic method for (2×2) games, Dominance principle to reduce size of game, solution of $(2 \times n)$ and $(m \times 2)$ games, Algebraic Solution to rectangular games.

Queuing Models: Structure of queuing models, characteristics of Queuing process, Kendall's notation, Single channel systems - $(M/M/1:\infty/FIFO)$ model and $(M/M/1:N/FIFO)$ model.

Network Analysis: Introduction, Project Scheduling by CPM and PERT, Network diagram representation, rules for drawing network diagram, Labeling by Fulkerson's rule, Network calculations - EST, EFT, LST, LFT, Float/Slack and critical path, PERT calculations.

Text Books:

1. Operations Research by S.D. Sharma (Kedarnadh & Ramnadh & Co., Meerut).
2. Operations Research by V.K.Kapoor (Sultan Chand & Sons, New Delhi).

Reference Books:

1. Operations Research by H.A. Taha (PHI, New Delhi).
2. Operations Research: Methods and Problems by M. Sasiene, A. Yespal and L. Friedman (John Wiley & Sons, Newyork).
3. Operations Research by Kanti Swaroop, Gupta, Manmohan (Sultan Chand & Sons, New Delhi)

FLUID MECHANICS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

1. Develop an appreciation for the properties of Newtonian fluids,
2. Study analytical solutions to variety of simplified problems,
3. Understand the dynamics of fluid flows and the governing non-dimensional parameters,
4. Apply concepts of mass, momentum and energy conservation to flows.

Course Outcomes

After the completion of the course, students are able to

1. Apply the Bernoulli equation to solve problems in fluid mechanics.
2. Apply the concepts of momentum equation for finding the forces acting on the vanes of the turbines.
3. Apply control volume analysis to problems in fluid mechanics.
4. Apply potential flow theory to solve problems in fluid mechanics.
5. Identify the recent developments in fluid mechanics, with application to aerospace systems.

SYLLABUS

Properties of fluids- Introduction-Viscosity- Pressure and its measurement, Absolute, Gauge, Atmospheric and Vacuum pressure – Manometers, Simple manometers, Differential manometers. Hydrostatic forces on surfaces- Total Pressure and Pressure Centre- Vertical, Horizontal, inclined and Curved plane surfaces submerged in liquid- Buoyancy and Floatation.

Fluid Kinematics & Fluid Dynamics: Types of fluid flow- Continuity equation- Velocity potential function and Stream Function- Types of Motion, Linear Translation, Linear deformation, Angular deformation, Rotation, Vorticity and circulation-Vortex flow, forced and Free Vortex – Equation of Motion- Euler's equation - Bernoulli's equation and its applications- Venturimeter, Orifice Meter, Pitot tube-Momentum Equation-Momentum of momentum Equation.

Viscous Flow: Favourable pressure gradient and adverse pressure gradient-Power absorbed in Viscous Flow- Flow through pipes & Plates,- Hagen Poiseuille flow- Darcy's Weisbach friction factor- Loss of head due to friction in pipes, Minor Losses and Major losses - Flow through branched pipes- Power transmission through pipes-.

Dimensional and Modeling Analysis: Fundamental and derived dimensions- Dimensionless groups- Rayleigh method- Buckingham π -theorem- Model Analysis - Types of similarity- Geometric, Kinematic and Dynamic similarities- Dimensionless numbers- Modal Laws.

Laminar Boundary Layer: Definition- Laminar Boundary Layer- **Turbulent Boundary Layer** -Laminar Sub layer- Boundary Layer thickness-Displacement thickness, Momentum thickness and Energy thickness-Momentum integral equation- Flow over a flat plate.

Turbulent Boundary Layer: Laminar- Turbulent transition- Momentum equations and Renold's stresses- Fully developed turbulent flow through a pipe- Turbulent boundary layer on a flat plate- Laminar sub-layer- Boundary layer separation and control.

Compressible Fluid Flow: Thermodynamic relations- Continuity, Momentum and Energy equations- Velocity of sound in a compressible fluid- Mach number and its significance- Limits of incompressibility- Pressure field due to a moving source of disturbance- Propagation of pressure waves in a compressible fluids- Stagnation properties- Stagnation pressure, Temperature and density- Area velocity relationship for compressible flow

Text Books:

1. Fluid Mechanics and Hydraulic Machines, by R. K. Bansal, Laxmi publications.
2. Fluid Mechanics, by A.K. Mohanty, Prentice Hall of India Pvt.Ltd.

References:

1. Fluid Mechanics and Fluid Power Engineering by Dr. D.S. Kumar, S.K. Kataria & Sons.
2. Foundations of Fluid Mechanics, by Yuan, Prentice Hall of India.
3. Fluid Mechanics and its Applications, by S. K.Gupta and A.K.Gupta, Tata McGraw Hill New Delhi.
4. Fluid Mechanics and Hydraulic Machines by R. K. Rajput, S.Chand & Co.
5. Fluid Mechanics by Kothandaraman and Rudramoorthy.

IC ENGINES & GAS TURBINES

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

1. To impart the knowledge of basic gas power cycles and combustion phenomena, various performance factors and design aspects of SI and CI Engines.
2. To impart the knowledge of combustion phenomenon taking place in SI engines
3. To impart the knowledge of combustion phenomenon taking place in CI engines.
4. To impart the knowledge of Reciprocating compressors and methods to improve their efficiency.
5. To impart the knowledge of Rotary compressors and their performance.
6. To impart the knowledge of gas turbine cycles, and their components and efficiency improving techniques, and jet propulsion.

Course Outcomes

After completion of the course the student will be able to

1. Apply the knowledge of gas power cycles adequately and can calculate their efficiencies.
2. Explain the processes involved in combustion in S.I Engines.
3. Explain the processes involved in combustion in C.I Engines.
4. Apply the knowledge of reciprocating compressors in engineering applications.
5. Calculate the performance of rotary compressors in various engineering applications.
6. Compute and develop various methods to improve the efficiency of gas turbine power plants, and can explain jet propulsions.

SYLLABUS

I.C. engines: classification-comparison of two stroke and four stroke engines- comparison of S.I. and C.I. engines-Air cycles-Otto, Diesel, Dual, Stirling, Ericson and Atkinson cycles and their analysis Valve timing and port timing diagrams- Efficiencies- air standard efficiency, indicated thermal efficiency, brake thermal efficiency, mechanical efficiency, volumetric efficiency and relative efficiency-Testing and performances of I.C. engines-Basic principles of carburetion and fuel injection.

Combustion in S.I. Engines- Normal combustion and abnormal combustion Importance of flame speed and effect of engine variables-types of abnormal combustion pre-ignition and knock, Fuel requirements and fuel rating, anti-knock additions- Combustion chamber requirements and Types of combustion chamber- Design principles of combustion chambers.

Combustion in C.I. Engines- Stages of combustion- Delay period and its importance- effect of engine variables, diesel knock, suction compression and combustion induced turbulence, open and divided combustion chambers.

Reciprocating Compressors: Reciprocating compressors-effect of clearance in compressors, volumetric efficiency-single stage and multi stage compressors-effect of inter cooling in multi stage compressors.

Rotary Compressors: Classification- Roots blower- Vane type blower-centrifugal compressor- Adiabatic efficiency- Diffuser - Axial flow compressors- Velocity diagrams, degree of reaction, performance characteristics.

Gas Turbines: Simple gas turbine plant- Ideal cycle, closed cycle and open cycle for gas turbines Efficiency, work ratio and optimum pressure ratio for simple gas turbine cycle- Parameters of performance- Actual cycle, regeneration, Inter-cooling and reheating, closed and semi-closed cycle, Jet propulsion and Rockets.

Text Books:

1. A Treatise on Heat Engineering by Vasandhani and Kumar.
2. Applied Thermodynamics-II by R. Yadav.

Reference Books:

1. I.C. Engines by V. Ganesan.
2. Thermal Engineering, by R.K.Rajput.
3. I.C. Engines, by Mathur and Nehata.
4. Gas Turbines, by Cohen and Rogers.
5. I.C. Engines by V. Ganesan.

Code: B16 ME 3104

INDUSTRIAL MEASUREMENTS & METROLOGY

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

The main objectives of this module are:

1. To provide to the students an understanding and appreciation of the science of Measurement.
2. To expose the students to various mechanical and electrical engineering measuring devices, and understand the different degree of accuracy obtained from different types of instruments.

Course Outcomes:

Students who successfully complete this course will be able to:

1. identify the uncertainties in dimensional metrology and the define the measurement standards;
2. describe the fundamentals of dimensional and geometrical tolerances;
3. measure length and angles using line-graduated instruments, i. e. verniercalipers, micrometers, bevel protractor, sine bar and surface plates;
4. use comparative length-measuring instruments, i.e. dial indicator, to measure variations in the distance between two or more surfaces;
5. use effective methods of measuring straightness, flatness, roundness, profile, screw threads and gear teeth;
6. measure dimensions of shafts, bearings and linear surfaces in metric and imperial units using calipers, micrometers, and scales;
7. use contour projector and coordinate measuring machine to record measurements of complex profiles with high sensitivity;
8. use gage blocks, fixed gages, pneumatic gages gage blocks to measure various workpieces;
9. explain the effect of environmental conditions on the accuracy of measurements;
10. demonstrate the correct methods for adjustment and calibration of various measuring instruments;
11. use appropriate method for determination of accuracy based on product function and manufacturing capability.

SYLLABUS

Instrumentations: Concepts of measurements, static performance, characteristics accuracy of measurement and its analysis. Instrumentation, for measurement: Force, torque, strain. pressure, flow, Temperature and vibration.

Optical Methods of Measurement: Introduction, Laser beam as a light pointer, length/displacement measurement, temperature sensors, seismographic measurement. Introduction to fiber optics, fiber types, properties of optical fibres and a fibre optic sensor configuration.

ISO system of limits, Fits and Tolerances, Interchangeability, Plain limit gauges, Measurement of screw threads, major diameters, Minor diameters and effective diameter, Pitch, Limit gauges for internal and external threads, Measurement of spur gears, pitch, profile, lead, backlash, tooth thickness.

Tool maker's microscope, Straightness measurement, Slip gauges, Twisted strip mechanical comparator, Optical lever comparator, Optical projector, Electric comparator, Pneumatic comparator, Squareness testing, Optical bevel protractor, Sine bar, Angle gauges, Precision level, Autocollimeter, Angle dekkor, Optical dividing heads and rotary tables, Flatness measurement, Roundness measurement.co-ordinate measuring machines.

Surface texture: Parameters, sampling length, Specification, Stylus instruments for surface roughness measurement. Acceptance tests on machine tools: Lathe, Milling machine, Radial drilling machine, laser equipment

Text Books:

1. Process & Materials of Manufacture, R.A.Lindberg, 4th edition, Prentice-Hall of India, New Delhi.
2. A Text Book of Engineering Metrology, I.C.Gupta, Dhanpat Rai & Sons, Delhi.
3. Mechanical Measurements, by R.S.Sirohi, H.G. Radha Krishna, Wiley Eastern, New Delhi.

Reference Books:

1. A.S.T.M.E., Hand book of Industrial Metrology, Prentice-Hall of India, New Delhi.
2. A.S.T.M.E., Hand book of Manufacturing Engineering.
3. Manufacturing Processes & Materials for Engineers, L.E. Doyle & others, Prentice-Hall of India, New Delhi.
4. Manufacturing Technology by Adithan, New age international.
5. Instrumentation for Engineering Measurements, by R.H. Cerni and L.E.Foster, J.Wiley & Sons, New York.

KINEMATICS OF MACHINES

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

The overall objective of the course is to teach the basic principles of mechanisms, turning moment diagrams, flywheel, friction and speed control systems.

Specific objectives include:

1. To familiarize students with basic types of mechanisms, kinematic and dynamic analysis of mechanisms different methods
2. To understand the concept of turning moment diagrams, flywheel and speed control systems such as governors
3. To understand the concept of bearings, clutches and effect of friction in motion transmission and in machine components.
4. To understand the effect of inertia in mechanisms and to determine inertia torque

Course Outcomes:

Upon completion of this course, students should able:

1. Understand the basic principles of mechanisms in mechanical engineering applications
2. Apply kinematic and dynamic analysis of various machine components
3. Understand the turning moment diagrams and flywheel in various applications
4. Understand the importance of governors, bearings, clutches and their applications
5. Analyze the effect of inertia in mechanism and inertia torque
6. Understand the importance and effect of friction in motion transmission in machine components

SYLLABUS

Mechanisms and Machines: Introduction; Mechanism and machine; Rigid and resistant bodies; Link; Kinematic pair; Degrees of freedom; Classification of kinematic pairs; Kinematic chain; Linkage, mechanism and structure; Mobility of mechanisms; The four-bar chain; Mechanical advantage; Transmission angle; The slider-crank chain; Double slider-crank chain; Miscellaneous mechanisms.

Velocity Analysis: Introduction; Absolute and relative motions; Vectors; Addition and subtraction of vectors; Motion of a link; Four-link mechanism; Velocity images; Angular velocity of links; Velocity of rubbing; Slider-crank mechanism; Crank and slotted lever mechanism; Algebraic methods; Instantaneous center (I-center); Kennedy's theorem; Locating I-centers; Angular velocity ratio theorem; centrode.

Acceleration Analysis: Introduction; Acceleration; Four-link mechanism; Four-link mechanism; Acceleration of intermediate and offset points; Slider-crank mechanism; Coriolis acceleration component; Crank and slotted lever mechanism; Algebraic methods; Klein's construction; Velocity and acceleration from displacement-time curve.

Lower Pairs: Introduction; Pantograph; Straight line mechanisms; Engine indicators; Automobile steering gears; Types of steering gears; Hooke's joint; Double Hooke's joint.

Friction: Introduction; Kinds of friction; Laws of friction; Coefficient of friction; Inclined plane; Screw threads; Wedge; Pivots and collars; Friction clutches; Rolling friction; Antifriction bearings; Greasy friction; Film friction; Mitchell thrust bearing.

Dynamic Force Analysis: Introduction; D'Alembert's principle; Equivalent offset inertia force; Dynamic analysis of four-link mechanism; Dynamic analysis of slider-crank mechanism; Velocity and acceleration of piston; Angular velocity and angular acceleration of connecting rod; Turning moment on crankshaft; Turning-moment diagrams; Fluctuations of energy; Flywheels.

Governors: Introduction; Types of governors; Watt governor (simple conical governor); Porter governor; Proell governor; Hartnell governor; Hartung governor; Wilson-Hartnell governor (radial-spring governor); Pickering governor; Spring-controlled gravity governor; Inertia governor; Sensitiveness of a governor; Hunting; Isochronism; Stability; Effort of a governor; Controlling force.

Text Book:

1. Theory of Machines by R.S. Khurmi & J.K. Gupta

Reference books:

1. Theory of Machines by Thomas Bevan.
2. Theory of Machines by S.S. Rattan.

**FINITE ELEMENT ANALYSIS
(Elective-I)**

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

1. To teach students the basic principles and basic implementation method of finite element methods.
2. To teach students how to perform structural analysis using finite element methods.

Course Outcomes

At the end of the course, the student shall be able to:

1. Understand the principles and concepts related to finite element methods.
2. Implement finite element methods for simple analysis of 1-D problems such as bar, truss and beam either by hand calculation or by programming.
3. Numerically solve for deformation, stresses and strains of a structural component subjected to axial and bending loads.
4. Use commercial software package to perform structural analysis and are able to conduct engineering design.

SYLLABUS

Introduction: stress and equilibrium, strain – displacement relations, stress – strain relations, plane stress and plane strain conditions, The potential energy approach; Rayleigh-Ritz method Finite Element Method: Discretization, Types of elements, band width, node numbering, interpolation functions, local and global coordinates, convergence requirements, Types of boundary conditions, Steps in Finite Element Method, Applications of Finite Element Method .

One Dimensional Bar Problems: 1-D bar element - shape functions – Stiffness matrix and load vector– assembly of Matrices – Treatment of boundary conditions One dimensional quadratic element.

Trusses: Introduction; Plane trusses; shape functions – Stiffness matrix and load vector– assembly of Matrices – Treatment of boundary conditions; simple problems on trusses.

Analysis of Beams: Beam Element - Shape functions and Element stiffness matrix, load vector for concentrated and Uniformly Distributed Load, simple problems on beams.

Two Dimensional Problems: Finite element modeling of two dimensional Problems - constant strain triangle Element - treatment of boundary conditions 2D four noded iso parametric element, numerical integration , Gaussian Quadrature Approach.

Axisymmetric Solids Subjected to Axisymmetric Loading: Introduction; Axisymmetric formulation; Finite element modeling - triangular element; Problem modeling and boundary conditions.

Text Book:

1. Introduction to Finite Elements in Engineering, Chandrupatla, Ashok and Belegundu, Prentice – Hall.

Reference Books:

1. The Finite Element Method by O.C. Zienkiewicz, Tata McGraw Hill Company Ltd.
2. The Finite Element Methods in Engineering by Rao, S.S.
3. Concepts and Applications of Finite Element Analysis by Cook, R.D.
4. Applied Finite Element Analysis by Segerland, L.J.

AUTOMATION IN MANUFACTURING (Elective-I)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

1. To make the students understand the principles of advanced manufacturing procedures by providing the knowledge of various automation strategies used in production systems

Course Outcomes

Students will be able to

1. Understand the basic principles of automation and its components which are implemented in production systems.
2. Identify the importance of material handling and various automatic identification methods used in production systems.
3. Understand the components of manufacturing systems and different production lines.
4. Understand cellular manufacturing, forming part families, group technology and their involvement in flexible manufacturing systems.
5. Understand various automated inspection methodologies and manufacturing support systems like CAPP, shop floor control, etc.

SYLLABUS

Overview of Manufacturing and Automation: Production systems, automation in production systems, automation principles and strategies, manufacturing operations, production facilities, basic elements of an automated system, levels of automation, hardware components for automation and process control, programmable logic controllers and personal computers.

Material Handling and Identification Technologies: Material handling, equipment, analysis, storage systems, performance and location strategies. automated storage systems, AS/RS, types, automatic identification methods, Barcode technology, RFID.

Manufacturing Systems and Automated Production Lines: Manufacturing systems-components of a manufacturing system, single station manufacturing cells; manual assembly lines, line balancing algorithms, mixed model assembly lines, alternative assembly systems, automated production lines, applications, analysis of transfer lines.

Automated Assembly Systems: Fundamentals, analysis of assembly systems, cellular manufacturing, part families, coding and production flow analysis, group technology and flexible manufacturing systems, quantitative Analysis.

Quality Control and Support Systems: Quality in design and manufacturing, inspection principles and strategies, automated inspection, contact vs non-contact, CMM, manufacturing support systems, quality function deployment, computer aided process planning, concurrent engineering, shop floor control, just in time and lean production.

Text Books:

1. Mikell. P. Groover, Automation, production systems and computer integrated manufacturing 3rd edition, Published by Prentice Hall, 2012.
2. P. Radha Krishnan & S. Subrahmanyarn and Raju, CAD/CAM/CIM, 3rd Edition New Age International Publishers-2003.

Reference Books:

1. Singh, Nanua, System Approach to Computer Integrated Design and Manufacturing, Published by Wiley.
2. Tien-Chien Chang, Richard A. Wysk and Hsu-Pin Wang, Computer Aided Manufacturing, Pearson Publications - 2009.
3. R Thomas Wright and Michael Berkeihiser Good Heart, Manufacturing and Automation Technology, Willcox Publishers

TOOL DESIGN (Elective-I)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To apply the principles of locating and clamping systems.
2. Apply the design of different tools and components, such as jigs and fixtures, press tools for sheet metal working, molds for plastic injection molding, and die casting.
3. Use of limit gauges and tolerances

Course Outcomes:

1. Designing Injection Moulding Dies and Die Casting Dies, Jigs & Fixtures
2. Design jigs and fixtures for conventional and NC machining.
3. Select and design progressive, compound or combination dies for press working operations.
4. Can apply knowledge of designing Limit gauges and the use of gauge materials.

SYLLABUS

Locating and Clamping Devices: Principles of Jigs and Fixtures design-Locating principles-Locating elements-Standard parts-Clamping devices-Mechanical actuation-Pneumatic & hydraulic actuation-Analysis of clamping forces-Tolerance and error analysis.

Jigs & Fixtures: Drill bushes-Different types of Jigs-Plate latch, channel, box, post, angle plate, angular post, turnover, pot jigs- Automatic drill jigs-Rack & Pinion Operated, Air operated Jigs Components. General principles of lathe, milling and broaching fixtures-Grinding, Drilling and shaping fixtures, Assembly, Inspection and Welding fixtures-Modular fixtures. Design and development of Jigs and fixtures for simple components.

Press Tools: Press working terminology-Presses and Press accessories-Computation of capacities and tonnage requirements-Design and development of various types of cutting, forming and drawing dies. Bending dies – Introduction, bend allowance, spring back, edge bending die design. Drawing dies – Single action, double action and triple action dies, factors affecting drawing, drawing die design.

Die Casting Dies: Basic Terminology, Types of Dies: Single cavity, multicavity dies, combination dies, unit dies, advantages and disadvantages of types of dies, defects in die casting.**Injection Moulding:**Basic Terminology, Injection moulding machine and its elements, general configuration of a mould. 2 plate and 3 plate mould. Introduction to compression, transfer, blow moulding, extrusion, forming and calendaring.

Design of Limit Gauges: Elements, types and application of limit gauges, Gauge materials, their selection, Taylor's principles of gauge design, Types and methods to provide gauge tolerances. Design steps and design of plug & ring / snap gauge for given dimension and application.

Text Books:

1. Donaldson. C, Tool Design, Tata McGraw-Hill, 1986
2. "ASTME Handbook of Fixture Design ".Prentice Hall of India Pvt. Ltd.
3. Basu, Mukherjee, Mishra, Fundamentals of Tool Engg. Design, Oxford & IBH Publishing, N. Delhi

References:

1. A. K. Goroshkin, " Jigs and Fixtures Handbook ", Mir Publishers, Moscow, 1983.
2. "Die Design Handbook ", Ivana Suchy, McGraw Hill Book Co., 2005.
3. Injection Moulding Design, RGW Pye, john.1998
4. Production technology, HMT,Tata McGraw Hill.
5. P. Eugene Ostergaard, "Basic Die Making" - Mc Graw Hill Book, 1963.

NON CONVENTIONAL ENERGY RESOURCES (Elective-I)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

1. To study various types of non-conventional sources of energy and techniques used in exploiting solar, wind, tidal and geothermal sources of energy and bio-fuels.

Course Outcomes

Students will be able to:

1. Analyze the significance of renewable energy.
2. Understand the principles of solar radiation and design the solar collectors.
3. Know the functioning of basic components of wind energy and understand the utilization of biomass in power generation.
4. Understand the working principles of geothermal, ocean, tidal and wave energy techniques.
5. Know the functioning of direct energy conversion techniques.

SYLLABUS

Introduction to Energy Sources: Energy sources and their availability-introduction, commercial or conventional energy sources, new energy technologies; renewable energy sources, prospects of renewable energy resources

Solar Energy: Solar radiation, solar constant, solar geometry, radiation measurements; orientation and different types of solar collectors, applications of solar energy.

Wind Energy: Introduction; basic principles of wind energy conversion, site selection, basic components, classification, advantages & disadvantages.

Energy from Biomass: Conversion technologies, photosynthesis, bio-gas generation, classification of biogas plants, advantages and disadvantages.

Geothermal Energy: Geothermal sources, hydrothermal resources, geopressured resources, applications, advantages and disadvantages of geothermal energy.

Energy from the Oceans: Ocean thermal electric conversion (OTEC)-methods, site selection.

Energy from Tides: Basic principle, components, site requirements advantages and limitation of tidal power generation.

Ocean Waves: Introduction, advantages & disadvantages of wave energy, energy and power from the waves, wave energy conversion devices.

Small scale hydro-electric: Introduction, nature, classification and components.

Alternate Energy Sources: MHD-generator, principles, MHD systems, advantages.

Thermo electric power: Basic principles, thermo electric power generator

Thermo nuclear fusion energy: Introduction, basic nuclear fusion and reactions, requirements for nuclear fusion, advantages of nuclear fusion.

Energy conservation: Principles of energy conservation, co-generation, waste heat utilization, combined cycle power generation, heat recuperators, regenerators, heat pipes, heat pumps, renewable energy sources/devices.

Text Books:

1. Tiwari and Ghosal, “Renewable energy resources”, Narosa.
2. G.D. Rai, “Non-Conventional Energy Sources”, Dhanpat Rai and Sons

Reference Books:

1. Twidell & Weir, “Renewable Energy Sources”
2. Sukhatme, “Solar Energy”, Tata McGraw-Hill Education.
3. B.S Magal Frank Kreith & J.F Kreith, “Solar Power Engineering”
4. Frank Kreith & John F Kreider, “Principles of Solar Energy”

**PRODUCTION PLANNING AND CONTROL
(Elective-I)**

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To make the student understand the concept of Production planning and production control. In this process the teacher will be attempting to make the student knowledgeable about the objectives of Production planning and production control.
2. To make the student understand the different types of production and their selection criteria and make the student knowledgeable to deal with production planning and control in different types of production.
3. To make the student to understand the concept of forecasting and its necessity in real world requirements and its linkage to other functions of production manager.
4. To make the student aware of different forecasting techniques and errors associated with each forecasting methods.
5. To make the student understand the concept of process planning and about computer aided process planning.
6. To explain the student to understand the steps of production control, types of control and about control organization and its significance.
7. To make the student understand the meaning of inventory and different types of inventories and about need and necessity of inventory control.
8. To make the student to understand EOQ and EBQ models without shortages and about necessary terminology of inventory control like buffer stock, safety stock, re-order level, lead time.
9. To make the student to understand the concept of MRP-I, MRP-II.
10. To make the student to understand Aggregate planning and Master processing instructions.
11. To make student understand the about the concept of routing, steps in Routing and about route sheet.
12. To make the student to understand the concept of scheduling and forward and backward scheduling, Master scheduling.
13. To explain the student to understand the evaluation of job shop schedules with reference to priority scheduling rules.
14. To make the student understand the sequencing techniques in production scheduling.
15. To make the student understand the concept of centralized and de-centralized Dispatching

Course Outcomes:

1. Student is able to participate and can interact in real world scenario regarding production planning and production control and suggest the type of production required for specific real world requirement.

2. Student can undertake the responsibility of doing forecasting in real world situation is able to suggest correct forecasting method/technique for a specific real world situation and can also able to judge the suitability of the method for a real world situation depending on the error associated with the method.
3. Student can understand the need of inventory control and can able to undertake activities relating to inventory management.
4. The student is knowledgeable about MRP-I &II, Aggregate planning can able to implement them in real world situation.
5. Student can understand and participate in the design of both forward and backward scheduling and Master scheduling and can able to evaluate different job shop schedules with reference to priority scheduling rules.

SYLLABUS

Introduction: Definition – Objectives of production Planning and Control – Functions of production planning and control – Types of production – Organization of production planning and control department.

Forecasting: Importance – Types of forecasting– Forecasting techniques – qualitative methods and quantitative methods

Inventory management: Functions of inventories – relevant inventory costs – EOQ model – Inventory control systems: Fixer order quantity system and Periodic review system - ABC analysis -VED analysis- Material Requirement Planning, Bill of material, MRP II - Master Production Scheduling.

Aggregate planning: Chase planning, Expediting, controlling aspects.

Routing: Definition – Routing procedure –Route sheets — Factors affecting routing, procedure – Difference with loading

Scheduling: Policies – Types of scheduling - Forward and Backward Scheduling – Gantt Charts – Flow shop Scheduling – n jobs and 2 machines, n jobs and 3 machines – Job shop Scheduling – 2 jobs and n machines – Line of Balance.

Dispatching: Activities of dispatcher – Dispatching procedure – follow up – priority rules for dispatching jobs - Applications of computer in production planning and control.

Text Books:

1. Elements of Production Planning and Control / Samuel Eilon.
2. Modern Production/ operation managements / Baffa & Rakesh Sarin

References:

1. Operations Management – S.N. Chary.
2. Inventory Control Theory and Practice / Martin K. Starr and David W. Miller.
3. Production Control A Quantitative Approach / John E. Biegel.
4. Operations Management / Joseph Monks.

**RAPID PROTOTYPING
(Elective-I)**

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

1. To introduce Rapid Prototype tools and techniques and additive manufacturing techniques for design and Manufacturing.

Course Outcomes

Students will be able to:

1. Assess the need of RPT in Product development.
2. Judge the correct RP Process for Product/Prototype development.
3. Predict the technical challenges in 3D printing.
4. List the applications of RPT

SYLLABUS

Introduction: Need - Development of RP systems-, – RP process chain - Impact of Rapid Prototyping and Tooling on Product Development – History of RP systems and their classification- Benefits Applications – Digital prototyping - Virtual prototyping.

Liquid Based And Solid Based Rapid Prototyping Systems: Stereo lithography Apparatus, Fused deposition Modeling, Laminated object manufacturing, three dimensional printing; Working Principles, details of processes, products, materials, advantages, limitations and applications.

Powder Based Rapid Prototyping Systems: Selective Laser Sintering, Direct Metal Laser Sintering, Three Dimensional Printing, Laser Engineered Net Shaping, Selective Laser Melting, Electron Beam Melting; Processes, materials, products, advantages, applications and limitations.

Reverse Engineering And Cad Modeling

Basic concept- Digitization techniques – Model Reconstruction – Data Processing for Rapid Prototyping; CAD model preparation, Data Requirements – geometric modeling techniques: Wire frame, surface and solid modeling – data formats – Data interfacing, Part orientation and support generation, Support structure design, Model Slicing and contour data organization, direct and adaptive slicing, Tool path generation.

Rapid Tooling: Classification: Soft tooling, Production tooling, Bridge tooling; direct and indirect – Fabrication processes, Applications.

Text Books:

1. Rapid prototyping: Principles and applications, second edition, Chua C.K., Leong K.F., and Lim C.S., World Scientific Publishers, 2003.
2. Rapid Tooling: Technologies and Industrial Applications, Peter D.Hilton, Hilton/Jacobs, Paul F.Jacobs, CRC press, 2000.

Reference Books:

1. Rapid prototyping, Andreas Gebhardt, Hanser Gardener Publications, 2003.
2. Rapid Prototyping and Engineering applications: A tool box for prototype development, Liou W.Liou, Frank W.Liou, CRC Press, 2007.
3. Rapid Prototyping: Theory and practice, Ali K. Kamrani, Emad Abouel Nasr, Springer, 2006.

INDUSTRIAL METROLOGY & MECHATRONICS LAB

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

Course Objectives:

1. To demonstrate the fundamentals of Metrology and Measurement Engineering concepts, different tools used for minute measurements, and calibrate various kinds of measuring instruments.
2. To demonstrate the involvement of sensors and electronic devices in the mechanical oriented industries by introducing PLC, and Mechatronics equipment.
3. To demonstrate the use of logic circuits in controlling various mechanical devices including material handling equipment, lift control systems, pneumatic controller systems etc.
4. To demonstrate the techniques and processes followed for calibrating measuring instruments like Micrometer, Mechanical Comparator, and Vernier Caliper etc. and educate students about the metrology equipment.
5. To train the students on the fundamentals of logic circuit design, to write ladder logic programs and execute them to control various mechanical devices.

Course Outcomes:

1. Students will be able to understand the various logics involved in controlling mechanical industry equipment.
2. The student will be able to operate measurement instruments on their own and test different components for their dimensional accuracy.
3. A project involving writing ladder logic for controlling a mechanical device, executing the program is required from each student and graded by the instructor, so that the student will be able to understand the Mechatronics concept, practically and from the application point of view.

LIST OF EXPERIMENTS**METROLOGY LAB EXPERIMENTS**

1. Calibration of the following instruments using slip gauges: (a) Calibration of Micrometer; (b) Calibration of Mechanical Comparator; (c) Calibration of Vernier Caliper; (d) Calibration of Dial Gauge.
2. Measurement of taper angle using: (a) Bevel Protractor; (b) Dial Gauge; (c) Sine-Bar; (d) Auto-Collimator.
3. Alignment tests: (a) Parallelism of the spindle; (b) Circularity & Concentricity of the spindle; (c) Trueness of running of the spindle.
4. Measurement of following Gear parameters: (a) diameter, pitch/module; (b) Pitch circle diameter; (c) Pressure angle; (d) Tooth thickness.

5. Check the flatness of a surface plate using: (a) spirit level; (b) Auto-collimator.
6. Study on flatness of slip gauges and finding the height of a slip gauge using light wave interference.
7. Measurement of following using Tool Maker's Microscope: (a) Thread details; (b) Cutting tool angles.
8. Determining the diameter of a cylindrical piece, taper angle of a V-block and central distance of two holes of a specimen.

MECHATRONICS LAB EXPERIMENTS

1. Training on Programmable Logic Controller to control the following using ladder logic program: (a) Lift Control; (b) Traffic Signal Control.
2. Training on Programmable Logic Controller Sensor Training Kit to study the operation of the following devices: (a) Proximity Switch; (b) Photo Electric Switch; (c) Limit Switch.
3. Training on Sensor and Transducer to study the operation of (a) Linear position or Force applications (LVDT and strain gauge transducer); (b) Rotational Speed or Position Measurement (The inductive Transducer); (c) Linear or Rotational Motion (D.C. Solenoid & D.C. Relay)
4. Training on actuation of hydraulic cylinder operation for Punching Machine application.
5. Training on control of conveyor operation for material handling equipment.

Reference Books:

1. Engineering Mechanics by S.Timoshenko and D.HYoung McGraw-Hill.
2. Engineering Mechanics by Singer.
3. Internal Combustion Engines by V. Ganesan, McGraw-Hill.

IC ENGINES LAB

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

Course Outcomes:

1. A This course comprehensively deals with practical approach to I.C Engines and four bar chain mechanisms
2. To expose students to different methods of finding friction power in single and multi cylinder engines
3. Understand the working of Gyroscopes

Course Objectives:

1. Students would appreciate the fundamentals of thermodynamics being extended to real time applications
2. Students might come out with innovative ideas which may be extended in the form of projects

LIST OF EXPERIMENTS

1. Load test on single cylinder diesel Engine.
2. Morse test on multi-cylinder engine.
3. Heat balance sheet on I.C. Engines
4. Study of multi-cylinder engines and determination of its firing order
5. Study of automobile mechanisms
6. Performance test on multi cylinder diesel engine
7. Verification of laws of balancing.
8. (a) Determination of ratios of angular speeds of shafts connected by Hooke's joint.
(b) Determination of the ratio of times and ram velocities of Whitworth quick return motion mechanism.
9. To draw curves of slider displacement and crank angle and linear velocities w.r.t. time for a slider crank mechanism and compare with theoretical values.
10. To determine the relation of gyroscopic couple and compare with the theoretical values.

Reference Books:

1. Thermal Engineering, by R.K.Rajput.
2. I.C. Engines, by Mathur and Nehata.

VERBAL & QUANTITATIVE APTITUDE – I
(Common to All Branches)

Theory	: 5 Periods(VA-3+QA-2)	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Part-A: Verbal and Soft Skills-I

Course objectives:

1. To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.
2. To familiarize the learner with high frequency words as they would be used in their professional career.
3. To inculcate logical thinking in order to frame and use data as per the requirement.
4. To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.
5. To familiarize students with soft skills and how it influences their professional growth.

Course Outcomes:

The student will be able to

1. Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.
2. Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.
3. Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.
4. Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.
5. Apply soft skills in the work place and build better personal and professional relationships making informed decisions.

SYLLABUS

Grammar: (VA)

Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns-number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.

Vocabulary: (VA)

Synonyms and synonym variants(with emphasis on high frequency words), antonyms and antonym variants(with emphasis on high frequency words), contextual meanings with regard to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.

Reasoning: (VA)

Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms).

Usage: (VA)

Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.

Soft Skills:

Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis –Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.

Text Books:

1. Oxford Learners's Grammar – Finder by John Eastwood, Oxford Publication.
2. R S Agarwal's books on objective English and verbal reasoning
3. English Vocabulary in Use- Advanced , Cambridge University Press.
4. Collocations In Use, Cmbridge University Press.
5. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. English Grammar in Use by Raymond Murphy, CUP
2. Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. The Art of Public Speaking by Dale Carnegie
5. The Leader in You by Dale Carnegie
6. Emotional Intelligence by Daniel Golman
7. Stay Hungry Stay Foolish by Rashmi Bansal
8. I have a Dream by Rashmi Bansal

Part-B: Quantitative Aptitude -I

Course objectives:

The objective of introducing quantitative aptitude-1 is:

1. To familiarize students with basic problems on numbers and ratio's problems.
2. To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.
3. To enable the students to work efficiently on percentage values related to shares, profit and loss problems.
4. To inculcate logical thinking by exposing the students to reasoning related questions.
5. To expose them to the practice of syllogisms and help them make right conclusions.

Course Outcomes:

1. The students will be able to perform well in calculating on number problems and various units of ratio concepts.
2. Accurate solving problems on time and distance and units related solutions.
3. The students will become adept in solving problems related to profit and loss, in specific, quantitative ability.
4. The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry.
5. The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies.

SYLLABUS

Numbers, LCM and HCF, Chain Rule, Ratio and Proportion

Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion.

Time and work, Time and Distance

Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.

Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends

Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.

Introduction, number series, number analogy, classification, Letter series, ranking, directions

Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.

Data sufficiency, Syllogisms

Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi

References:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

Websites:

1. www.m4maths.com
2. www.Indiabix.com
3. www.800score.com
4. Official GRE site
5. Official GMAT site

INTRODUCTION TO ABRASIVE MACHINING AND FINISHING PROCESSES (MOOCS-I)

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

About the Course:

This course will define the areas of application of traditional as well as non-traditional abrasive finishing processes in the manufacturing industry. The lectures will introduce the basic principles of material removal by use of abrasives particles and material removal mechanism of different abrasive process. The effects of various input parameters on the outputs as well as the use of cutting fluids in various finishing process will be discuss. A variety of numerical problems and MCQs, discussions will also be included.

Course Layout:

Week 1: Introduction conventional abrasive processes, Introduction to abrasive processes, Grinding Process

Week 2: Conventional abrasive finishing processes (CAFP): Honing & Wire Brushing,CAFP: Lapping, Buffing & Super finishing,Practical Conventional abrasive finishing processes

Week 3 : Adv. abrasive machining processes (AAMP),AAMP

Week 4 : Hybrid Adv. Abrasive Machining Processes,Advanced Finishing

Week 5 : Adv. Finishing: Abrasive Flow Finishing

Week 6 : Adv. Finishing: Magnetic Abrasive Finishing

Week 7 : Adv. Finishing: Magnetic Rheological Finishing

Week 8 : Hybrid abrasive finishing,Finishing of Advanced Materials

Suggested Reading Materials:

1. M. C. Shaw, Principles of Abrasive Processing, Oxford University Press, 1996.
2. VK Jain, Micromanufacturing Processes, CRC press, 2012.
3. Jain VK, Nanofinishing Science and Technology: Basic and Advanced Finishing and Polishing Processes, CRC Press, 2016.
4. J. A McGeough, Advanced methods of machining, Springer Science & Business Media, 1988.
5. G. K. Lal, Introduction to Machining Science, New Age International Publishers, 2007.
6. A. Ghosh and A. K. Malik, Manufacturing Science, East West Press, 2010.
7. Metalworking Fluids (MWFs) for Cutting and Grinding, Edited by: V.P. Astakhov and S. Joksch

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_me70/preview

PRINCIPLES OF CASTING TECHNOLOGY (MOOCS-I)

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

About the course :

The course focuses on understanding the basics of science and technology of casting processes. Metal casting industries have evolved during the past hundred years because of advancements in technologies. The properties of the cast metals significantly depends upon the type of molding, melting, solidification and post treatment practices. This needs to be understood by the young students as well as practicing shop floor engineers so that products with superior qualities can be cast. The basic purpose of this course is to provide a sound understanding of concepts and principles of casting technology so as to enable them to be conversant with advances in these methods in the long run towards increasing the productivity of casting industries.

Course Plan :

Week 1: Introduction to Casting technology, Solidification analysis for metals and alloys

Week 2: Technology of patternmaking, study of molding sands and their testing methods

Week 3: Technology of mould making and core making, Special sand moulding processes

Week 4: Principles of gating design for castings

Week 5: Principles of risering design for castings

Week 6: Special casting methods, Melting furnaces

Week 7: Melting and pouring practices for production of Cast Iron family, steel and non-ferrous metals and alloys

Week 8: Fettling and Heat treatment of castings, Casting defect and its diagnostic methods

Suggested reading:

1. Heine, R.W., Loper, C.R., and Rosenthal, P.C., "Principles of Metal Casting", TMH
2. Ghosh, A., and Mallik, A.K., "Manufacturing Science", Affiliated East-West Press Pvt. Ltd.
3. Jain P. L., "Principles of Foundry Technology", TMH
4. Chakrabarti, A. K., "Casting Technology and Cast Alloys", PHI

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_me50/preview

**PRINCIPLES OF METAL FORMING TECHNOLOGY
(MOOCS-I)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

About the Course:

The course focuses on understanding the science and technology of different forming processes. Most of the metallic objects undergo at least one of the metal forming operations, except the cast ones. Understanding basic principles of metal forming and further being applied by engineers and metallurgists directly contribute towards improvement in production in the industries. The concept of stress, deformation and failure, mechanics of metalworking and analysis of different metal working processes will be covered during the whole course. Introduction and working principle of powder metallurgy forging will be presented in the end. The course will enable the students be conversant with working principles so that they can use the knowledge gained towards increasing the productivity of manufacturing industries in the long run.

Course Layout:

Week 1 : Introduction and classification of metalworking processes, Behavior of materials

Week 2 : Concept of stress and strain, Hydrostatic and deviatoric stresses

Week 3 : Flow curve Yield criteria for ductile materials, plastic stress strain relationships

Week 4 : Yielding and ductility during instability, Effect of strain rate and temperature on flow properties

Week 5 : mechanics of metalworking, Analysis methods, Hot and cold working

Week 6 : Introduction, classification and analysis of forging and rolling operations

Week 7 : Defects in rolled and forged components, Analysis of extrusion process

Week 8 : Classification and analysis of wire and tube drawing and sheetmetal working, Powder metallurgy forming

Suggested Reading Materials:

1. Ghosh, A., and Mallik, A.K., Manufacturing Science, Affiliated East-West Press Pvt. Ltd.
2. Rowe, Geoffrey W., An introduction to the principles of Metal Working, TMH.
3. Dieter George E., Mechanical Metallurgy, McGrawHill

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_me49/preview

MECHANICS OF MACHINING (MOOCS-I)

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

About the Course:

Machining is a metal removal process, which can be accomplished by applying force on raw material by means of a cutting tool. This course aims at explaining the physics of the cutting process. The course will contain discussion of statics, kinematics and kinetics of the cutting process. Experimental findings relevant to mechanics of the process will also be discussed. The course will also include introductory discussion on non-traditional machining processes.

Course Layout:

Week 1 : Deformation of metals, Mechanism of plastic deformation, Machining processes: Single edge tool, types of chips

Week 2 : Tool geometry: single point cutting tool specifications, Tool specifications, conversion of tool angles, Multi-point cutting tools, Mechanics of orthogonal cutting, force relationships

Week 3 : Determination of stress, strain, and strain rate, Measurement of shear angle, Other analysis for force relationships

Week 4 : Mechanics of oblique cutting, Measurement of cutting forces

Week 5 : Thermal aspects of machining: Temperatures in orthogonal cutting, Tool wear and tool life and tool life equations, Economics in machining

Week 6 : Practical machining operations: Turning and shaping & planning operation, Practical machining operations: milling and drilling, Grinding of metals and mechanics of grinding process

Week 7 : Abrasive machining and finishing operations, CNC machines and CNC programming

Week 8 : Introduction to advanced machining processes

Suggested Reading Materials:

1. G K Lal, Introduction to Machining Science, 3rd edition, New Age International Pvt Ltd., 2007.
2. A Ghosh and A K Mallik, Manufacturing Science, 2nd edition, Affiliated East-West Press Pvt. Ltd., 1986.
3. G Boothroyd and W. A. Knight, Fundamentals of Machining and Machine Tools, CRC-Taylor and Francis, 2006.
4. V K Jain, Advanced Machining Processes, Allied Publishers, 2009.
5. P M Dixit and U S Dixit, Modeling of Metal Forming and Machining Processes: By Finite Element and Soft Computing Methods, Springer, 2010.
6. A Bhattacharya, Metal Cutting: Theory and Practice, New Central Book Agency, 2012.

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_me56/preview

MANUFACTURING OF COMPOSITES (MOOCS-II)

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

About the course

Selecting manufacturing technique has emerged as one the paramount challenge in the field of composites. Composites are now being used in almost every field of industry, and students working in the area of the composites need to learn the basics, and progressive techniques of composites manufacturing. This course covers the important aspects of composites manufacturing: process selection guidelines, thermoset ad thermoplastic Composites manufacturing processes, process parameters and characterizations. Applications and use of each manufacturing process is focused and this is represented separately.

Course Plan:

Week 1- Introduction to Composites

- Introduction to Composites
- Function of the Matrix and Reinforcement in Composites
- Matrices: Thermosets and Thermoplastic
- Fiber Reinforcement

Week 2- Properties and testing composites

- Properties of Composites
- Composites testing
- Composites design: Laminate theory, Rule of mixtures, symmetry and balance

Week 3- Thermoset Composites manufacturing processes

- Material selection process cont.
- Material selection process cont.
- Design for manufacturing.

Week 4- Thermoset composite manufacturing processes

- Thermoset Composite manufacturing:Lay-up processes,Spray up process
- Thermoset Composite manufacturing:Fiber placement process
- Thermoset Composite manufacturing:Resin transfer moulding

Week 5- Thermoplastic composite manufacturing processes

- Thermoset Composite manufacturing:Vaccum assisted resin transfer moulding
- Thermoset Composite manufacturing:Compression molding process
- Thermoset composites manufacturing:Filament winding

Week 6- Thermoplastic composite manufacturing processes

- Thermoplastic Composite manufacturing: Sheet moulding
- Thermoplastic Composite manufacturing: Injection moulding, sheet moulding, Calendaring
- Thermoplastic Composite manufacturing: Extrusion, Blow molding, rotational molding, Thermoforming

Week 7- Metal and ceramic matrix composites

- Metal Matrix Composites: Metal matrix and reinforcement
- Manufacturing processes for Metal Matrix Composites: Dispersion hardened and particle composite
- Manufacturing processes for Metal matrix composites: Layer composites and infiltration method

Week 8- Prevention of Damage, repair of Composites and selection of processes

- Ceramic matrix composites: Hot isostatic processing
- Non – destructive testing of Composites
- Manufacturing process selection: Cost, performance, size shape, rate of production. Steps for process selection

Suggested reading

1. Mazumdar, S., 2001. Composites manufacturing: materials, product, and process engineering. CRC press.
2. Balasubramanian, M., 2013. Composite materials and processing. CRC press.
3. Campbell Jr, F.C. ed., 2003. Manufacturing processes for advanced composites. Elsevier.

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_me59/preview

PROCESSING OF POLYMERS AND POLYMER COMPOSITES (MOOCS-II)

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

About the Course

The main objective of the current MOOC on Processing of Polymers and Polymer Composites course is to impart an understanding of the manufacturing science and engineering of polymers and polymer composites which is usually not covered at the UG level. The course deals with the study of the basic nature of different polymers and polymer composites and the manufacturing processes associated thereof. The classification of engineering materials and processing techniques, the structure and mechanical properties of plastics, thermoplastics and thermosets, the various processing techniques of polymers such as Extrusion, Injection molding, Thermoforming, Compression molding and Transfer molding have been explained with the relevant and specific examples. The fundamental concept and classification of composite materials, properties of composites and the primary as well as secondary processing methods of polymeric matrix composites have been explained in detail. The various issues, challenges and opportunities in primary and secondary processing of polymers and polymer composites has also been explained.

Course Plan

Week1: Introduction to course, Engineering materials and processing techniques, Thermoplastics and thermosets, Processing of polymers, Thermoforming process.

Week 2: Extrusion, Compression molding, Injection molding.

Week 3: Transfer molding, Rotational molding, Blow molding, Composite materials: basic concepts, Classification of composite materials.

Week 4: Processing of polymer composites, Hand-layup, Spray-layup, Compression molding Injection molding.

Week 5: Reaction injection molding, Autoclaving, Resin transfer molding, Filament winding, Pultrusion.

Week 6: Sheet molding, Pre-pegging and challenges in primary processing of composites, Secondary processing of polymer composites, Joining of polymer composites, Adhesive joining.

Week 7: Mechanical joining, Microwave joining, Induction and resistance welding, Drilling of polymer composites.

Week 8: Conventional vs ultrasonic drilling, Remedies for reducing drilling induce damages, Research tools for secondary processing, Numerical problems and case studies.

Suggested Reading

1. Plastic Materials and Processing : A. Brent Strong, Prentice Hall, ISBN 0-13-021626-7
2. Composite Materials: Engineering and Science: F.L. Mathews and R.D. Rawlings, CRC press, 084930251X
3. Handbook of Composites: S.T. Peters, ISBN 978-1-4615-6389-1.

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_me57/preview

ROBOTICS (MOOCS-II)

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

About the Course:

The course will start with a brief introduction to robots and robotics. The motivation behind keeping robots in modern industries will be discussed. After providing a brief history of robotics, different components of a robotic system will be identified. The method of determining degrees of freedom of a robotic system will be discussed with some examples. After classifying the robots based on certain criteria, workspace analysis of manipulators will be carried out. Applications of robots in different areas like in manufacturing units, medical science, space, and others, will be discussed. Various methods of robot teaching will be explained with some suitable examples. Economic analysis will be conducted to decide whether we should purchase a robot. Both forward and inverse kinematics problems will be solved with the help of some suitable examples. To ensure smooth variation of joint angles of the robot, trajectory planning schemes will be explained. After carrying out velocity analysis with the help of Jacobian matrix, inverse dynamics problems of robots will be solved using Lagrange-Euler formulation. Control scheme used in robots to realize the joint torques will be discussed. Besides manipulators, analysis will be carried out on wheeled and multi-legged robots. The working principles of various sensors used in robots will be explained in detail. The steps to be followed in robot vision will be discussed with some suitable examples. The principles of motion planning algorithms will be explained in detail. Thus, this course will deal with all the issues related to kinematics, dynamics, control schemes and robot intelligence.

Course Layout:

- Week 1** : Introduction to Robots and Robotics
- Week 2** : Introduction to Robots and Robotics; Robot Kinematics
- Week 3** : Robot Kinematics
- Week 4** : Robot Kinematics; Trajectory Planning
- Week 5** : Robot Dynamics
- Week 6** : Control Scheme; Sensors; Robot Vision
- Week 7** : Robot Vision; Robot Motion Planning
- Week 8** : Intelligent Robot; Biped Walking; Summary

Suggested Reading Materials:

1. Fundamentals of Robotics by D.K. Pratihar, Narosa Publishing House, New-Delhi, 2017.
2. Robotics by K.S. Fu, R.C. Gonzalez, C.S.G. Lee, McGraw-Hill Book Company, 1987
3. Introduction to Robotics by J.J. Craig, Addison-Wesley Publishing Company, 1986

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_me61/preview

**ANALYSIS AND MODELING OF WELDING
(MOOCS-II)****Theory : 4 Periods**
Exam : 3 Hrs.**Sessionals : 100**
Credits : 2

About the Course

Welding is an important fabrication process in manufacturing industries. Complex parts fabricated using forming, machining and metal removal processes are assembled into a larger part using welding. The welding process involves a number of physical processes that play a role in determining the performance of the joint. With newer and dissimilar combinations required to be joined as the technological advances take place in the area of manufacturing, it is important for practicing engineers, graduate students and prospective researchers to understand welding to take advantage of the developments taking place. This course offers detailed analysis and modeling techniques that apply to the different physical phenomena that take place during welding.

COURSE LAYOUT**Week 1**

- 1 Introduction to fusion welding processes
- 2 Heat sources
- 3 Heat removal

Week 2

- 4 Thermal modeling
- 5 Single domain approach - handling phase change
- 6 Analytical solutions

Week 3

- 7 Fluid flow in the weld pool
- 8 Zones in a weldment
- 9 Numerical Solutions

Week 4

- 10 Conduction mode and Keyhole mode
- 11 Solute transfer – Macroscale
- 12 Solute transfer – Microscale

Week 5

- 12 Solute segregation profiles
- 13 Microstructure evolution
- 14 Defects in fusion welds

Week 6

- 15 Effects of dilution
- 16 Weld Cladding
- 17 Distortion in welding

Week 7

- 18 Dissimilar welding
- 19 Solutions to Dissimilar welding
- 20 Integrated approach

Suggested Reading

1. Welding Metallurgy – Sindo Kou, Wiley Interscience 2nd Edition (2002)
2. Contemporary Literature – will be listed on the course site.

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_mm15/preview

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)
III/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards)
Under Choice Based Credit System
MECHANICAL ENGINEERING
II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ME 3201	Industrial Engineering & Management	4	3	1	--	4	30	70	100
B16 ME 3202	Fluid Machinery & Systems	4	3	1	--	4	30	70	100
B16 ME 3203	Design of Machine Elements	4	3	1	--	4	30	70	100
B16 ME 3204	Refrigeration and Air Conditioning	4	3	1	--	4	30	70	100
B16 ME 3205	Dynamics of Machines	4	3	1	--	4	30	70	100
# ELE-II	Elective-II	4	3	1	--	4	30	70	100
B16 ME 3212	Industrial Engineering Lab	2	--	--	3	3	50	50	100
B16 ME 3213	Fluid Mechanics & Machines Lab	2	--	--	3	3	50	50	100
B16 ENG 3202	Verbal & Quantitative Aptitude-II	2	5	--	--	5	100	--	100
B16 ME 3214	Mini Project	2	--	--	3	3	50	--	50
# M-III	MOOCS-III	2	4	--	--	4	100	--	100
Total		34	27	6	9	42	530	520	1050

	Code No.	Name of the Course
# ELE-II	B16 ME 3206	Control Systems
	B16 ME 3207	Optimization Techniques
	B16 ME 3208	Automobile Engineering
	B16 ME 3209	Supply Chain Management
	B16 ME 3210	Nano Technology
	B16 ME 3211	Computational Fluid Dynamics
# M-III	B16 ENG 3203	Basic Coding
	B16 ME 3215A	Design For Quality, Manufacturing and Assembly
	B16 ME 3215B	Introduction to Research
	B16 ME 3215C	Nature and Properties of Materials

Code: B16 ME 3201

INDUSTRIAL ENGINEERING & MANAGEMENT

Theory	: 4 Periods	Sessionals	: 30
Exam	: 3 Hrs.	Ext. Marks	: 70
		Credits	: 4

Course Objectives:

Students learn

1. To know management practices in industry
2. To acquire capacity to handle industrial disputes
3. To know much about production activities and to improve productivity
4. To learn the work study procedures and quality concepts to get more productivity
5. To have exposure on some maintenance practices in industry.

Course Outcomes:

Students will be able to

1. Students will be able to apply management theories in organization
2. They will know personal management techniques to motivate the workers.
3. They are able to settle the industrial disputes in organization.
4. They are also acquire full knowledge on production planning and control procedures
5. They understand the economics of plant layout
6. Students are aware of materials handling principles and equipment
7. They will be able to apply maintenance practices
8. They will have knowledge of materials management
9. They will be able to improve the productivity by applying work study procedures and quality concepts

SYLLABUS

Concepts of Industrial Management: Principles of management- Growth of management thought, Functions of management, Principles of organization, Types of organization and committees.

Introduction to personnel management- Functions, Motivation, Theories of motivation, Hawthorne studies, Discipline in industry, Promotion, Transfer, lay off and discharge, Labour turnover.

Industrial relations- Trade unions, Industrial disputes, Strikes, Lock-out, Picketing, Gherao, Settlement of industrial disputes, Collective bargaining, Industrial dispute act 1947 and factories act 1948.

Production Planning and Control: Types of productions, Production cycle, Product design and development, Process planning, Forecasting, Loading, Scheduling, Dispatching, Routing, Progress, Control, Simple problems.

Plant Layout: Economics of plant location, Rural Vs Suburban sites, Types of layouts, Types of building, Travel chart technique, Assembly line balancing simple problems.

Materials Handling- Principles, Concept of unit load, Containerization, Pelletization, Selection of material handling equipment, Applications of belt conveyors, Cranes, Forklift trucks in industry.

Work Study: Concept of productivity, Method Study - Basic steps in method study, Process charts, Diagrams, Models and Templates, Principles of motion economy, Micro motion study, Therbligs, SIMO chart. Work Measurement - Stop watch procedure of time study, Performance rating, allowances, Work sampling, Simple problems.

Materials Management: Introduction, Purchasing, Objectives of purchasing department, Buying techniques, Purchase procedure, Stores and material control, Receipt and issue of materials, Store records. Inventory Control, EOQ model(Simple problems).

Quality Control - Control charts of variables and attributes (Use of formulae only). Single and Double sampling plans.

Text Book:

1. Industrial Engineering Management, by Dr. O. P .Khanna.

Reference Books:

1. Principles of Management by Koontz & Donnel.
2. Production and Operations Management by Everette Adam & Ronald Ebert.
3. Operations Management by John McClain & Joseph Thames.
4. Industrial Engineering and Production Management by Telsay, S. Chand & Co.

FLUID MACHINERY & SYSTEMS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Develop an appreciation for the properties of Newtonian fluids,
2. Study analytical solutions to variety of simplified problems,
3. Understand the dynamics of fluid flows and the governing non-dimensional parameters,
4. Apply concepts of mass, momentum and energy conservation to flows,
5. To teach the student about the fundamental of fluid dynamic equations and its applications fluid jets
6. To impart the knowledge of various types of pumps and turbines.

Course Outcomes:

After the completion of the course, students are able to

1. Apply the Bernoulli equation to solve problems in fluid mechanics.
2. Apply the concepts of momentum equation for finding the forces acting on the vanes of the turbines.
3. Apply control volume analysis to problems in fluid mechanics.
4. Apply potential flow theory to solve problems in fluid mechanics.
5. Identify the recent developments in fluid mechanics, with application to aerospace systems.
6. To impart the knowledge of various types of turbines and the performance of hydraulic turbines and pumps.

SYLLABUS

Impact of jet and jet propulsion: Impact of jet on stationary surfaces- Impact of jet on hinged surfaces- moving curved vane with tangential entry of water- Radial flow over the vanes- Jet propulsion.

Hydraulic Turbines: Classification- Pelton wheel- Reaction turbines- Inward and outward radial flow reaction turbines- Francis turbine- Axial flow reaction turbine- Kaplan turbine- Draft tube- Types- Theory- and efficiency of draft tube.

Specific Speed: Determination- Significance- Unit quantities- Unit speed- Unit discharge and unit power- Characteristic curves of hydraulic turbines- Constant heat curves- Constant speed curves and Iso-efficiency curves- Governing of turbines.

Centrifugal Pumps: Main parts- Efficiency- Minimum speed for starting- Multi-stage centrifugal pumps- Specific speed of a centrifugal pump- Priming of a centrifugal pump-

Characteristic curves- Main, Operational and constant efficiency curves- Cavitation- Effects- Cavitation in Hydraulic machines.

Reciprocating Pumps: Main parts- Classification- Velocity and acceleration variation in suction and delivery pipes due to piston acceleration- Effect of variation of velocity on friction in suction and delivery pipes- Effect of acceleration in suction and delivery pipes on indicator diagram- Effect of friction- Maximum speed of reciprocating pump- Air vessels.

Hydraulic Press- Hydraulic accumulator- Differential hydraulic accumulator- Hydraulic intensifier- Hydraulic ram- Hydraulic lift- Hydraulic crane- Fluid coupling- Hydraulic torque converter. Air lift pump Servo systems- Open and closed loop systems- Hydraulic and Pneumatic systems.

Text Books:

1. Fluid Mechanics and Hydraulic Machinery, by R.K.Bansal, Laxmi publications.
2. Fluid Mechanics and Hydraulic Machinery, by R.K.RAJPUT.

Reference Books:

1. Fluid Flow Machines, by N.S.Govinda Rao, Tata McGraw Hill publishing company Ltd.
2. “Hydraulics & Fluid Mechanics”, P.N. Modi and S.M. Seth, TEXT BOOKS House, Delhi.

DESIGN OF MACHINE ELEMENTS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Students will be acquainted with standards like ASTM, ASME etc., safety, reliability, importance of dimensional parameters, manufacturing aspects in mechanical design.
2. Students will understand to formulate and analyze stresses and strains in machine elements like shafts, springs etc. And structures under static and/ or dynamic load conditions

Course Outcomes:

Students will be able to

1. Identify the need for design process and enumerate different design models
2. Design the machine elements subjected to static and fatigue loading.
3. Determine the dimensions of shafts and keys subjected to different types of loadings.
4. Design and Analyze bolted, riveted and welded joints subjected to different types of loadings.
5. Classify Design and Analyze the Helical, torsion and Leaf springs for the given loading.
6. Design of various types of joints and couplings subjected to different types of loadings.

SYLLABUS

Introduction to Mechanical engineering design: traditional design methods, different design models, Problem formulation, Design considerations, engineering materials and processes and their selection, BIS designation of steels, Mechanical properties, manufacturing considerations in design.

Design against static loads: Modes of failure, Factor of safety, Axial, bending and torsional stresses, Stress concentration factors. Static failure theories.

Fluctuations and fatigue stresses, Soderberg, Goodman and modified Goodman diagrams, fatigue failure, design consideration in fatigue.

Threaded and welded joints: forms of threads, basic types of screw fastenings, ISO metric screw threads, design of eccentrically loaded bolted joints, different types of welded joints and strength of weld joints subjected to bending and fluctuating loads, welding inspection, Design of cotter and knuckle joints.

Shafts, keys and couplings: shafts design on strength basis, torsional rigidity basis, Design of hollow shafts, flexible shafts, ASME codes for shafts and Keys design, Flat, square keys, Splines, Rigid and flange couplings, Flexible couplings.

Spring Design: classification and spring materials, Design of helical compression springs and helical springs subjected to fatigue loading, Design of laminated springs, Equalized stress in spring leaves. Multi - leaf springs. Surge in springs, Nipping and shot peening.

Text Book:

1. Design of Machine Elements by V.B.Bhandari, TMH Publishing Co. Ltd., New Delhi

References Books:

1. Machine Design by Jain, Khanna Publications.
2. Machine Design by Pandya and Shaw, Charotar publications
3. Machine design, an integrated approach by R.L.Norton, 2nd edition, Pearson Education

REFRIGERATION AND AIR CONDITIONING

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students

1. Understand the principles of refrigeration and air craft refrigeration systems.
2. Have the complete knowledge of the Vapour Compression Refrigeration System and its performance.
3. Understand the working of Vapour Absorption refrigeration system.
4. Understand the working of Steam Jet Refrigeration system.
5. Understand the principles and processes of psychrometry.
6. Develop the knowledge of students in selecting the right equipment for a particular application of Air- conditioning.

Course Outcomes:

Students will be able to

1. Explain necessary, applications of refrigeration, ideal cycle and also air craft refrigeration system.
2. Demonstrate the working principle of vapor compression refrigeration system and its performance parameters.
3. Describe working principle of vapor absorption refrigeration system.
4. Describe working principle of steam jet refrigeration system.
5. Demonstrate different psychrometric properties in air conditioning and requirements of comfort air conditioning.
6. Explain cooling and heating loads in air conditioning and describe the various components of air conditioning system.

SYLLABUS

Introduction to Refrigeration: Necessity of refrigeration and air conditioning, applications, unit of refrigeration. Carnot cycle, Bell Coleman cycle, Open and Dense air systems, Actual air refrigeration system –numerical problems. Refrigeration needs of air craft's, methods of air refrigeration systems.

Vapour Compression Refrigeration System - Basic Cycles- Working principle and Essential components of the plant – COP – Representation of cycle on T-S and P-h charts- Effect of sub cooling and super heating - cycle analysis. Actual cycle, Influence of various parameters on system performance -numerical Problems. Refrigerants- Classifications- Desirable properties.

Vapour Absorption Refrigeration (VAR) System – Description and Working of NH₃ – Water System and Li Br – Water (Two Shell & Four Shell) System – Calculation of Max COP, Principle of Operation of Three Fluid Absorption System.

Steam Jet Refrigeration System: Working Principle and Basic Components – Nonconventional refrigeration methods: Principle and operation (i) Thermoelectric refrigerator (ii) Vortex tube or Hilsch tube.

Psychrometry: Psychrometric properties and relations- Psychrometric chart- Psychrometric processes- Human comfort and comfort chart- Effective temperature and factors governing effective temperature.

Air Conditioning: Summer, Winter and year round air conditioning- Different types of Air conditioning load - By pass factor, RSHP, GSHF- Fresh air quantity- Cooling coils and Dehumidity- Air washers.

Text Books:

1. A Course in Refrigeration and Air Conditioning, S. C Arora & Domkundwar, Dhanpatrai Publications
2. Refrigeration and Air conditioning, by R. S. Khurmi and J. K. Gupta, S Chand publications.

References:

1. Refrigeration and Air conditioning, by Jordan R.C. and Priester G.B.
2. Principles of Refrigeration, by Dossat.
3. Refrigeration and Air-conditioning, by W.P.Stoecky.
4. Refrigeration and Air Conditioning, CP Arora, TMH, 15th edition 2013.

DYNAMICS OF MACHINES

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

Students learn

1. To understand the balancing procedure for rotating and reciprocating masses, rotors and engines
2. To understand the concept theory of toothed gears, analysis of gear trains and gyroscope effect on various vehicles
3. To understand the fundamentals of free and forced vibrations
4. To develop competency in drawing the cam profile understand the follower motions

Course Outcomes

Students will be able to

1. Perform balancing of rotating and reciprocating masses
2. Understand the importance of toothed gears, gear trains and their practice application
3. Calculate gyroscopic couple on various vehicles
4. Analyze free and forced vibrations of machines, engines and structure
5. Design cam and follower for specific motion profile

SYLLABUS

Gyroscopic Couple and Precessional Motion: Precessional and angular motion- gyroscopic couple- effect of gyroscopic couple on an aero plane and on a naval ship, stability of a four wheel vehicle moving in a curved path, stability of a two-wheel vehicle taking a turn.

Cams: Classification of followers and cams- Definitions- Motions of the follower- Uniform velocity-Simple harmonic motion- Uniform acceleration and retardation- Displacement-Velocity and acceleration diagrams. Construction of cam profiles- Cam with knife edged follower and roller follower.

Toothed gearing: Classification of toothed wheels, technical terms, Law of gearing, Length of contact, arc of contact, interference in involute gears, minimum number of teeth required on pinion to avoid interference- Helical gears, Spiral gears- Efficiency of spiral gears.

Gear Trains: Types of gear trains- Simple, compound, reverted and epicyclic gear trains- Velocity ratio of epicyclic gear train- Tabular method- Algebraic method- Torques in epicyclic gear trains.

Balancing of Rotating and Reciprocating Masses: Balancing of a single rotating mass in the same plane and by two masses in different planes, balancing of several masses revolving in the same plane- Balancing of several masses revolving in different planes Primary and secondary unbalanced forces of reciprocating masses, Partial balancing of unbalanced primary forces in a reciprocating engine, Partial balancing of locomotives- Effect of partial balancing of reciprocating parts of two cylinder locomotives- Variation of tractive force, Swaying couple and hammer blow- Balancing of primary and secondary forces in multi cylinder in-line engines- Direct and reverse cranks- Balancing of V- Engines.

Vibrations:

Definitions- Types of vibrations- Natural frequencies of free longitudinal vibrations of systems having single degree of freedom- Equilibrium method- Energy method and Rayleigh's method. Frequency of damped vibration and forced vibration with damping- Magnification factor

Transverse and Torsional Vibrations: Natural frequency of free transverse vibrations due to point load and uniformly distributed load acting over a simply supported shaft- Transverse vibrations for a shaft subjected to number of point loads- Energy method- Dunkerley's method, Critical speed of a shaft. Natural frequency of free torsional vibrations- Free torsional vibrations of single rotor system, two rotor system.

Text Book:

1. Theory of Machines by R.S.Khurmi & J.K.Gupta.

Reference books:

1. Theory of Machines by Thomas Bevan.
2. Theory of Machines by S.S. Rattan.

CONTROL SYSTEMS (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To introduce the basic principles of control systems to develop mathematical models for physical systems.
2. To familiarize the students on the basic concepts of feedback characteristics of control systems for standard test signals.
3. To familiarize the students on finding stability of control systems using time and frequency domain techniques.

Course Outcomes:

Students will be able to

1. Develop mathematical models for physical systems using the knowledge of fundamental principles of mathematics and control systems.
2. Apply the knowledge of various controlling techniques to develop suitable controller to meet specific requirements.
3. Describe and determine the various time and frequency domain specifications.
4. Select appropriate stability techniques to determine performance characteristics of physical systems.

SYLLABUS

Introduction: Control systems, Classification of Control systems, Feedback and its effects. Transfer Function, Block Diagram and Signal Flow Graph, Impulse response and Transfer functions of: linear systems, Block diagrams.

Mathematical Modeling of Physical Systems: Modeling of mechanical system elements, Equations of mechanical systems, Electrical analogous of mechanical systems. Matrix representation of state equations, State transition matrix, State transition equation, Characteristic equation, eigen values and eigen vectors.

Time Response Analysis: Time response, Typical test signals for the time response of control systems, Order of a system, Time - domain performance of control systems. Type number of control systems, Steady state error: Static error constants.

Stability of control systems: Stability, Characteristic equation, Methods of determining stability of linear control systems, Routh- Hurwitz criterion,

Frequency Response Analysis: Frequency response, Frequency-domain Analysis of Control Systems: Gain margin, Phase margin, Polar/Nyquist plot. Nyquist stability criterion, Application of the Nyquist criterion. Stability of multi loop systems.

Text Books:

1. Control Systems by A. Nagoor Kani, RBA Publications.
2. Advanced Control Theory by A. Nagoor Kani, RBA Publications.

Reference Books:

1. Control Systems Engineering by Nagrath/Gopal , New Age International.
2. Automatic Control Systems by Benjamin C Kuo

OPTIMIZATION TECHNIQUES (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

1. To understand the theory of optimization methods and algorithms developed for solving various types of optimization problems.
2. To apply the mathematical results and numerical techniques of optimization theory to concrete Engineering problems.
3. To develop and promote research interest in applying optimization techniques and reliability concepts in problems of Engineering and Technology.

Course Outcomes

Students will be able to

1. Have a basic understanding of conventional, unconventional optimization algorithms and concepts of reliability.
2. Formulate engineering design problems as mathematical optimization problems and solve them by using suitable optimization technique(s).
3. Use mathematical software for the solution of engineering problems.
4. Several homework assignments delving on core concepts and reinforcing analytical skills learned in class.

SYLLABUS

Classical Optimization Techniques: Single variable optimization - local and global minimum, multi variable optimization without constraints, multi variable optimization with constraints – method of Lagrange multipliers, Kuhn-Tucker conditions, merits and demerits of classical optimization techniques.

Numerical Methods For Single Variable Optimization Problems: Unimodal function, Dichotomous search, Fibonacci method, Golden section method, Quadratic interpolation method, Newton method, Secant method.

Numerical Methods For Multi Variable Optimization Problems: Nelder Mead's Simplex search method, Pattern Search methods – Hooke and Jeeve's method, Powell's method, Gradient of a function, Steepest descent method, Sequential linear programming for constrained problems, Exterior and interior penalty function methods for handling constraints.

Genetic Algorithms: Differences and similarities between conventional and evolutionary algorithms, working principle, reproduction, crossover, mutation, termination criteria, different reproduction and crossover operators, GA for constrained optimization, draw backs of GA.

Multi-Objective Optimization: Utility function method, Inverted utility function method, Global criterion method, Bounded objective function method, Lexicographic method, Goal Programming method, Example of Minimization of weight and deflection of a cantilever beam subject to stress constraints.

Text Book:

1. Engineering Optimization – S.S.Rao, New Age Publishers.

Reference Books:

1. Optimization for Engineering Design – Kalyanmoy Deb, PHI Publishers.
2. Multi objective optimization by Kalyanmoy Deb, PHI Publishers.

AUTOMOBILE ENGINEERING (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

This course deals with the basic components of automotive vehicles, which includes the elements of power transmission system and vehicle control systems as follows:

1. First chapter deals with the technological advancements of modern automotive vehicles/engines and the techniques of exhaust emission reduction.
2. Chapter 2 & 3 describes the constructional features of power transmission components and their working principles.
3. Chapter 4 & 5 describe the electrical circuits, and ignition systems including battery.
4. Chapter 6 & 7 describes the constructional features of differential and alignment of axles (live & dead), including wheels.
5. Chapter 8 provides the information on various aspects of vehicle maintenance. Also the driving aspects and Motor Vehicle Act of the Indian Government.

Course Outcomes:

After completing this course, students can learn and understand the following:

1. Students are familiar with the basic knowledge of automotive vehicles, and various modes of vehicle driving, including the safety aspects.
2. Students can understand the affects of automotive exhaust emissions on the environment and the health of human beings. And also the techniques of exhaust emission control/reduction techniques used in modern vehicles.
3. Students are able to understand the principle of electronic fuels injection and spark ignition.
4. Students are familiar with the various design aspects of chassis and power transmission components, including the design of wheels and tires.
5. Students are exposed to different maintenance procedures of automotive vehicles and the safety aspects while driving, including traffic regulations.

SYLLABUS

Introduction: Definition of automobile, Automobile Layout, Chassis and Transmission: Introduction to Drive Train: Clutch, Gearbox, Hook's Joint, Propeller /Drive Shaft, Slip Joint, Final Drive and Differential, Front and Rear Axles, Wheels and Tires, Control systems: Introduction to Steering, and Brakes. Electrical system: Introduction to Starting System, Ignition, dynamo/alternator, cut-out and wiring. Automobile Body: Parts and Stream lining, Automobile types: Front, Rear and Four wheel drive and Automotive materials

Engine (Power Plant): Multi cylinder engine parts, Classification: ‘In-line’ and ‘V’ type, Multi-Valve Engines, Super Charging/Turbo charging, Air filters, Fuel Systems: Petrol Engines: Carbureted and MPFI, Ignition Systems: Conventional and Electronic, Diesel Engines: Conventional, CRDI, and Dual Fuel engines., Performance, Combustion and Exhaust Emissions, Air pollution and their control: EGR and Catalytic Converters, EURO/Bharat Stage Norms: I, II, III, IV and V., Manifolds and Mufflers, Engine Cooling and Lubrication.

Clutch: Necessity, Clutch Assembly: Construction and Working Principle, Types: Single and Multiple Plates, Free-Play, Fluid coupling/Torque converter, Clutch Troubles and Remedies.

Gearbox: Necessity of Transmission, Construction and Working Principle, Selector Mechanism, Types: Sliding mesh, Constant mesh, Synchromesh, and Epicyclical. Three, Four and Five- Speed Gearbox, Overdrive, Automatic Gearbox, Gearbox Troubles and remedies.

Drive shaft and Final Drive: Drive Shaft: Constructional Features: Universal/Hooks Joints, Slip Joint, and Working Principle., Types of Propeller shafts, Final drive and Differential: Necessity, Constructional Features and Working Principle., Front/Rear Axles: Constructional Features and Types of Rear Axle Floating, Wheels: Disc and Drum type, Tires: Tire Construction, Tube and Tubeless Tires, Radial Tires, Tire specification, Tire rotation and Tire Maintenance.

Suspension System and Vehicle Control: Coil and Leaf Springs, Shock absorbers, Wheel alignment: Kingpin angle, Caster, Camber, Toe-in, and Toe-out., Necessity of vehicle control, Steering Mechanism and its Elements: Steering gear box and its types, Steering gear ratio, Constant Velocity Joints and linkages. Power Steering, Brake system: Necessity, Parking and Power Brakes, Parts and Working Principle of Mechanical, Air and. Hydraulic Brakes: Master and Wheel cylinder, Properties of Brake Fluids, Brake Diagnostics and Service: Brake Bleeding, Anti-lock Braking System, Automobile Accessories and Tips for Safe Driving.

Electrical and Electronic Systems: Battery, Starting system, ECU/ ECM.

Trouble shooting and Maintenance: Engine and Vehicle Troubles: Diagnostic Information: Symptom descriptions and their Causes and Remedies, Periodic, Preventive and Break down Maintenance: Engine tuning, Fuel and Air filters, Lubricants, Maintenance of Battery and Electrical/Electronic System, and Tires. The Motor Vehicle Act (India).

Text Books:

1. Automotive Mechanics (10/e) - William H. Crouse and Donald L. Anglin, Tata McGraw-Hill Publishing Company Limited, ISBN: 0-07-059054-0.
2. Automobile Engineering – KK Jain/ RB Asthana, Tata McGraw-Hill Publishing Company Limited, ISBN: 0-07-044529-X.
3. Internal Combustion Engines and Air Pollution- E.F. Obert, Harper & Row International Publishers Inc., ISBN: 0-06-350561-4.

Reference Books:

1. Automotive Mechanics – S. Srinivasan, Tata McGraw-Hill Publishing company Limited, ISBN: 0-07-044941-6.
2. Internal Combustion Engines – Heywood, John, B. McGraw-Hill Publications Limited.
3. Automotive Engines- S Srinivasan, Tata McGraw-Hill Publishing Company Limited, ISBN: 0-07-040265-5

**SUPPLY CHAIN MANAGEMENT
(Elective-II)**

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Supply Chain Management involves the flows of materials and information among all of the firms that contribute Value to a product, from the source of raw materials to end customers.
2. It will integrate issues from finance (investments in productive assets), marketing (channels of distribution), logistics, and operations management to develop a broad understanding of a supply chain.
3. By taking a strategic perspective, it will focus on relatively long term decisions involving the investment in productive resources, configuration of processes, product Designs and development of partnerships with suppliers and channels of distribution.
4. Although the development of analytical tools is not the primary objective of the course, students should be Comfortable with quantitative analysis.
5. By the end of the course, you should have enhanced your ability to use Analytical tools and conceptual frameworks to make decisions in supply chain contexts as well as a better Understanding of the major strategic issues and trade-offs that arise in supply chain management

Course Outcomes:

Students will be able to

1. Provide students with the requisite knowledge and skills to design and manage Supply chain. Analyze and improve the supply chain performance.
2. Align appropriate supply chain strategies with product characteristics.
3. To engage students in case studies based on real world logistics and supply chain decisions.
4. Acquaint the student with various Supply Chain Strategies; the differences between efficient and responsive supply chains and the correct strategies to use based on product type and location in the product life cycle.
5. Causes of Bullwhip Effect by playing a version of the well known "Beer Game" simulation.
6. The student will be able to explore three fundamental design concepts: component commonality, modularity vs. integral design, and universality, and a cost/benefit framework
7. Learn process improvements such as postponement, mass customization, resequencing production operations, and shifting the push-pull point; these design changes can significantly improve the performance of your supply chain.
8. Understand the importance of technology in supply chain optimization

SYLLABUS

Role of supply chain management in Economy and Organization- Introduction to SCM, Evolution, Key concepts, Decisions and Importance of SCM.

Supply chain strategy and Performance Measures- Competitive supply chain strategies, CRM strategy, Supplier relationship strategy- Performance Measures (Financial, Productivity, Quality and cycle time).

Supply chain drives- Introduction, Facilities, Inventory, Transportation and Information. Supply chain design- Network design and operation models. Sourcing and Transportation- Role of sourcing, Supplier selection and contracts, Procurement process, Role of Transportation, Design options for transportation network.

Planning and Managing Inventories-Introduction, cycle/safety/seasonal stock, Inventory for short life cycle products, Multi echelon inventory.

Information Technology in SCM- Role of IT, E-business and future trends. Supply chain innovations- Introduction, Supply chain integration, Restructuring, Agile supply chains.

Text Book:

1. Designing and Managing the Supply Chain: David Simchi-Levi, Philip Kaminsky, Edith Simchi – Levi, Ravi Shankar, Mc Graw Hill Education, 2008

Reference Books:

1. Supply chain management text and cases: Janat Shah, Pearson Education, 2009.
2. Supply chain management strategy, planning and operation, Sunil Chopra, Peter Meindl, PHI.
3. Supply chain management: Chopra, Pearson Education, 2009.
4. Business logistics/ Supply chain management, 5/e: Ballou, Pearson Education.

NANO TECHNOLOGY (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course objectives

The main objectives of this course are:

1. to provide the fundamental principles of nanotechnology
2. to understand the properties, production and characterization of nanomaterials
3. to understand the applications of nanotechnology in various science, engineering and technology fields.

Course Outcomes

On successful completion of this course the student shall be able to:

1. Identify the essential concepts used in nanotechnology.
2. Identify the materials, properties, synthetic and fabrication techniques, characterization techniques and applications in various fields.

SYLLABUS

Introduction: History of nano science. Definition of nano size, nano materials, nano technology. Classification of nano materials. Crystal symmetries, crystal directions, crystal planes. Band structure.

Properties Of Materials: Mechanical properties, electrical properties, dielectric properties, thermal properties, magnetic properties, opto electronic properties. Effect of size reduction on properties, electronic structure of nano materials.

Synthesis and Fabrication: Synthesis techniques for preparation of nano particle – Bottom Up Approach – sol gel synthesis, hydro thermal growth, thin film growth, PVD and CVD; Top Down Approach – Ball milling, micro fabrication, lithography.

Characterization Techniques: X-Ray diffraction and Scherrer method, scanning electron microscopy, transmission electron microscopy, scanning probe microscopy, atomic force microscopy, piezoresponse microscopy, X-ray photoelectron spectroscopy, UV-Visible and Infra red spectroscopy, Raman spectroscopy, Thermogravimetric analysis, Differential thermal analysis, Differential Scanning Calorimetry.

Carbon Nano Technology: Characterization of carbon allotropes, synthesis of diamond – nucleation of diamond, growth and morphology. Applications of nano crystalline diamond films, graphene, carbon nano tubes.

Applications of Nano Technology: Applications in material engineering, biology and medicine, surface science, energy and environment. Applications of nano structured thin films, applications of quantum dots.

Text Books

1. Nano science and nano technology by M.S Ramachandra Rao, Shubra Singh, Wiley publishers.
2. Introduction to Nano Technology by Charles P. Poole, Jr., Frank J.Owens, Wiley publishers.
3. Nanotechnology by Jermy J Ramsden, Elsevier publishers.

Reference Books

1. Nano Materials- A.K.Bandyopadhyay/ New Age Introduction.
2. Nano Essentials- T.Pradeep/TMH.
3. Nanotechnology the Science of Small by M.A Shah, K.A Shah, Wiley Publishers.
4. Principles of Nanotechnology by Phani Kumar, Scitech.

COMPUTATIONAL FLUID DYNAMICS (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. The course aims at providing required numerical techniques for solving various engineering problems involving fluid flow.
2. The course aims at providing required software techniques for solving various engineering problems involving fluid flow

Course Outcomes:

1. After undergoing the course the student shall be able to apply various numerical tools like finite volume, finite difference etc for solving the different fluid flow problems.

SYLLABUS

Introduction: Finite difference method, finite volume method, finite element method, Philosophy of Computational Fluid Dynamics: Computational fluid dynamics: Why? – Computational fluid dynamics as a research tool– Computational fluid dynamics as a design tool – The impact of Computational fluid dynamics—some other examples; Automobile and engine applications; Industrial manufacturing applications; Civil engineering applications; Environmental engineering applications; Naval architecture applications(submarine example) - Computational fluid dynamics: What is it?. Governing equations and boundary conditions. Derivation of finite difference equations.

Solution methods: Solution methods of elliptical equations – finite difference formulations, interactive solution methods, direct method with Gaussian elimination. Parabolic equations - explicit schemes and Von Neumann stability analysis, implicit schemes, alternating direction implicit schemes, approximate factorization, fractional step methods, direct method with tridiagonal matrix algorithm.

Hyperbolic equations: explicit schemes and Von Neumann stability analysis, implicit schemes, multi step methods, nonlinear problems, second order one-dimensional wave equations.

Burgers equations: Explicit and implicit schemes, Runge-Kutta method.

Formulations of incompressible viscous flows: Formulations of incompressible viscous flows by finite difference methods, pressure correction methods, vortex methods.

Treatment of compressible flows: potential equation, Euler equations, Navier-stokes system of equations, flow field-dependent variation methods, boundary conditions, example problems.

Finite volume method: Finite volume method via finite difference method, formulations for two and three-dimensional problems.

Standard variational methods: Linear fluid flow problems, steady state problems, Transient problems

Text Books:

1. Computational fluid dynamics- T. J.Chung, Cambridge University press, 2002.
2. Introduction to Computational Fluid Dynamics – Niyogi.P, Chakrabartty S.K. and Laha.M.K., Pearson Education, India, 2005.

Reference Books:

1. Computational Fluid Dynamics – The basics with Applications, John D. Anderson, Jr (TMH) & Mc Graw Hill, 1995.
2. An Introduction to Computational Fluid Dynamics – FVM Method – H.K. Versteeg, W.Malalasekhara, Longman Scientific and Technical, 1995 & PHI.
3. Text book of fluid dynamics, Frank Chorlton, CBS Publishers & distributors, 1985.

INDUSTRIAL ENGINEERING LAB

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

Course Outcomes:

1. To make student acquainted with the control charts and measure the quality of the product.
2. To make aware of the different types of process charts used for improving the method of doing the work.
3. To help students to learn the different methods of finding the standard time for a job.
4. To make the students acquainted with the probability distributions.
5. To make the students learn the impact of work on the human physiology and the physiological constraints of the body.

Course Objectives:

1. Students will be able to find the quality of the product using different charts.
2. Can improve the method of doing work by applying principle of motion economy and method study charts.
3. Can find the standard time required for completing a job by different methods.
4. Understands the basic probability distributions.
5. Understands the impact of work on the human body and also the physiological constraints of the body.

LIST OF EXPERIMENTS

1. To measure the skill and dexterity in the movement of Wrist and Fingers using pin board.
2. To measure the Heart beat using Stethoscope.
3. To show that the sample means from a normal universe follow a normal distribution.
4. To draw the control chart for fraction defective for a given lot of marble balls.
5. To determine the cycle time using PMTS.
6. To draw two handed process charts for
 - i. Bolt, Washer and nut assembly
 - ii. Assembly of electric tester.
7. To study the changes in heart rate for different subjects using Tread mill.
8. To draw Multiple Activity chart using an electric toaster.
9. To determine the percentage utilization using work sampling.
10. To study the process capability of a given process.
11. To measure the Heart rate during working and recovery periods of the subjects under different loads, using Bicycle ergometer.
12. To draw flow process charts on activities in Workshop/ Laboratory/Office.
13. To determine the time required to perform motion sequence using work factor system.

14. To draw SIMO charts for
 - i. Ball point pen assembly
 - ii. Electric plug assembly.
15. To conduct time study of the bulb holder assembly operation of the existing method.
16. To collect the anthropometrics data using 'Anthropolometer'.

Reference Books:

1. Industrial Engineering Management, by Dr. O. P .Khanna.
2. Industrial Engineering and Production Management by Telsay, S. Chand & Co.

FLUID MECHANICS & MACHINES LAB

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

Course Objectives:

1. To understand the working principles and fundamentals of Fluid mechanics, turbines and pumps.
2. To prepare the student for industrial applications.
3. The students are taught with testing and performance of hydraulic machines.

Course Outcomes:

1. The student gets complete knowledge on fluid mechanics, hydraulic turbines and pumps.
2. The student learns the complete calculation procedures for designing hydraulic turbines, and pumps.
3. The student is prepared to work in industry immediately after this course.

LIST OF EXPERIMENTS

1. Calibration of flow meters: (a) Venturi meter; (b) Orifice meter; (c) Nozzle meter.
2. Determination of coefficient of discharge for: (a) small orifice; (b) cylindrical mouth piece.
3. Finding coefficient of discharge for: (a) rectangular notch; (b) triangular notch; (c) trapezoidal notch.
4. To draw the performance characteristics of Centrifugal pump.
5. To find the specific speed of: (a) Pelton turbine; (b) Francis turbine.
6. To draw the characteristic curves for reciprocating pump.
7. To draw the pressure distribution and finding coefficient of drag for a bluff body and an Aero foil.
8. To draw the characteristic curves for the hydraulic ram.

Reference Books:

1. Fluid Mechanics and Hydraulic Machinery, by R.K.Bansal, Laxmi publications.
2. Fluid Mechanics and Hydraulic Machinery, by R.K.RAJPUT.
3. Fluid Flow Machines, by N.S.Govinda Rao, Tata McGraw Hill publishing company Ltd.
4. "Hydraulics & Fluid Mechanics", P.N. Modi and S.M. Seth, TEXT BOOKS House, Delhi.

VERBAL & QUANTITATIVE APTITUDE – II
(Common to All Branches)

Theory	: 5 Periods(VA-3+QA-2)	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Part-A: Verbal Aptitude and Soft Skills-II

Course objectives:

1. To expose the students to bettering sentence expressions and also forming equivalents.
2. To instill reading and analyzing techniques for better comprehension of written discourses.
3. To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.
4. To inculcate conversational skills, nuances required when interacting in different situations.
5. To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.

Course Outcomes:

The students will be able to

1. Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.
2. Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences
3. Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).
4. Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.
5. Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.

SYLLABUS

UNIT -I (VA)

Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing .

UNIT- II (VA)

Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author's intention of writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.

UNIT- III (VA)

Punctuation, discourse markers, general Essay writing, writing Issues and Arguments(with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a 'Statement of purpose', 'Letters of Recommendation', business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.

UNIT-IV (VA)

Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.

UNIT-V (SS)

Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.

Reading/ Listening material:

1. Guide to IELTS, Cambridge University Press
2. Barron's GRE guide.
3. Newspapers like 'The Hindu', 'Times of India', 'Economic Times'.
4. Magazines like Frontline, Outlook and Business India.
5. News channels NDTV, National News, CNN

Text Books:

1. Objective English and Verbal Reasoning by R S Agarwal.
2. Communication Skills by Sanjay Kumar and Pushpa Latha, Second Edition, OUP.
3. Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. Oxford Guide to Effective Writing and Speaking by John Seely.
2. Collins Cobuild English Grammar by Collins
3. The Art of Public Speaking by Dale Carnegie
4. The Leader in You by Dale Carnegie
5. Emotional Intelligence by Daniel Golman
6. Stay Hungry Stay Foolish by Rashmi Bansal

Part-B: Quantitative Aptitude-II

Course objectives:

The objective of introducing quantitative aptitude-II is:

1. To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages.
2. To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.
3. To familiarize the students with finding out accurate date and time related problems.
4. To enable the students solve the puzzles using logical thinking.
5. To expose the students to various problems based on geometry and menstruation.

Course Outcomes:

1. The students will be able to perform well in calculating different types of data interpretation problems.
2. The students will perform efficaciously on analytical and logical problems using various methods.
3. Students will find the angle measurements of clock problems with the knowledge of calendars and clock.
4. The students will skillfully solve the puzzle problems like arrangement of different positions.
5. The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.

SYLLABUS

UNIT I: Averages, mixtures and allegations, Data interpretation

Understanding of AM,GM, HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.

UNIT II :Puzzle test, blood Relations, permutations, Combinations and probability

Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.

UNIT III :Periods,Clocks, Calendars, Cubes and cuboids

Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.

UNIT IV: Puzzles

Selective puzzles from previous year placement papers, sitting arrangement, problems-circular arrangement, linear arrangement, different puzzles.

UNIT V: Geometry and Menstruation

Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals-Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of menstruation-Introduction of cylinder, cone, sphere, hemi sphere.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi
4. More puzzles by shakunataladevi
5. Puzzles by George summers.

Reference Books:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

MINI PROJECT**Lab : 3 Periods****Sessionals : 50****Credits : 2**

Course Objectives:

1. Aims to provide students with the opportunity to synthesis knowledge from various areas of learning, and critically and creatively apply it to real complex engineering problems.
2. It enhances students' knowledge and enables them to acquire skills like collaboration, communication and independent learning, prepares them for lif long learning and the challenges ahead.
3. To provide students an opportunity to exercise their creative and innovative qualities in a project group environment.

Course Outcomes:

1. Demonstrate a sound technical knowledge of their selected project topic.
2. Undertake problem identification, formulation and solution.
3. Design engineering solutions to complex problems utilising a systems approach.
4. Conduct an engineering project.
5. Communicate with engineers and the community at large in written an oral forms.
6. Demonstrate the knowledge, skills and attitudes of a professional engineer.

BASIC CODING
(Common to CE & ME)
(MOOCS-III)

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course Objectives:

1. To develop programming skills among the students.
2. To familiarize the student with Control Structures, Loop Structures.
3. To familiarize the student with Basic searching and sorting Methods.
4. To familiarize the student with Functions, Recursions and Storage Classes.
5. To familiarize the student with Structures and Unions.
6. To familiarize the student with Operating System concepts.
7. To familiarize the student with Networking concepts.

Course Outcomes:**At the end of the course students will be able to**

1. Know about Control Structures, Loop Structures and branching in programming.
2. Know about various searching and sorting methods.
3. Know about Functions, Recursions and Storage Classes.
4. Know about Structures and Unions.
5. Know different Operating System concepts.
6. Differentiate OSI Model Vs. TCP/IP suite.

SYLLABUS

UNIT I Review of Programming constructs

Programming Environment, Expressions formation, Expression evaluation, Input and Output patterns, Control Structures, Sequential branching, Unconditional branching, Loop Structures, Coding for Pattern Display.

UNIT II Introduction to Linear Data, strings and pointers

Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding, Working on character data, Compiler defined methods, Substitution coding for defined methods, Row Major representation, Column Major representation, Basic searching and sorting Methods.

UNIT III Functions, Recursions and Storage Classes

Functions – Introduction to modular programming – Function Communication - Pass by value, Pass by reference – Function pointers – Recursions – Type casting – Storage classes

Practice: programs on passing an array and catching by a pointer, function returning data, comparison between recursive and Iterative solutions.

Data referencing mechanisms: Pointing to diff. data types, Referencing to Linear data, Runtime-memory allocation, Named locations Vs pointed locations, Referencing a 2D-Matrix.

UNIT IV User-defined data types, Pre-processor Directives and standard storage

Need for user-defined data type – structure definition – Structure declaration – Array within a Structure – Array of Structures – Nested Structures - Unions – Declaration of Union data type, StructVs Union - Enum – Pre-processor directives , Standard storage methods, Operations on file, File handling methods, Orientation to Object oriented programming

Practice: Structure padding, user-defined data storage and retrieval programs

UNIT V Operating system principles and Database concepts

Introduction to Operating system principles, Process scheduling algorithms, Deadlock detection and avoidance, Memory management, Networking: Introduction to Networking, OSI Model Vs. TCP/IP suite, Datalink layer, Internet layer, DVR Vs. LSR, Transport Layer, Application Layer

References:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific .
4. Object Oriented Programming in C++: N. Barkakati, PHI.
5. Object Oriented Programming through C++ by Robat Laphore.
6. <https://www.geeksforgeeks.org/>.
7. <https://www.tutorialspoint.com/>

**DESIGN FOR QUALITY, MANUFACTURING AND ASSEMBLY
(MOOCS-III)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

ABOUT THE COURSE:

In the context of product design, it is very important to appreciate the limitations of a design from manufacturing and assembly perspective and to produce high quality products at low cost. This course will introduce methods that can provide guidance to design teams in simplifying product structure to reduce manufacturing and assembly costs, quantify improvements and how robust design concepts can be used for ensuring quality. This course aims at introducing the need to account for variability, mathematically represent it, formulate it and control it. Concepts such as quality, robustness, six sigma and orthogonal array will be discussed.

COURSE LAYOUT:

Week 1 : Introduction, course expected outcomes, discussion on quality

Week 2 : Measuring quality: Quality loss function. Discussion on robustness, six sigma concepts

Week 3 : Quantifying robustness: Signal to Noise Ratio, problem formulation using SNR. Design of experiment discussions

Week 4 : Orthogonal array, linear graphs, triangular tables, finding optimum combinations. Case studies

Week 5 : Design for Manufacturing: over the wall design, most influential phase in design, best practices in injection molding and sheet metal working

Week 6 : Design for additive manufacturing, single point and multipoint tools

Week 7 : Design for Assembly: Boothroyd Dewhurst method, theoretical minimum number of parts, Xerox producibility index (XPI) method

Week 8 : Do's and don't's in manual assembly, assembly time estimation, design for robotic assembly considerations

Week 9 : Design for sustainability

SUGGESTED READING MATERIALS:

1. Madhav Phadke, "Quality Engineering Using Robust Design", Prentice Hall, NJ, USA.
2. Phillip J. Ross, Taguchi Techniques for Quality Engineering, McGraw-Hill, 1989.
3. Geoffrey Boothroyd, Peter Dewhurst, "Product Design for Manufacturing and Assembly", Winston Knight, Marcel Dekker, 2001.
4. Daniel E. Whitney "Mechanical Assemblies" Oxford University Press, 2004.

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_me53/preview

INTRODUCTION TO RESEARCH (MOOCS-III)

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

ABOUT THE COURSE

Large numbers of students are actively considering and taking up research and associated higher studies. This course aims to introduce students to the important aspects of research. The intent of the course is to make students aware of the details associated with formal research and to help students overcome common misconceptions that may be present in their minds. By going through this course, students are likely to be able to take up research activities in a more systematic and formal manner right from the beginning.

COURSE LAYOUT

Week 1: A group discussion on what is research; Overview of research;

Week 2: Literature survey , Experimental skills;

Week 3: Data analysis, Modelling skills;

Week 4: Technical writing; Technical Presentations; Creativity in Research

Week 5: Creativity in Research; Ethics in Research

Week 6: Design of Experiments

Week 7: Intellectual Property

Week 8: Department specific research discussions

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ge12/preview

NATURE AND PROPERTIES OF MATERIALS (MOOCS-III)

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

ABOUT THE COURSE

This course introduces to the basics of metals and metallic alloys, polymers, composites and smart materials which have extensively broadened the scope of engineering design in the fields of Civil, Mechanical, Aerospace and other structural applications.

After learning this course, students will be well-versed with the underlying principle governing the material properties and should be able to select proper material for their application.

COURSE LAYOUT

Module 1: Introduction to Engineering materials & Mechanical properties

Module 2: Atomic bonding and crystal structure

Module 3: Metals and Ceramics

Module 4: Polymers

Module 5: Composite Materials

Module 6: Smart Materials

Module 7: Materials selection in Engineering design

Module 8: Non-mechanical properties and Laboratory demonstration

SUGGESTED READING

1. Material Science and Engineering: An Introduction by W.D. Callister, Wiley publication.
2. Engineering materials: Michael F. Ashby and David Jones, Elsevier publication
3. Materials selection in Mechanical design by Michael F. Ashby, Elsevier publication
4. Mechanics of Composite by Materials Robert M. Jones, Taylor & Francis publication.
5. Smart Structures Theory by Inderjit Chopra & Jayant Sirohi, Cambridge press

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_mm10/preview



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

IV/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards)

Under Choice Based Credit System

MECHANICAL ENGINEERING

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ME 4101	Computer Aided Design	4	3	1	-	4	30	70	100
B16 ME 4102	Machine Design	4	3	1	-	4	30	70	100
B16 ME 4103	Heat and Mass Transfer	4	3	1	-	4	30	70	100
# ELE-III	ELECTIVE-III	4	3	1	-	4	30	70	100
B16 ME 4110	Heat Transfer Lab	2	-	-	3	3	50	50	100
B16 ME 4111	Project Phase-I	2	--	--	3	3	50	--	50
Total		20	12	4	6	22	220	330	550

# ELE-III	B16 ME 4104	Mechanical Vibrations
	B16 ME 4105	Project Management
	B16 ME 4106	Non-Destructive Testing
	B16 ME 4107	Power Plant Engineering
	B16 ME 4108	Mechatronics
	B16 ME 4109	Design for Manufacturing

COMPUTER AIDED DESIGN

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

The overall objective of the course is to teach the basic principles of CAD which includes computer graphics, application of computers in FEM and development algorithms of different mechanical elements. Specific objectives include:

1. This course integrates theoretical approaches and advanced tools for geometric modelling in mechanical engineering applications.
2. This course will help students actively involve to build on practical techniques in exploring issues of current interest in engineering computer aided design and analysis.
3. Student will be using advanced modelling approaches and tools for real - world mechanical design problems that will enable him to execute comprehensive and professional engineering projects.
4. Through topics in this course addressing the impact of CAD in engineering design and analysis, student will develop the ability to establish the CAD techniques appropriate for mechanical engineering applications.
5. The interdisciplinary nature of computer graphics, geometric modelling and engineering design is emphasized in the wide variety of examples and applications presented with both broad work and addressed through the hands - on nature of the course work.
6. The course material is structured to meet the needs of Automotive, Mechanical and manufacturing engineering students focussing on CAD and its applications to problem solving in engineering design and analysis.

Course Outcomes:

Upon completion of this course, students should be able:

1. Analyze and use engineering computer graphics and geometric modelling techniques for mechanical engineering applications.
2. Able to understand and apply theories, methods and procedures for complex-shapes part design.
3. Apply advanced modelling and computational tools for complex part and shape design and analysis.
4. Select and use various engineering design procedures for mechanical design problems involving complex shapes.
5. Execute professional engineering CAD projects for mechanical engineering applications in the current industrial practice.

SYLLABUS

Fundamentals of CAD - Introduction - The design process - Application of computers for design - Operating systems - Hardware in CAD: The design work station - I/O Devices - CAD system configuration - Creating database for manufacturing - Benefits of CAD.

Interactive Computer Graphics - Graphic display devices- Graphics system- Graphics standards - Graphical user interface- Transformation systems- windowing - clipping - 2D and 3D transformations - Linear transformation- Display files for 3D data - Geometric Modeling - Modeling Techniques - Wire frame Modeling - Surface Modeling - 3 D Solid Modeling.

Introduction to Finite Element Analysis - CAD techniques to finite element data preparation- Automatic mesh generation- presentation of results - 3-dimensional shape description and mesh generation- CAD applications of FEM.

CAD applications and exposure to CAD packages: Simple examples of computer aided drafting, design and analysis - Introduction to simple machine elements - Analysis of cross sectional area, centroid & moment of inertia- Kinematics of crank- slider mechanism and other simple design applications. Introduction to CAD packages like ANSYS, NASTRAN, NISA-II.

Introduction to Artificial Intelligence - Applications of AI in design and CAD.

Text Books:

1. CAD/CAM- Computer Aided Design & Manufacturing, by M.D.Groover & E.W.Zimmer.
2. CAD/CAM/CIM by Radhakrishna, New age international.

Reference Books:

1. Computer Aided Design in Mechanical Engineering, by V.Rama Murthy.
2. Elements of Computer Aided Design & Manufacturing, by Y.C.Pao.
3. Computer Aided Kinetics for Machine Design, by D.L.Ryan.
4. Computer Aided Design and Manufacturing, by C.B.Besant & C.W.K.Lui.
5. Computer-Aided Analysis & Design by S. Ghosal, Prentice Hall of India.
6. CAD CAM by IBRAHIM ZEID

MACHINE DESIGN

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

1. To acquaint with standards like ASTM, ASME etc.,
2. To Design and formulate, to analyze stresses and strains in machine elements like gears, clutches, bearings, etc. under static and dynamic load conditions.
3. To Design of Internal combustion engine parts like cylinder, pistons and connecting rod
4. To understand the stresses on miscellaneous machine elements like wire ropes, chain drives etc.
5. To understand the Selection and application of composite materials

Course Outcomes:

Up on the completion of course student will be able to

1. Classify different types of gears and apply the design concepts to evaluate the strength of gears.
2. Design various parts of IC Engines such as cylinders, pistons and connecting rods
3. Apply the design concepts to determine the various parameters of clutches.
4. Apply the design concepts to determine the torque and dimensions related to brakes
5. Design the sliding and roller contact bearings under various environmental and service Conditions
6. Classify and Analyze different types of stresses induced in wire ropes and chain drives

SYLLABUS

Gears: Classification of gears, Standard tooth systems. Spur, Helical and Bevel gears. Terminology of each. Tooth failure. Face width and beam strength. Lewis equation. Design for dynamic and wear loads. Force analysis of Bevel gears

Engine parts: I.C. engine design. Design of cylinders and heads. Design of pistons. Design of cross-head, connecting rods

Clutches: Torque capacity, single and multi-plate clutches. Design considerations. Energy considerations and Temperature rise friction materials. Centrifugal clutches.

Brakes: Energy equations. Band and block brakes. Internal expanding shoe brakes, self locking, brake design.

Bearings: Sliding contact bearings, Lubrication modes. Temperature effect on viscosity. Journal bearing design. Bearing modulus. McKee equations. Heating of bearings. Collar and thrust bearings. Roller and ball bearings. Static and dynamic load capacity. Equivalent bearing load. Load-life relationships. Load factor. Selection of bearings from manufacturers catalogue.

Wire Ropes: Construction and classification. Stresses in wire ropes, Design of Wire rope

Chain drives: Nomenclature, Design of chain drives

Text Books:

1. Design of Machine Elements by V.B. Bhandari, TMH publishing Co. Ltd., New Delhi.
2. A Textbook of Machine Design by R.S.Khurmi and J.K.Gupta, Eurasia publishing house PVT Ltd, New Delhi

Reference Books:

1. Machine Design by R.K. Jain, Khanna publications.
2. Mechanical Engineering Design by Joseph E. Shingley.

HEAT AND MASS TRANSFER

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

The overall objective of the course is to

1. Understand the fundamentals of heat transfer mechanisms in fluids and solids and their applications in various heat transfer equipment in process industries.
2. Formulate models for heat conduction processes.
3. Apply analytical and numerical methods to solve one- and two-dimensional conduction problems.
4. Combine thermodynamics and fluid mechanics principles to analyze heat convection processes.
5. Integrate radiation aspects into real-world global heat transfer problems.
6. Analyze and design complex heat transfer applications, such as heat exchangers.
7. To develop problem solving skills in energy-related areas and understand the role that heat transfer plays in everyday life.

Course Outcomes:

Upon completion of this course, students should able to:

1. Understand the basic laws of heat transfer.
2. Apply principles of heat and mass transfer to basic engineering systems.
3. Will be able to do basic calculations involving heat and mass transfer as is typical for a mechanical engineer. This includes conduction, convection and radiation heat transfer as well as heat exchanger design.
4. Calculate fluid temperatures, mass flow rates, pressure drops, heat exchange and effectiveness during parallel, counter and cross flow in simple and baffled-shell and tube type heat exchangers, condensers, evaporators, etc.

SYLLABUS

Introduction: Basic modes of heat transfer- Rate equations- Generalized heat conduction equation in Cartesian, Cylindrical and Spherical coordinate systems.

Steady state heat conduction solution for plain and composite slabs, cylinders and spheres- Critical thickness of insulation- Heat conduction through fins of uniform cross section- Fin effectiveness and efficiency.

Unsteady steady state heat conduction- Transient heat conduction- Lumped system analysis, and use of Heisler charts.

Convection: Continuity, momentum and energy equations- Dimensional analysis- Boundary layer theory concepts- Free, and Forced convection- Approximate solution of the boundary layer equations- Laminar and turbulent heat transfer correlation- Momentum equation and velocity profiles in turbulent boundary layers- Application of dimensional analysis to free and forced convection problems- Empirical correlation.

Radiation: Black body radiation- radiation field, Kirchoff's laws- shape factor- Stefan Boltzmann equation- Heat radiation through absorbing media- Radiant heat exchange, parallel and perpendicular surfaces- Radiation shields.

Heat Exchangers: Types of heat exchangers- Parallel flow- Counter flow- Cross flow heat exchangers- Overall heat transfer coefficient- LMTD and NTU methods- Fouling in heat exchangers- Heat exchangers with phase change.

Boiling: Different regimes of boiling-Nucleate, Transition and Film boiling. Condensation: Laminar film condensation- Nusselt's theory- Condensation on vertical flat plate and horizontal tubes- Dropwise condensation.

Mass Transfer: Conservation laws and constitutive equations- Isothermal equimass, Equimolar diffusion- Fick's law of diffusion- diffusion of gases, Liquids- Mass transfer coefficient.

Text Books:

1. Heat Transfer, by J.P.Holman, Int. Student edition, McGraw Hill book company.
2. Analysis of Heat transfer, by Eckert and Drake, Int.Student edition, McGraw Hill Kogakusha Ltd.

Reference Books:

1. Heat and Mass Transfer by R.K. Rajput, S. Chand & Co.
2. Heat and mass transfer by Sachdeva.
3. Heat and mass transfer by Kothandaramanna, New Age International.

MECHANICAL VIBRATIONS
(Elective-III)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To gain the knowledge of mathematical modeling of a physical system and applying the principles of Newton's Second Law and conservation of energy to derive the equations of motion.
2. To study the response of a vibrating system with periodic excitation and understand the principle of vibration isolation.
3. To develop the equations of motion for a continuous system in elongation, bending and torsion to find the natural frequencies and mode shapes.

Course Outcomes:

Students will be able to

1. Develop a mathematical model for a physical system and derive the governing differential equations.
2. Determine the natural frequencies of single and two degrees of freedom systems without and with damping.
3. Determine and analyze the response of machine members or structures in forced vibration with different excitation frequencies.
4. Apply the techniques of vibration isolation to minimize the transmission of vibrating forces.
5. Determine the natural frequencies and mode shapes of bars in elongation and torsion and beams in bending.

SYLLABUS

Single degree of Freedom systems: Undamped and damped free vibrations: forced vibrations ; coulomb damping; Response to harmonic excitation; rotating unbalance and support excitation, Vibration isolation and transmissibility, Vibrometers, velocity meters & accelerometers.

Two Degrees of Freedom Systems: Introduction, Principal modes of vibration, Natural Frequencies of a tightly stretched string, having two masses, Double pendulum, Torsional System, Systems with damping, undamped forced vibration with harmonic excitation.

Multi-degree of Freedom Systems – Exact Analysis: Introduction, free vibrations, Equation of motion, Influence Co efficient, Natural frequencies and mode shapes (Eigen values and Eigen vectors)

Numerical Methods: Rayleigh's, Stodola's, Matrix iteration, Rayleigh-Ritz Method and Holzer's methods

Application of concepts: Free vibration of strings – longitudinal oscillations of bars-transverse vibrations of beams- Torsional vibrations of shafts. Critical speeds without and with damping, secondary critical speed.

Text Books:

1. Elements of Vibration Analysis by Meirovitch.
2. Mechanical Vibrations by G.K. Groover.

Reference Books:

1. Vibrations by W.T. Thomson
2. Mechanical Vibrations – Schaum series.
3. Vibration problems in Engineering by S.P. Timoshenko.
4. Mechanical Vibrations – V.Ram Murthy.

**PROJECT MANAGEMENT
(Elective-III)**

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

Students

1. Will understand the importance of Project Management (PM) across all avenues of work.
2. Learn and apply the specific tools, models and processes of PM at the four distinct stages in the projects life cycle. The Definition, Planning, Execution and Controlling phase.
3. Learn various project planning and scheduling tools.
4. Understand the sources of risks in projects and learn how to manage the risks through various tools and methods.
5. Learn the cost accounting systems for project control; understand the key levers for measurement and key performance indicators.
6. Learn to control the changes that occur during the course of project progress and the related problems.
7. Learn how to evaluate the project and terminate the project.

Course Outcomes:

Students will

1. Understand that PM skills are critical to most careers and they can be applied at most businesses and professions.
2. Acquire thorough knowledge on various analytical tools required during different stages of project life cycle.
3. They will be able to apply various tools and techniques for planning and scheduling the projects.
4. Learn how to be proactive to the risks and be able to manage them that occur during the progressive stages of the projects.
5. Acquire thorough knowledge on cost accounting systems and key performance indicators.
6. Learn all possible practical situations that lead to different changes during the course of project execution and the problems related to controlling the changes.
7. Possess full knowledge on how to evaluate the projects, terminate the projects and finally how to close the contract.
8. Finally, students will acquire all the key skills to become effective project managers across various industries.

SYLLABUS

Overview of Project Management: Characteristics of projects, Need and evolution of project management, Definition and Objectives of project management, Project management: the person, the team, the system; The Project Life Cycle, Stages and different forms of Project Management.

Project Planning and Scheduling: Work breakdown structure, Gantt charts, CPM/PERT networks, Resource allocation: resource loading and levelling, GERT.

Project Risk Management: Risk concepts, Risk identification: Sources of risks and identification techniques, Risk Assessment, Risk response planning, Risk analysis methods.

Project Control: Introduction, Cost accounting systems for project control, Project control process, Project control emphasis, Performance analysis, Controlling changes, and Control problems.

Project Evaluation, Reporting and Termination: Project evaluation, Project review meetings, Reporting, Terminating the project, Closure of contract, Project extensions.

Text Books:

1. John M Nicholas, Project Management for Business and Technology: Principles and Practice, Prentice Hall of India, 2002.
2. Shtub, Bard and Globerson, Project Management: Engineering, Technology, and Implementation, PH Inc.
3. Lock, Gower, Project Management Handbook.
4. S. Choudhury, Project Scheduling and Monitoring in Practice.
5. P. K. Joy, Total Project Management: The Indian Context, Macmillan India Ltd.

Reference Books:

1. Larson, E.W. and Gray, C.F. (2018), Project management the managerial process, Seventh Edition, McGraw-Hill
2. Harvey Maylor (2007), Project management, Third Edition, Pearson Education.
3. N. J. Smith (Ed), Project Management, Blackwell Publishing, 2002.
4. Robert K. Wysocki, Robert Back Jr. and David B. Crane, Effective Project Management, John Wiley, 2002.
5. Jack R Meredith and Samuel J Mantel, Project Management: A Managerial Approach, John Wiley, 4th Edition, 2000.

NON-DESTRUCTIVE TESTING (Elective-III)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

This course objective is to inculcate fundamentals, methodology and skill to use various techniques for defect detection and estimation so as to ensure desired properties and long product service life

Course Outcomes:

After learning the course the students should be able to:

1. Classify various non-destructive testing.
2. Check different metals and alloys by visual inspection method.
3. Explain and perform non-destructive tests like: Liquid penetrant test, Magnetic particle test, Ultrasonic test, X-ray and Gamma ray radiography, Leak Test, Eddy current test.
4. Identify defects by using relevant NDT methods

SYLLABUS

INTRODUCTION: Introduction to destructive and non-destructive testing. Scope and limitations of NDT.

VISUAL INSPECTION - Physical Principles, Methodology, Limitations, Applications.

DYE PENETRANT TESTING/ LIQUID PENETRANT TESTING: Principle, procedure, characteristics of penetrant, types of penetrants, penetrant testing materials, different washable systems, fluorescent penetrant testing method– sensitivity, Developers, application and limitations

MAGNETIC PARTICLE TESTS: Important terminologies related to magnetic properties of material, principle, magnetizing technique, procedure, equipment, Methods of production of magnetic fields, fluorescent magnetic particle testing method, sensitivity, application and limitations

RADIOGRAPHIC TESTING: X-ray and Gamma-Ray radiography, Their principles, methods of generation, Types of films, screens and penetrometers, Interpretation of radiographs, Safety in industrial radiography Sources of ray-x-ray production, properties of d and x rays, film characteristics, exposure charts, contrasts, operational characteristics of x ray equipment, Industrial radiography techniques, inspection techniques, applications, limitations applications.

ULTRASONIC TESTING: Basic principles of sound propagation, types of sound waves, Production of ultrasonic waves, general characteristics of waves, Principle of UT, methods of UT, their advantages and limitations, Piezoelectric Material, Various types of transducers/probe, technique for normal beam inspection, flaw characterization technique, defects in welded products by UT, Thickness determination by ultrasonic method, Study of A, B and C scan presentations, principles of AET and techniques, Ultrasonic and acoustic emission techniques - pulse echo method advantage, limitations acoustic emission testing

EDDY CURRENT TESTING: Principle, instrument, techniques, sensitivity, application, limitation.

THERMAL METHODS - Principles of thermography and approaches in NDT, Sources and detectors, capabilities and limitations, measurement of diffusivity and wall thickness.

OPTICAL METHODS - Principles of Shearography and holography, applications in NDT.

Text Books:

1. Practical Non-Destructive Testing- Baldevraj, Jayakumar T., Thavasimuthu M., Narosa Publishers (2008).
2. Nondestructive Evaluation - Theory, Techniques, and Applications, by P.J. Shull, Marcell Decker Inc., NY 2002.

Reference Books:

1. JAIN, R.K., Engineering Metrology, Khanna Publishers, 1997.
2. Barry Hull and Vernon John, " Non Destructive Testing ", Mac Millan, 1988.
3. Non-destructive Evaluation - A tool in Design, Manufacturing and Service by D.E. Bray and R. K. Stanley, Revised Edition CRC Press, 1996.
4. NDT Handbooks Vol 1-17, ASNT Press, OH, USA.

**POWER PLANT ENGINEERING
(Elective-III)**

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

The course is aimed at providing knowledge of power generation through different prime movers viz steam, ICGT, Hydro, nuclear and hybrid systems along with their economics and environmental considerations.

Course Outcomes:

After undergoing this course the student can understand

1. principle of operation and performance of steam power plant along with their economics and their impact on environment.
2. principle of operation and performance of internal combustion and gas turbine power plants along with their economics and their impact on environment.
3. principle of operation and performance of hydro electric power plant along with their economics and their impact on environment.
4. principle of operation and performance of nuclear power plant along with their economics and their impact on environment.

SYLLABUS

Introduction to the sources of energy – resources and development of power in India.

STEAM POWER PLANT: Plant layout, working of different circuits, fuel and handling equipments, types of coals, coal handling, choice of handling equipment, coal storage, ash handling systems. Combustion: properties of coal – overfeed and underfeed fuel beds, traveling grate stokers, spreader stokers, retort stokers, pulverized fuel burning system and its components, combustion needs and draught system, cyclone furnace, design and construction, dust collectors, cooling towers and heat rejection. corrosion and feed water treatment.

INTERNAL COMBUSTION AND GAS TURBINE POWER PLANTS:

DIESEL POWER PLANT: Plant layout with auxiliaries – fuel supply system, air starting equipment, super charging.

GAS TURBINE PLANT: Introduction – classification - construction – layout with auxiliaries, combined cycle power plants and comparison.

HYDRO ELECTRIC POWER PLANT: Water power – hydrological cycle / flow measurement – drainage area characteristics – hydrographs – storage and pondage –classification of dams and spill ways. **HYDRO PROJECTS AND PLANT:** Classification – typical layouts – plant auxiliaries – plant operation pumped storage plants.

NUCLEAR POWER STATION: Nuclear fuel – breeding and fertile materials – nuclear reactor – reactor operation. **TYPES OF REACTORS:** Pressurized water reactor, boiling water reactor, sodium-graphite reactor, fast breeder reactor, homogeneous reactor, gas cooled reactor, radiation hazards and shielding – radioactive waste disposal.

POWER PLANT ECONOMICS AND ENVIRONMENTAL CONSIDERATIONS: Capital cost, investment of fixed charges, operating costs, general arrangement of power distribution, load curves, load duration curve, definitions of connected load, maximum demand, demand factor, average load, load factor, diversity factor – related exercises. Effluents from power plants and Impact on environment – pollutants and pollution standards – methods of pollution control.

Text Books:

1. A course in Power Plant Engineering –Arora and Domkundwar, Dhanpatrai & Co.
2. Power Plant Engineering – P.C.Sharma / S.K.Kataria Pub

Reference Books:

1. Power Plant Engineering: P.K.Nag/ II Edition /TMH.
2. Power station Engineering – ElWakil / McHill.
3. An Introduction to Power Plant Technology / G.D. Rai.

MECHATRONICS (Elective-III)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To develop an ability to identify, formulate, and solve engineering problems.
2. To develop an ability to design a system, component, or process to meet desired needs within realistic constraints.
3. To develop an ability to use the techniques, skills, and modern engineering tools necessary for engineering practice.

Course Outcomes:

Upon completion of this course, students should be able to:

1. Model and analyze electrical and mechanical systems and their interconnection.
2. Integrate mechanical, electronics, control and computer engineering in the design of mechatronics systems.
3. Do the complete design building, interfacing and actuation of a mechatronic system for a set of specifications.

SYLLABUS

Mechatronics system design: Introduction to Mechatronics: What is mechatronics, Integrated design issues in mechatronics, Mechatronics key elements, the mechatronics design process, Advanced approaches in mechatronics.

Modelling and simulation of physical systems: Simulation and block diagrams, Analogies and impedance diagrams, Electrical systems, Mechanical translational systems, Mechanical rotational systems, Electromechanical coupling, Fluid systems.

Sensors and transducers: An introduction to sensors and transducers, Sensors for motion and position measurement, Force, torque and tactile sensors, Flow sensors, Temperature-sensing devices. Actuating devices: Direct current motor, Permanent magnet stepper motor, Fluid power actuation.

Signals, systems and controls: Introduction to signals, systems and controls, System representation, Linearization of nonlinear systems, Time delays.

Real time interfacing: Introduction, Elements of a data acquisition and control system, Overview of the I/O process, Installation of the I/O card and software.

Advanced applications in mechatronics: Sensors for condition monitoring, Mechatronic control in automated manufacturing, Artificial intelligence in mechatronics, Microsensors in mechatronics.

Text Books:

1. Mechatronics System Design by Devdas Shetty and Richard A. Kolk, P.W.S. Publishing Company, 2001.
2. Mechatronics by W. Bolton, Pearson Education, Asia, II-Edition, 2001.

Reference Books:

1. Introduction to Mechatronics and Measurement Systems by Michael B. Histan and David G. Alciatore, Tata McGraw Hill Company Ltd.

DESIGN FOR MANUFACTURING (Elective-III)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. Understand the design rules and considerations with reference to various manufacturing processes.
2. To discuss capabilities and limitations of each manufacturing process in relation to part design and cost.
3. To examine DFM principles including how the design affects manufacturing cost, lean manufacturing, six sigma, etc.

Course Outcomes:

1. Select the design principle, suitable material, mechanism, fit and tolerance for designing a product/component.
2. Select the appropriate material, proper working principle and a feasible design.
3. Design (optimum) a component which requires less material removal, easy to machine, assemble, access and cost effective.
4. Redesign the uneconomical casting design and know the applications of DFMA.
5. Incorporate the Environmental Objectives, issues and guidelines into the design.

SYLLABUS

Introduction: Design philosophy-steps in design process-general design rules for manufacturability-basic principles of designing for economical production – creativity in design.

Machining processes: Overview of various machining processes-general design rules for machining-dimensional tolerance and surface roughness- Design for machining – ease – redesigning of components for machining ease with suitable examples. General design recommendations for machined parts.

Metal casting: Appraisal of various casting processes, selection of casting process,-general design considerations for casting-casting tolerance-use of solidification, simulation in casting design-product design rules for sand casting.

Metal joining: Appraisal of various welding processes, factors in design of weldments – general design guidelines-pre and post treatment of welds- effects of thermal stresses in weld joints-design of brazed joints. Forging: Design factors for forging – closed die forging design – parting lines of dies – drop forging die design – general design recommendations.

Extrusion & Sheet metal work: Design guide lines extruded sections- design principles for punching, blanking, bending, deep drawing-Keeler Goodman forging line diagram – component design for blanking.

Plastics: Visco elastic and creep behavior in plastics-design guidelines for plastic components- design considerations for injection moulding – design guidelines for machining and joining of plastics.

Text Books:

1. Design for manufacture, John cobert, Adisson Wesley 1995
2. Design for Manufacture by Boothroyd

Reference Books:

1. Design for manufacture, James Bralla
2. ASM Hand book Vol.20.

HEAT TRANSFER LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. This course is designed to introduce a basic study, the phenomena of heat and mass transfer, and to provide useful information concerning the performance and design of particular systems and processes.
2. A knowledge-based design problem requiring the formulations of solid conduction and fluid convection and the technique of numerical computation.
3. Examine the basic concepts of heat transfer models - thermal gradients, conduction, convection, and radiation.
4. To help the student develop skills that would apply to lifelong learning.

Course Outcomes:

Upon successful completion of this course, the student will be able to

1. Understand the basic laws of heat transfer, account for the consequence of heat transfer in thermal analyses of engineering systems.
2. Will be able to apply their knowledge of Dimensional Analysis to forced and free convection.
3. Analyze heat exchanger performance by using the method of log mean temperature difference, heat exchanger, effectiveness.
4. Calculate radiation heat transfer between black body surfaces and gray body surfaces.
5. Demonstrate and explain mechanism of boiling and condensation.

LIST OF EXPERIMENTS

1. Determination of Heat Transfer through Lagged Pipe.
2. Measurement of Thermal Conductivity for a given Asbestos Insulating powder.
3. Determination of Thermal Conductivity for a Given Copper Metal Rod.
4. Determination of Heat Transfer through Pin-Fin.
5. Experimentation on Transient Heat Conduction.
6. Determination of Heat Transfer through Forced Convection.
7. Determination of Heat Transfer through Natural Convection.
8. Determination of overall heat transfer coefficient for Parallel and Counter Flow Heat Exchanger.
9. Emissivity Measurement.
10. Measurement of Stefan Boltzmann constant.
11. Determination of Heat Transfer through Drop Wise and Film Wise Condensation.

12. Determination of Two phase heat Transfer.
13. Determination of Overall Heat Transfer Co-Efficient for Composite Wall.
14. Study of Refrigeration Test Rig.
15. Study of Air Conditioning Test Rig.

Reference Books:

1. Heat Transfer, by J.P.Holman, Int. Student edition, McGraw Hill book company.
2. Analysis of Heat transfer, by Eckert and Drake, Int.Student edition, McGraw Hill Kogakusha Ltd.
3. Heat and Mass Transfer by R.K. Rajput, S. Chand & Co.
4. Heat and mass transfer by Sachdeva.
5. Heat and mass transfer by Kothandaramanna, New Age International.

PROJECT PHASE-I**Lab : 3 Periods****Sessionals : 50****Credits : 2**

Course Outcomes:

1. Identify a current problem through literature/field/case studies and define the background objectives and methodology for solving the same.
2. Write report and present it effectively.

The phase-I of the project shall comprise of

- Problem identification in close collaboration with industry.
- Literature survey.
- Deriving work content and carry out of project requirement analysis.
- Submission of interim report.
- Presentation to an expert committee.

(Note: Sessionals 50 marks will be awarded based on
Continuous evaluation - 25 Marks
Seminar and Viva voce - 25 Marks.)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

IV/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards)
Under Choice Based Credit System
MECHANICAL ENGINEERING

II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ME 4201	Computer Aided Manufacturing	4	3	1	-	4	30	70	100
B16 ME 4202	Quality Control and Assurance	4	3	1	-	4	30	70	100
B16 ME 4203	CAD/CAM Lab	2	-	-	3	3	50	50	100
B16 ME 4204	Project Phase-II	12	-	-	9	9	50	100	150
	Total	22	6	2	12	20	160	290	450

COMPUTER AIDED MANUFACTURING

Theory : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 4

Course Objectives:

In this course students can acquire knowledge in the following areas:

1. Knowledge of various NC machine tools, Manual part programme and APT language
2. CNC Retrofitting
3. Group Technology, Various types of CAPP systems
4. Robots, AGV'S, Automated conveyors systems
5. Computer Aided PPC, MRP-I, MRP-II
6. Elements of FMS centre and CIM

Course Outcomes:

1. After completion of the course students can
2. Handle various NC machines,
3. Write G.T codes for any complex component, a
4. apply techniques on various old machines and converted into retrofitted,
5. Gain knowledge on various types of robots, AGV'S, Automated conveyors systems, FMS centers.

SYLLABUS

Introduction to CNC and CAM, CNC retrofitting, Adoptive control machining, NC part program preparation through computer languages. Group technology: Merits & demerits, Organisation, Classification and Coding systems, Facilities layout.

Computer aided process planning: Introduction to process planning, Methods of process planning, Computer aided process planning, CAPP systems, case studies.

Computer aided material handling and production planning: Robots: Structure and operation of Robots, robot sensors and applications. Automatic conveyor systems. Automated guided vehicles. Aid of computer in production planning and control, Inventory control and material requirement planning.

Computer aided inspection and quality control: Developments and practice, Quality assurance and quality control. Coordinate measuring machine. Non-contact inspection.

FMS & CIMS: Building blocks of Flexible Manufacturing Systems (FMS), Machining systems of FMS, Tool management systems, Advantages of FMS, Computer integrated manufacturing systems (CIMS).

Text Books:

1. Computer Aided Manufacturing, by P.N.Rao, N.K.Tewari&T.K.Kundra, Tata McGraw-Hill publishing company Ltd, NewtDelhi.
2. Automation, Production Systems and Computer Integrated Manufacturing, by MikellP.Groover, Prentice-Hall of India Pvt. Ltd.

Reference Books:

1. Computer Integrated Design and Manufacturing, by David D.Bedworth, Mark R.Henderson& Philip M.Wolfe, McGraw-Hill Book Company, Singapore.

QUALITY CONTROL AND ASSURANCE

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

The overall objective of the course is to teach the basic principles of Quality management which includes Taguchi's loss function, Deming's philosophy.

1. To understand the purpose and function of statistical quality control.
2. To understand the difference between attributes and variables.
3. To become familiar with basic methods of statistical process control.

Course Outcomes:

1. Stewart's normal bowl, control charts for variables, $\bar{X}$, R and sigma control charts.
2. Control charts for attributes, p-chart, standardized p-chart, np-chart, c-chart, u-chart, demerit control chart.
3. Type-I and Type-II errors, Process capability analysis.
4. Sampling plans: single, double, multiple and sequential sampling plans, rectifying inspection, AOQ, AOQL, and ATI. Use of Dodge Romig Tables, Design of single and sequential sampling plans.

SYLLABUS

Quality control in Perspective -Introduction to quality, quality assurance, quality control; quality of design, quality of conformance and quality of performance; quality characteristics – variables and attributes, growth of quality control, Statistical quality control, Taguchi's loss function, examples of off-line and on-line quality control techniques, quality costs, Deming's philosophy, introduction to six sigma concept.

Control charts for Variables: Shewart's norm bowl, $\bar{X}$ and R charts, $\bar{X}$ and σ charts, Statistical control of processes, group control chart, $\bar{X}$ chart with linear trend, warning limits.

Control charts for Attributes: Defect and defective, fraction defective and percent defective, p-chart, 100p-chart, np-chart, c-chart, u-chart, ku-chart, demerit control charts.

Process capability analysis: Determination of process capability, PCR, Design specifications and tolerances, PCR for nominal the better type, smaller the better type and larger the better type product specifications; Tolerances for sub assemblies, setting tolerances for intermediate steps in production.

Acceptance sampling plans: Single, double, multiple and sequential sampling plans, OC curve, rectifying inspection, AOQ, AOQL, ASN and ATI, Use of Dodge Romig Tables, Design of single and sequential sampling plans.

Text Books:

1. Statistical Quality Control by E.L.Grant and Leavenworth, McGraw Hill
2. Quality control and application by Bertrand.L.Hansen and P.M.Ghare, PHI.

Reference Books:

1. Introduction to Statistical Quality Control by D.C.Montgomery, Wiley.
2. Principles of Quality control by Jerry Banks, John Wiley.
3. Quality control hand book by Juran, McGraw Hill.

CAD/CAM LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. Introduction to AUTOCAD package in design and drafting of the different parts by using computer aided modelling.
2. Develop the design skills of the students to practice the different 2D/3D engineering drawings.
3. Application of CAD packages to solving the simple problems in modeling and analysis.
4. Motivation of the students towards the good position in automated and software industries as a design engineer.

Course Outcomes:

1. Students will be able to know to produce the industrial drawings by using CAD/CAM software's.
2. After successful completion of this laboratory student can do the job in CAD/CAM industry as a design engineer.

LIST OF EXERCISES**CAD Exercises:**

1. Initiating the graphics package; Setting the paper size, space; setting the limits, units; use of snap and grid commands.
2. Drawing of primitives (line, arc, circle, ellipse, triangle etc.)
3. 3D GEOMETRIC MODELING: Creation of 3D Models, Wire Frame, Surface, Solid modeling Techniques Using CAD Packages – CSG, B-Rep Approaches in Solid Modeling Feature Based Modeling Technique – Assembly – Detailing Exposure to Industrial Components – Application of GD&T.
4. Drawing a flange.
5. Drawing a Bushing assembly.
6. Dimensioning the drawing and adding text.
7. Setting the layers and application of the layers.
8. Isometric and orthographic projections.
9. Viewing in Three dimensions.
10. Removal of hidden lines - Shading and rendering.

CAM Exercises:

1. Preparation of manual part programming for CNC turning/Milling.
2. Part programming preparation through AutoCAD.

3. APT part programming for 2D - contour.
4. Machining of one job on CNC machine tool.
5. Robot programming through Teaching Box method.
6. Robot programming through computer.

Reference Books:

1. Solid Works Reference Guide by CADD Centre.
2. CAD/CAM Theory and Practice by Ibrahim Zeid.
3. CAD/CAM Principles and Applications by P.N. Rao, Tata McGraw Hill Publishing Company Ltd.
4. CAD/CAM Computer Aided Design and Manufacturing by Mikell P. Groover and Emory W. Zimmer, Jr.
5. Computer Integrated Design and Manufacturing by David D. Bedworth, Mark R. Henderson, Philip M. Wolfe.

PROJECT PHASE-II**LAB : 9 Periods**

Sessionals	: 50
Ext. Marks	: 100
Credits	: 12

Course Outcomes:

1. Identify a current problem through literature/field/case studies and define the background objectives and methodology for solving the same.
2. Analyze, design and develop a technology/ process.
3. Implement and evaluate the technology at the laboratory level.
4. Write report and present it effectively.

The phase-II of the project shall consists of

Implementing, Testing and validation.
Report Writing.

Sessionals (50 Marks) will be awarded by the Project Guide based on continuous evaluation. External Evaluation (100 marks) of project report and viva voce will be conducted by a committee consisting of HOD, Guide and External Examiner.

May be carried out using in-house facilities or in an industry by specified number of students in a group.

Format for Preparation of Project Thesis for B. Tech:

1. Arrangement Of Contents: The sequence in which the project report material should be arranged and bound should be as follows:
 1. Cover Page & Title Page .
 2. Bonafide Certificate
 3. Abstract.
 4. Table of Contents
 5. List of Tables
 6. List of Figures
 7. List of Symbols, Abbreviations and Nomenclature
 8. Chapters
 9. Appendices
 10. References

*The table and figures shall be introduced in the appropriate places.